PEARL Power and Energy Analysis and Research Laboratory

Escuela de Ingeniería Industrial y Aeroespacial Toledo
Universidad de Castilla-La Mancha

UCLM Universidad Castilla-La Mancha

Tools for the optimal operation and planning of power systems

Lecture 5: Further applications

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Universidad de Castilla-La Mancha, Spain

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Contents

1. Manual frequency restoration reserve-based demand response program
2. Single imbalance pricing scheme via stochastic robust optimization
3. Students' Projects Supervised
4. Topics beyond this course

1. Manual frequency restoration reserve (mFRR)-based demand response program

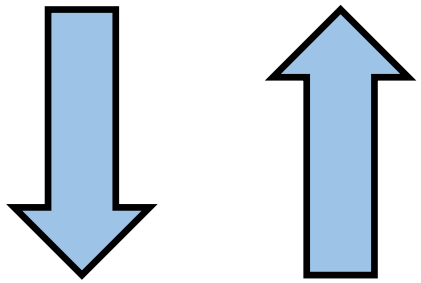
mFRR-based demand response program

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

1.1. Context

Balance of energy

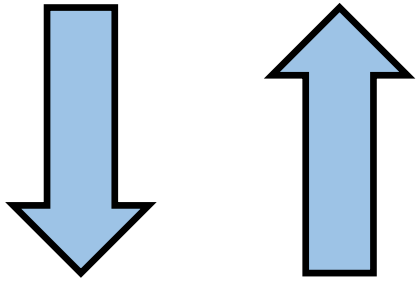
GENERATION = CONSUMPTION

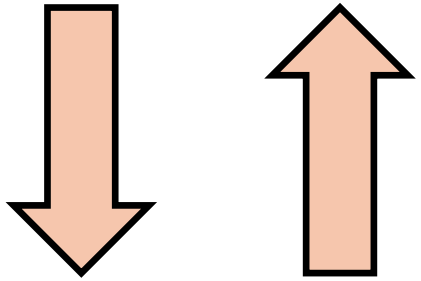


A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Balance of energy

GENERATION = CONSUMPTION





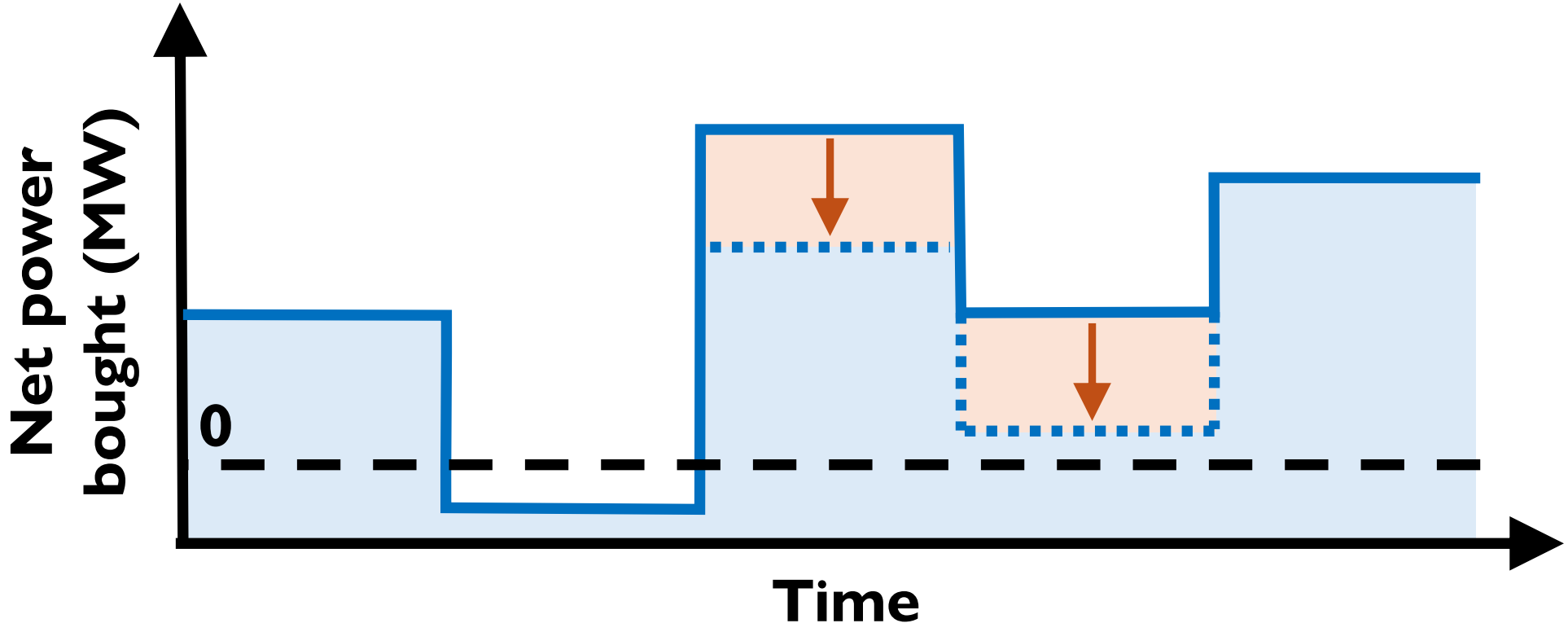
Demand Response

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Demand Response (DR)

Grid-Balancing Demand Response programme

Active Demand Response Service (ADRS)



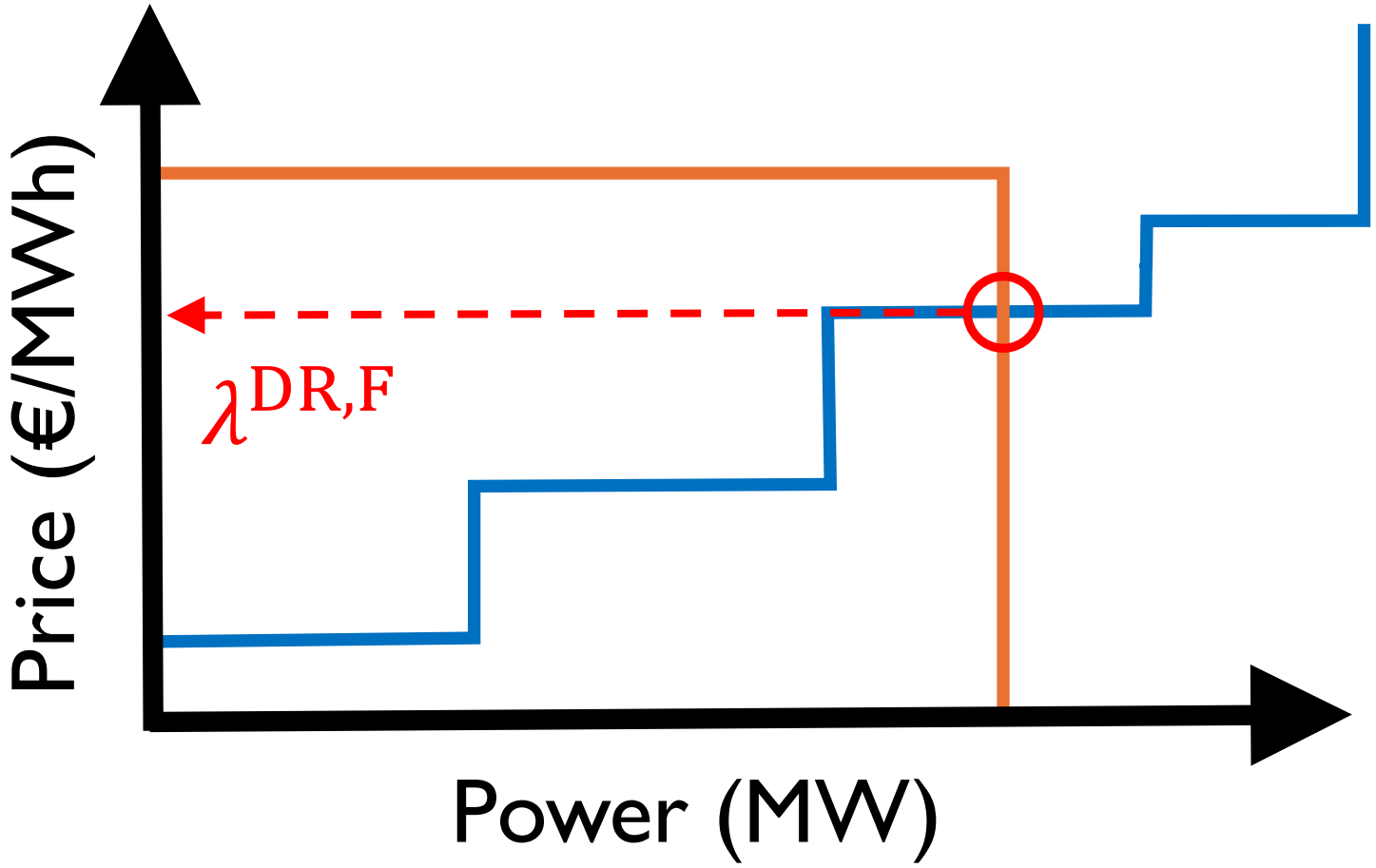
A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

ADRS – Rules

- Be prepared in certain hours: approx. 50%.
- Maximum duration of the activation: 3 h (once per day).
- Remuneration for participants
  - Fixed (be prepared)
  - Variable (activations)
- Yearly auction.

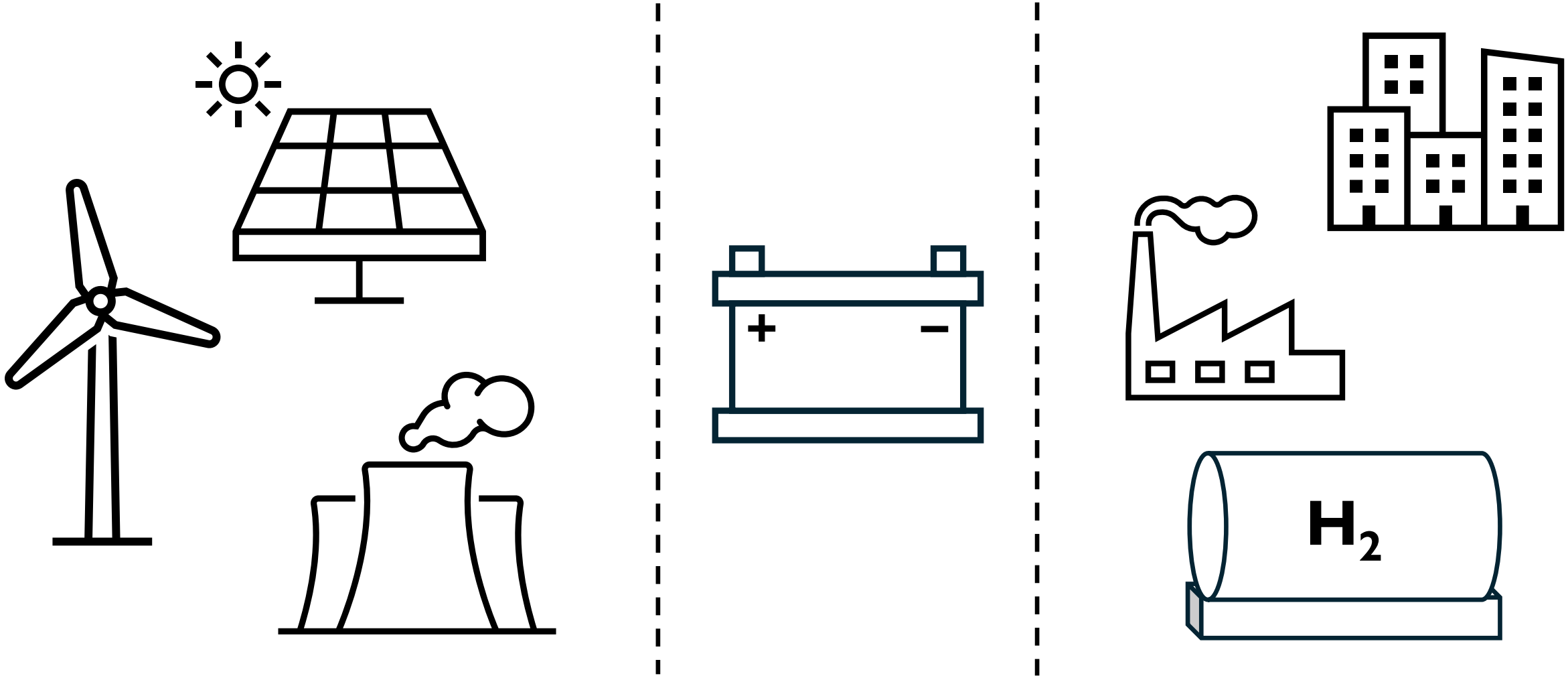
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ADRS – Rules



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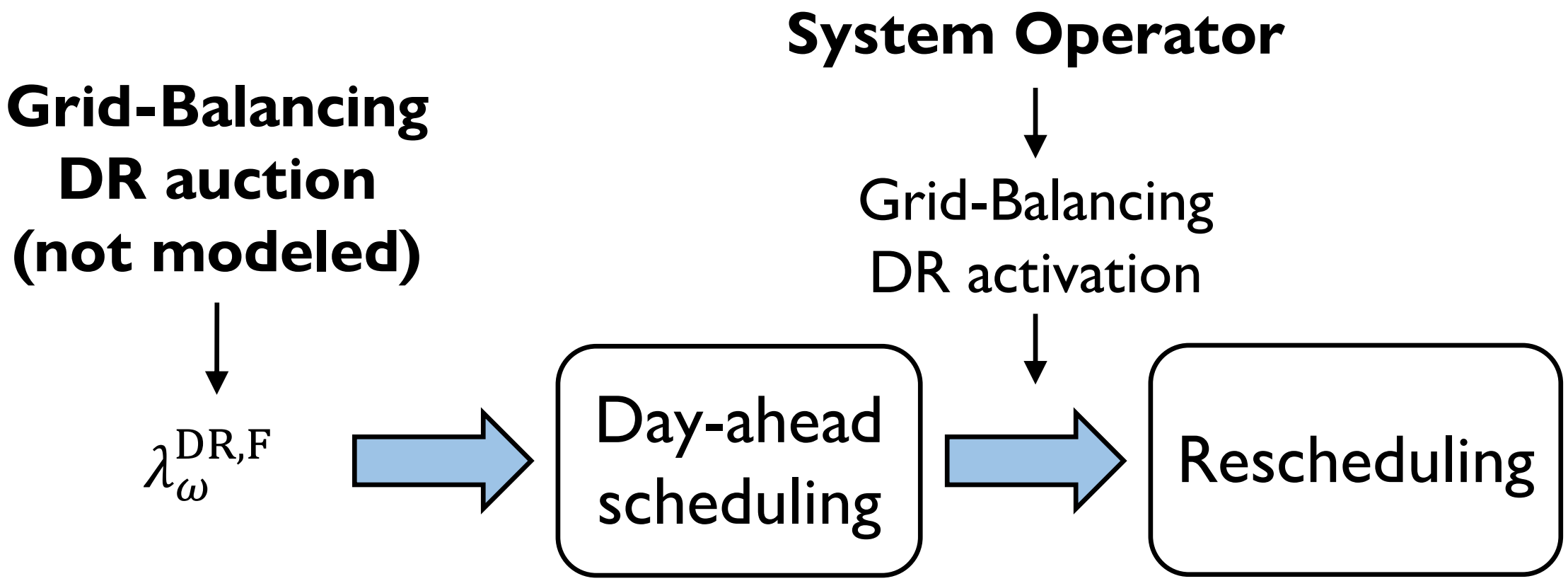
Virtual Power Plant (VPP)



A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

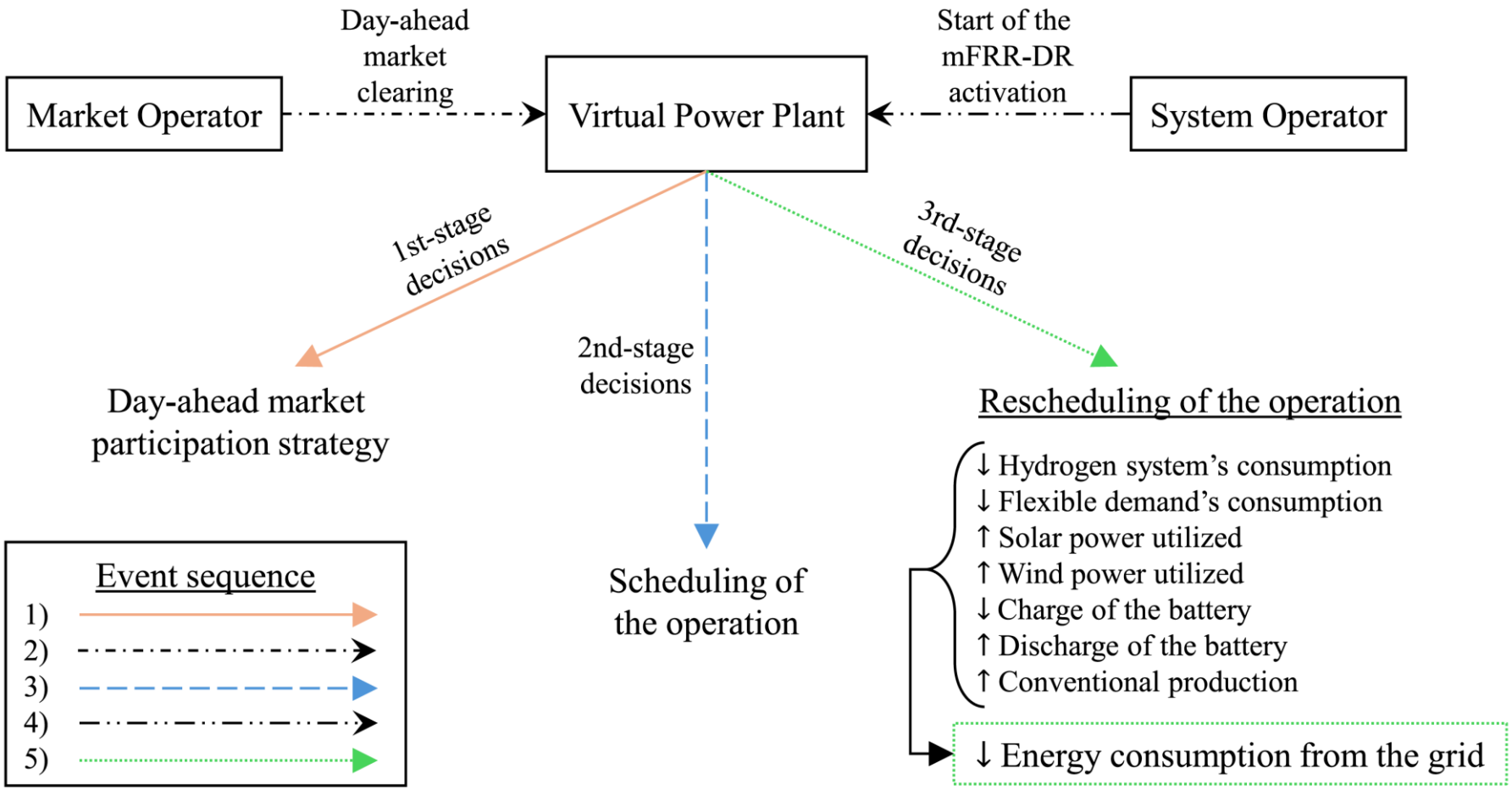
1.2. Proposed Optimization Model

Problem addressed



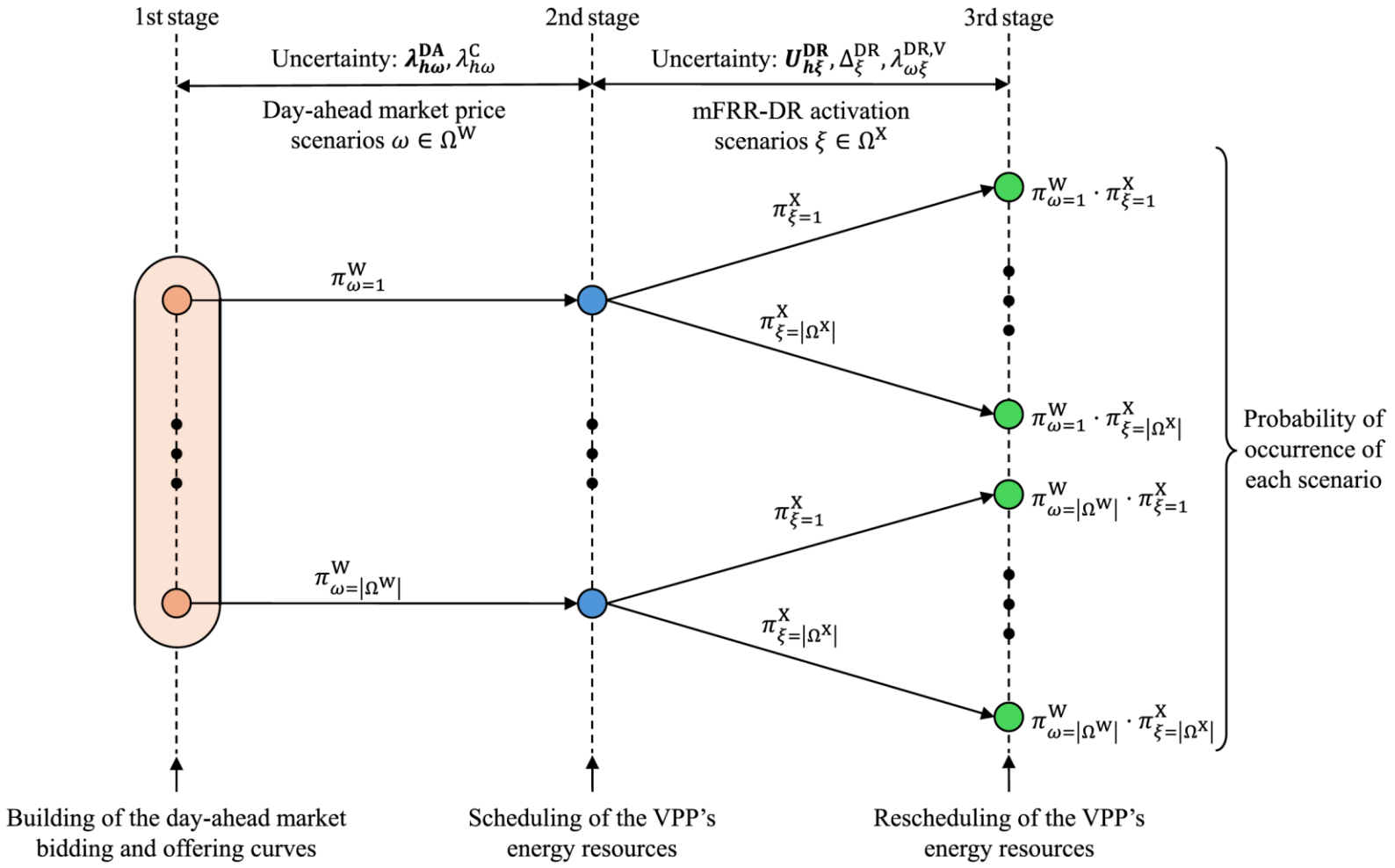
A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Problem addressed



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Problem addressed



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Formulation

The detailed formulation (70 equations) can be consulted in the paper.

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Objective function

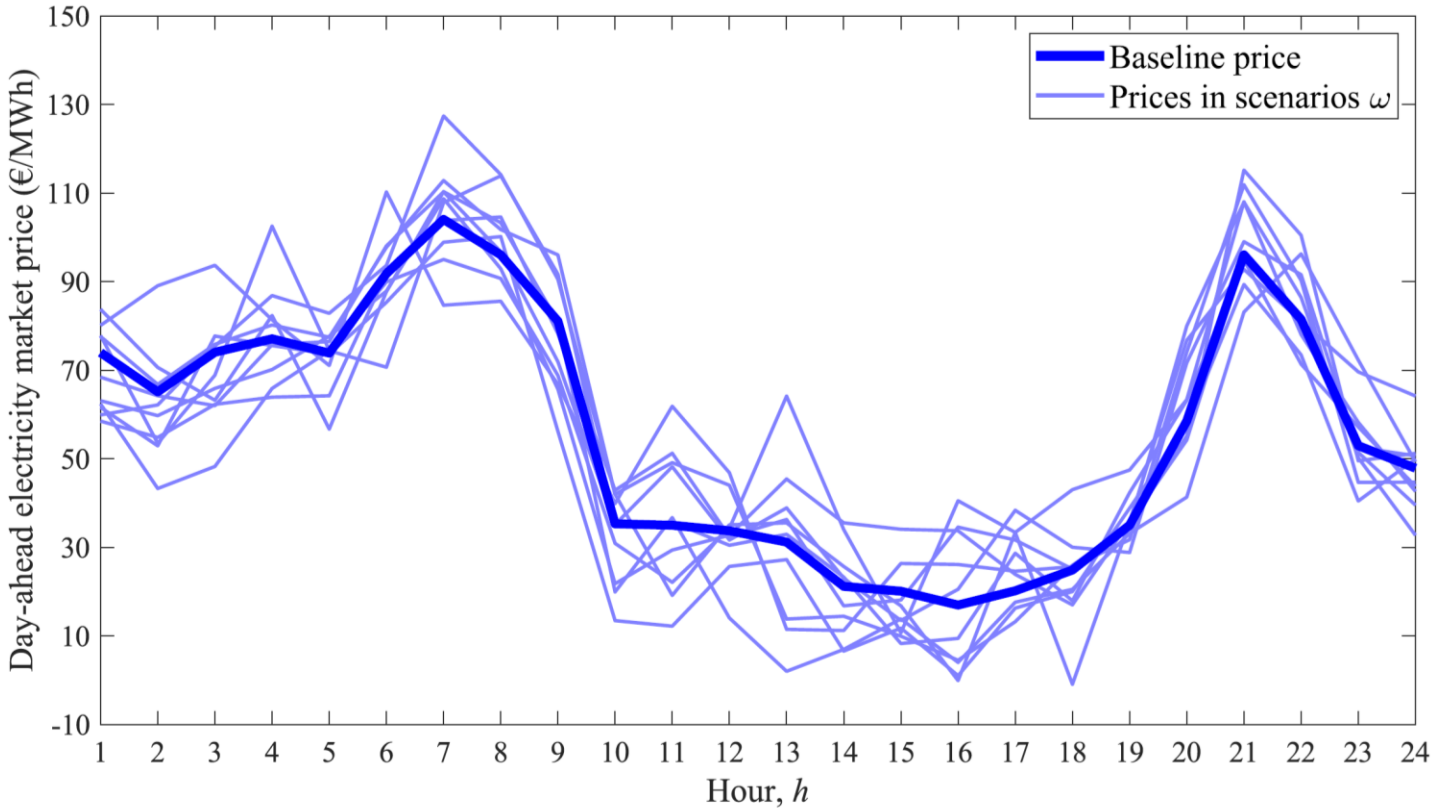
Objective function (min) terms:

- DA purchases.
- DA sales.
- Scheduling costs.
- Rescheduling costs.
- DR revenue (fixed and variable remunerations).

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Uncertainty

DA electricity market price scenarios ω.



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Formulation – Uncertainty

mFRR-DR activation scenarios ξ.

$U_{h\xi}^{\mathrm{DR}} \rightarrow$ Binary parameter set to 1 when the System Operator requests a reduction in the power consumed from the grid due to an mFRR-DR activation, being equal to 0 otherwise.

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Uncertainty

	mFRR-DR activation scenario																
Hour h	ξ_1	ξ_2	ξ_3	ξ_4	ξ_5	ξ_6	ξ_7	ξ_8	ξ_9	ξ_{10}	ξ_{11}	ξ_{12}	ξ_{13}	ξ_{14}	ξ_{15}	ξ_{16}	ξ_{17}
h_1		■															
h_2		■	■														
h_3		■	■	■													
h_4																	
h_5																	
h_6																	
h_7					■												
h_8					■	■											
h_9					■	■	■										
h_{10}						■	■	■									
h_{11}							■	■	■								
h_{12}								■	■	■							

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Formulation – Uncertainty

	mFRR-DR activation scenario																
Hour h	ξ_1	ξ_2	ξ_3	ξ_4	ξ_5	ξ_6	ξ_7	ξ_8	ξ_9	ξ_{10}	ξ_{11}	ξ_{12}	ξ_{13}	ξ_{14}	ξ_{15}	ξ_{16}	ξ_{17}
h_{13}									■	■	■						
h_{14}																	
h_{15}																	
h_{16}																	
h_{17}																	
h_{18}																	
h_{19}												■					
h_{20}												■	■				
h_{21}												■	■	■			
h_{22}													■	■	■		
h_{23}														■	■	■	
h_{24}															■	■	■

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Objective function

$$min\ \beta \cdot \mathbb{E} + (1 - \beta) \cdot \mathrm{CVaR}_{\alpha}$$

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Formulation – Scheduling

Net power bought in the DA electricity market:

$$p_{h\omega}^{\mathrm{DA}} = p_{h\omega}^{\mathrm{DA,P}} - p_{h\omega}^{\mathrm{DA,S}}, \forall h, \forall \omega$$

$$0 \leq p_{h\omega}^{\mathrm{DA,P}} \leq \overline{P}^{\mathrm{DA,P}}, \forall h, \forall \omega$$

$$0 \leq p_{h\omega}^{\mathrm{DA,S}} \leq \overline{P}^{\mathrm{DA,S}}, \forall h, \forall \omega$$

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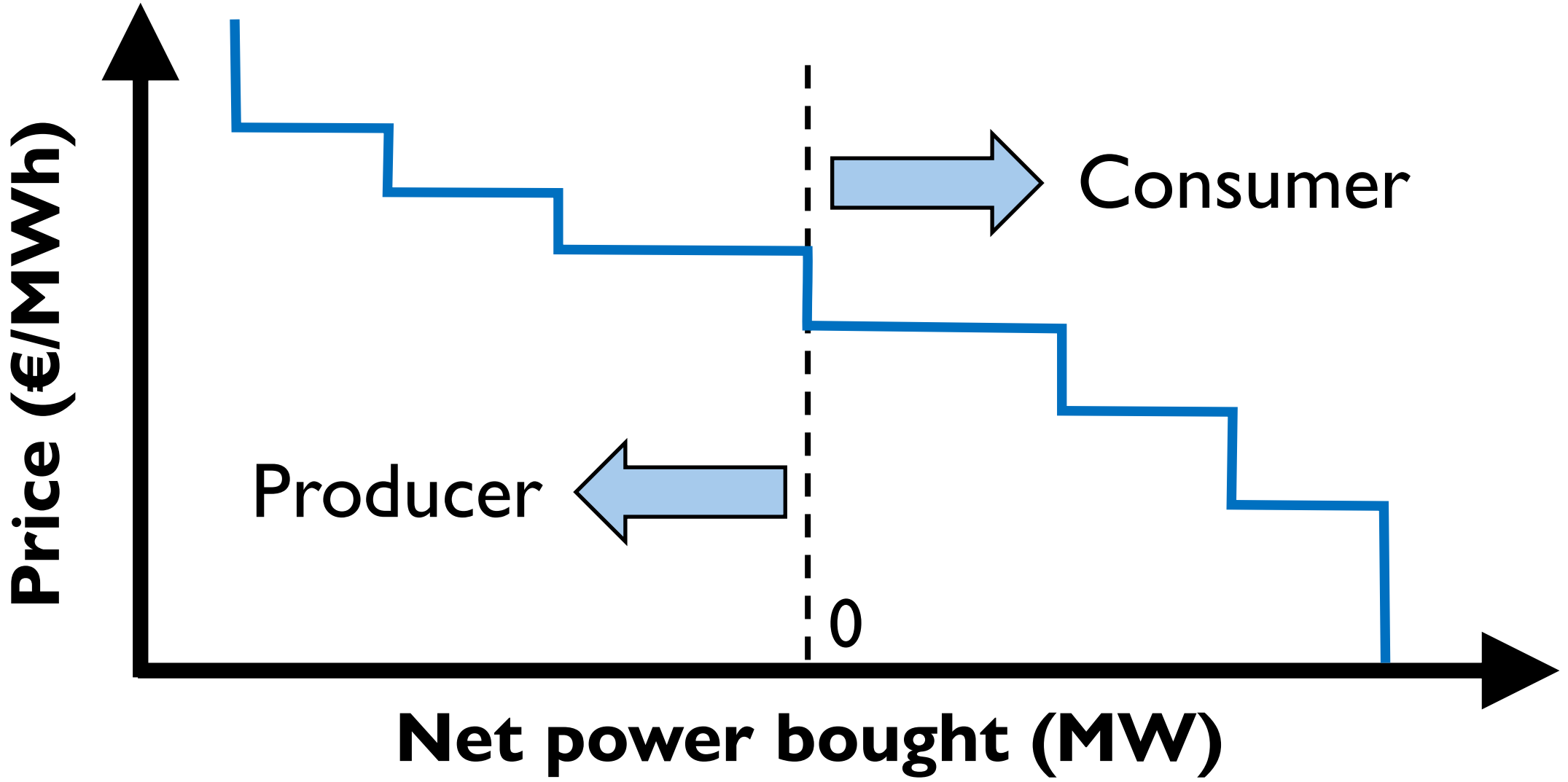
Formulation – Scheduling

Net power bought in the DA electricity market:

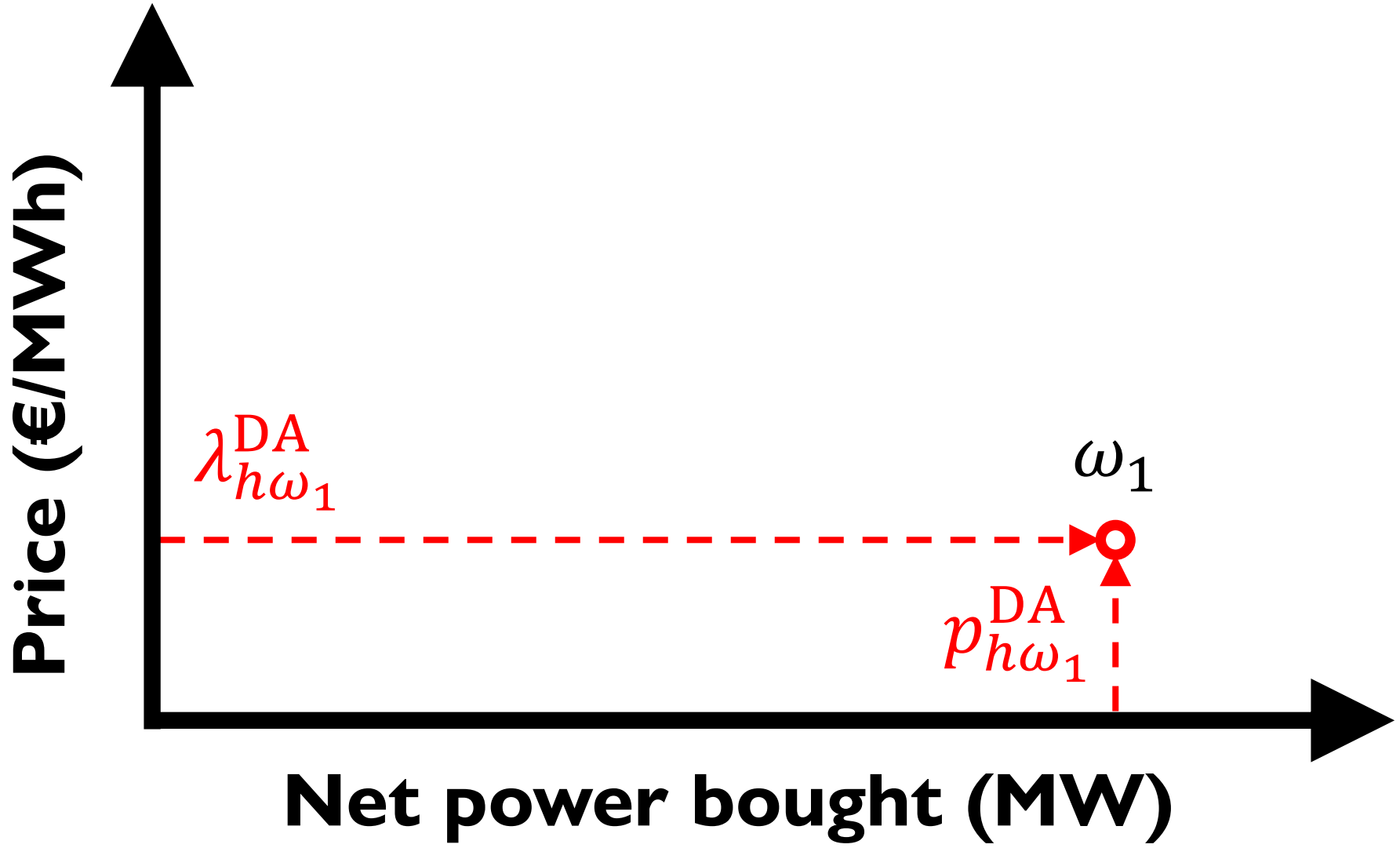
$$p_{h\omega}^{\mathrm{DA}} \geq p_{h\tilde{\omega}}^{\mathrm{DA}} \text{ if } \lambda_{h\omega}^{\mathrm{DA}} \leq \lambda_{h\tilde{\omega}}^{\mathrm{DA}}, \forall h, \forall \omega, \tilde{\omega}$$

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

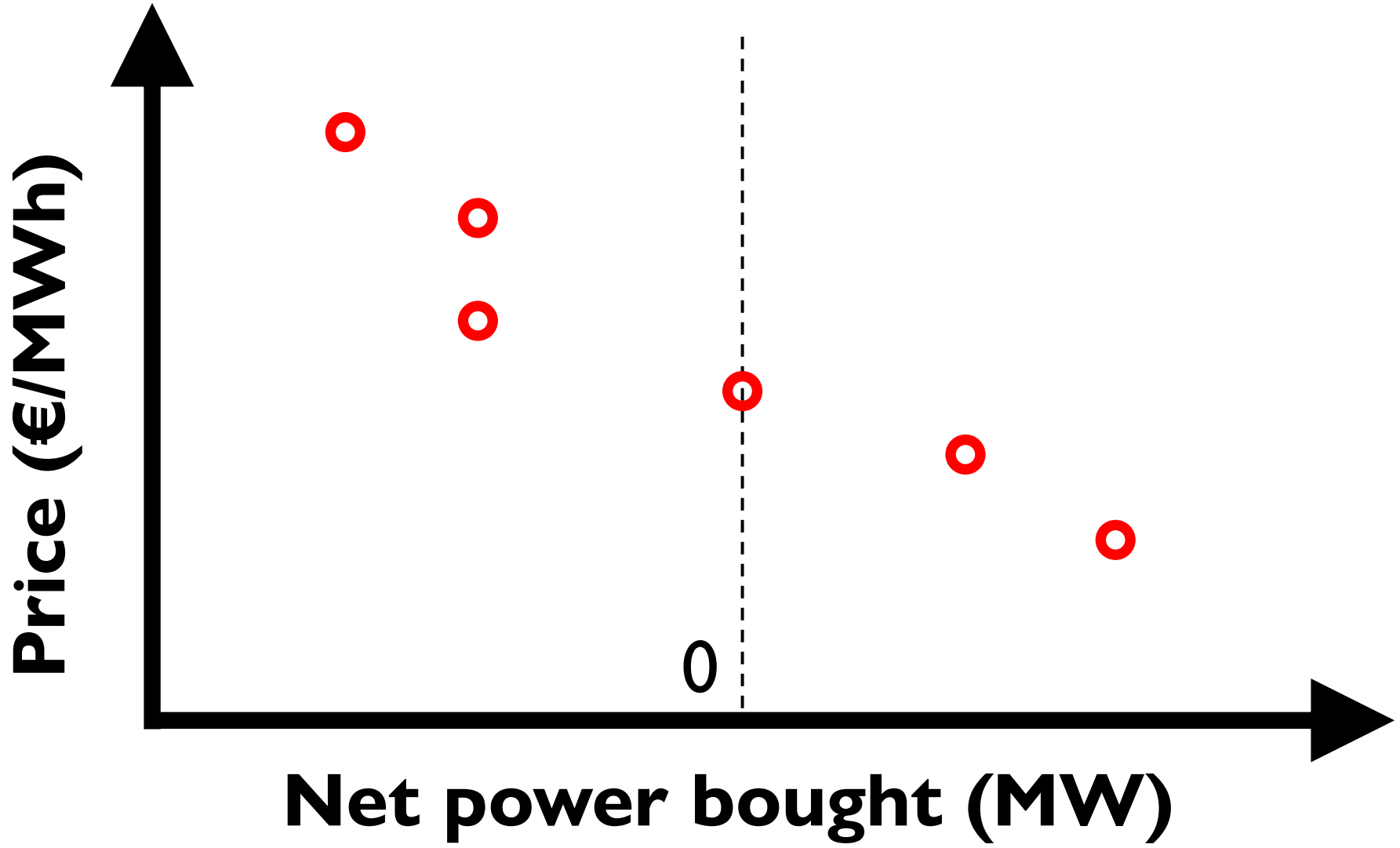
DA bidding and offering curve



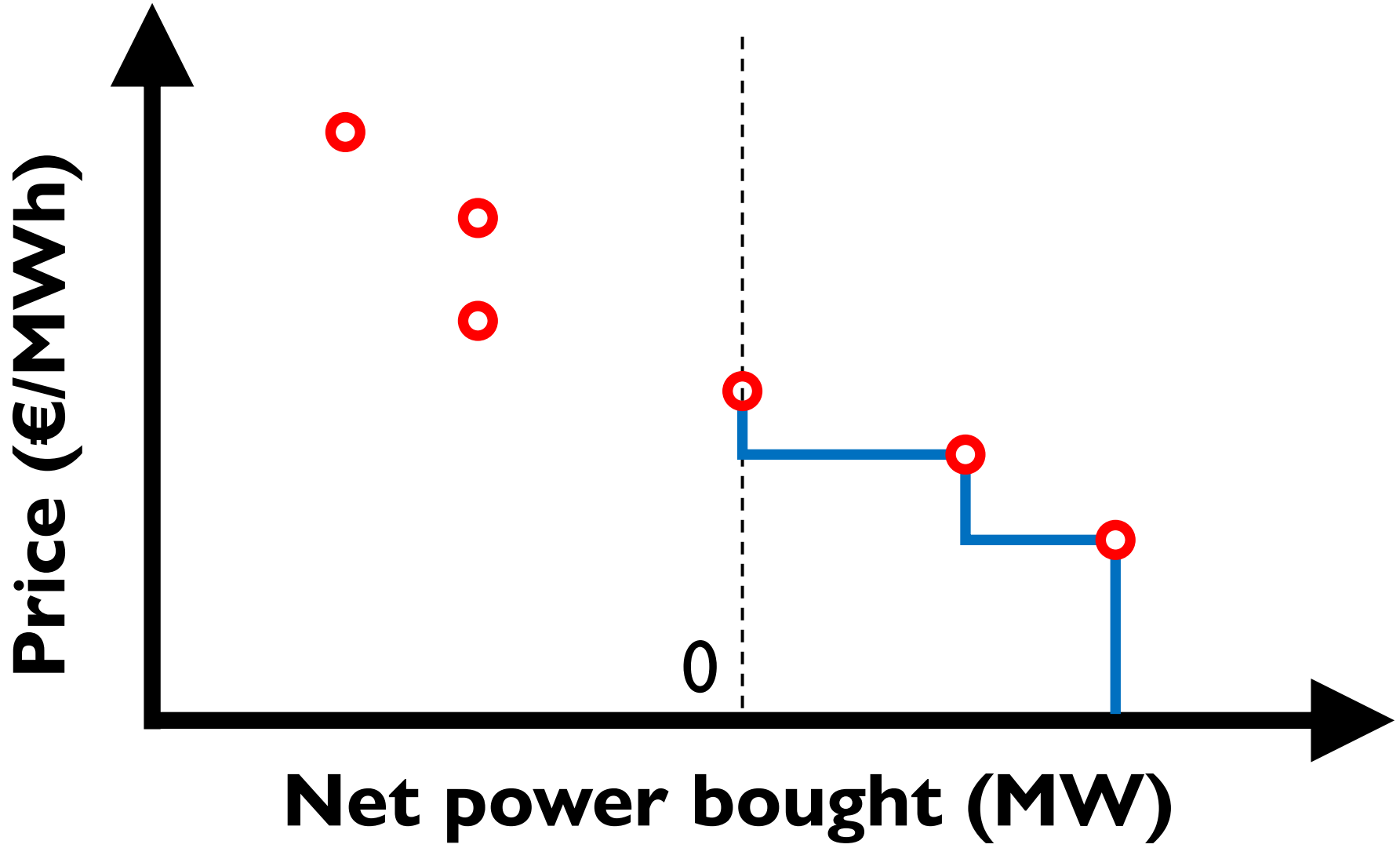
DA bidding and offering curve



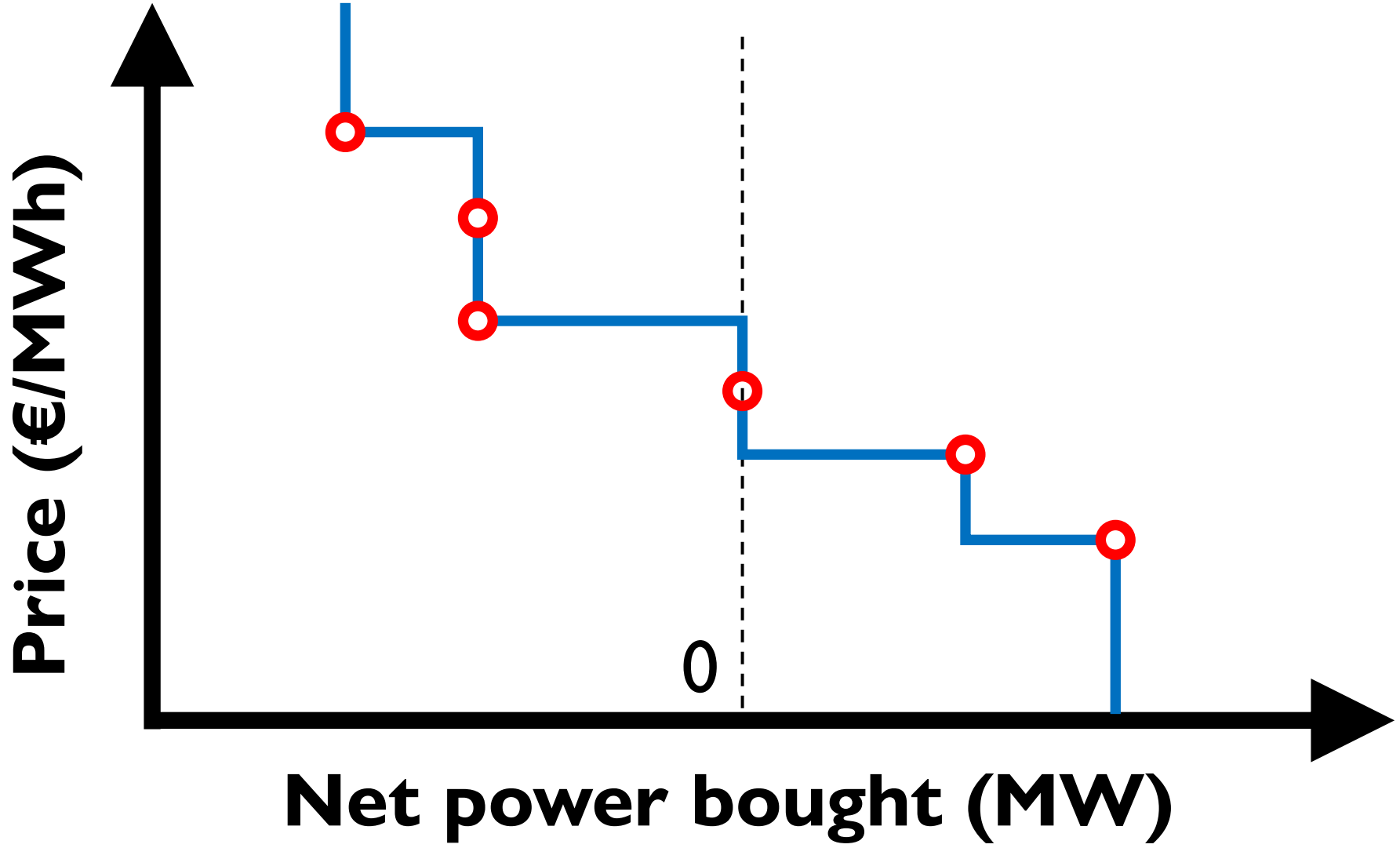
DA bidding and offering curve



DA bidding and offering curve



DA bidding and offering curve



Formulation – Scheduling

Balance of energy:

$$p_{h\omega}^{\mathrm{DA}} + p_{h\omega}^{\mathrm{C}} + p_{h\omega}^{\mathrm{PV}} + p_{h\omega}^{\mathrm{W}} + p_{h\omega}^{\mathrm{B,D}} - p_{h\omega}^{\mathrm{B,C}}$$

$$= p_{h\omega}^{\mathrm{D}} - p_{h\omega}^{\mathrm{D,U}}, \forall h, \forall \omega$$

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Formulation – Scheduling

Demand:

$$p_{h\omega}^{\mathrm{D}} = P_h^{\mathrm{D,I}} + p_{h\omega}^{\mathrm{D,F}} + p_{h\omega}^{\mathrm{H}}, \forall h, \forall \omega$$

Inelastic demand

Flexible demand

Hydrogen system's demand

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Scheduling

Hydrogen system:

$$p^{\mathrm{H}}_{h\omega} = p^{\mathrm{H,E}}_{h\omega} + p^{\mathrm{H,C}}_{h\omega}, \forall h, \forall \omega$$

Electrolyzer

Compressor

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Formulation – Scheduling

Scheduling of the resources:

- Solar-power generating unit.
- Wind-power generating unit.
- Battery.
- Conventional generating unit.
- Flexible demand.
- Hydrogen system.

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Formulation – Rescheduling

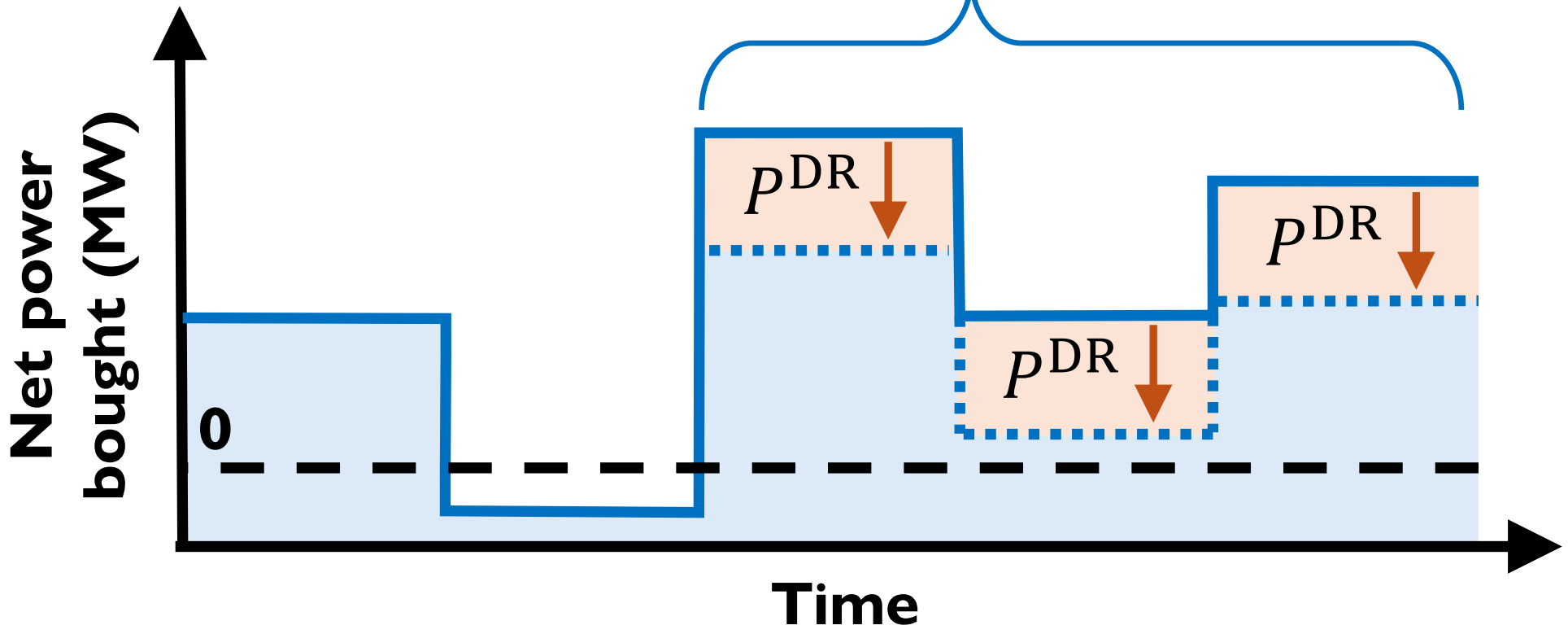
Rescheduling of the resources:

- Solar-power generating unit.
- Wind-power generating unit.
- Battery.
- Conventional generating unit.
- Flexible demand.
- Hydrogen system.

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Formulation – Rescheduling

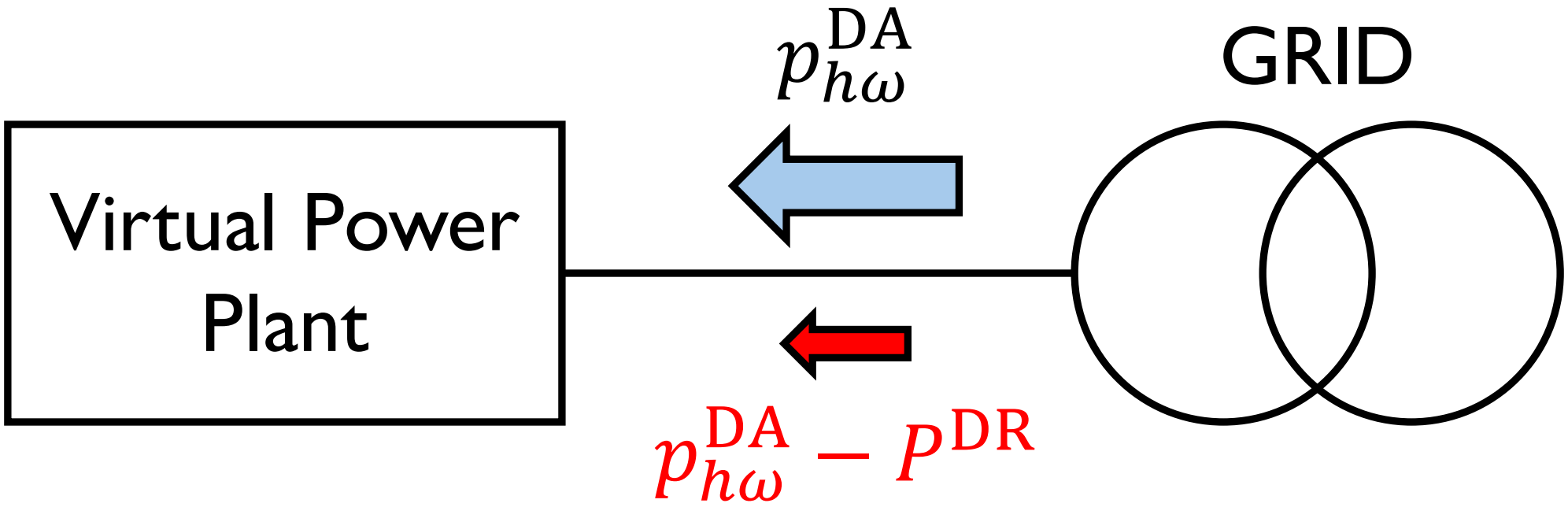
$$p_{h\omega}^{\mathrm{DA}} \geq P^{\mathrm{DR}}, \forall h \in \boldsymbol{\Omega^{H,DR}}, \forall \omega$$



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Formulation – Rescheduling

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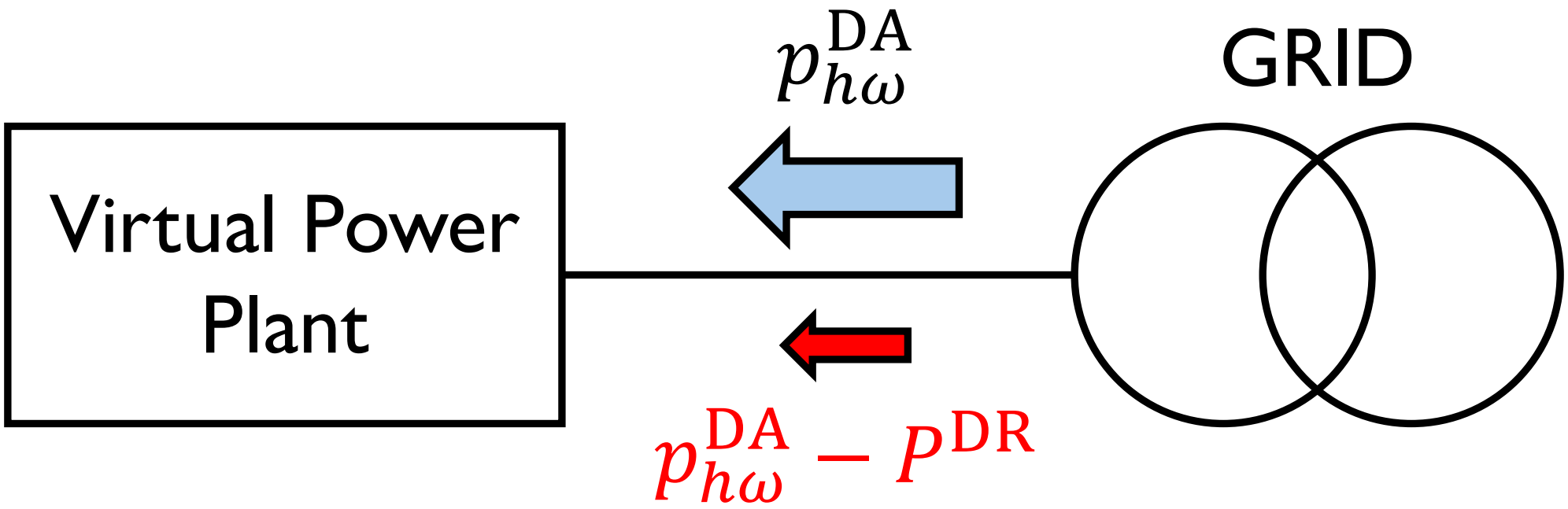


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Formulation – Rescheduling

Net consumption

- Increase internal production
- Reduce internal consumption



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Formulation – Rescheduling

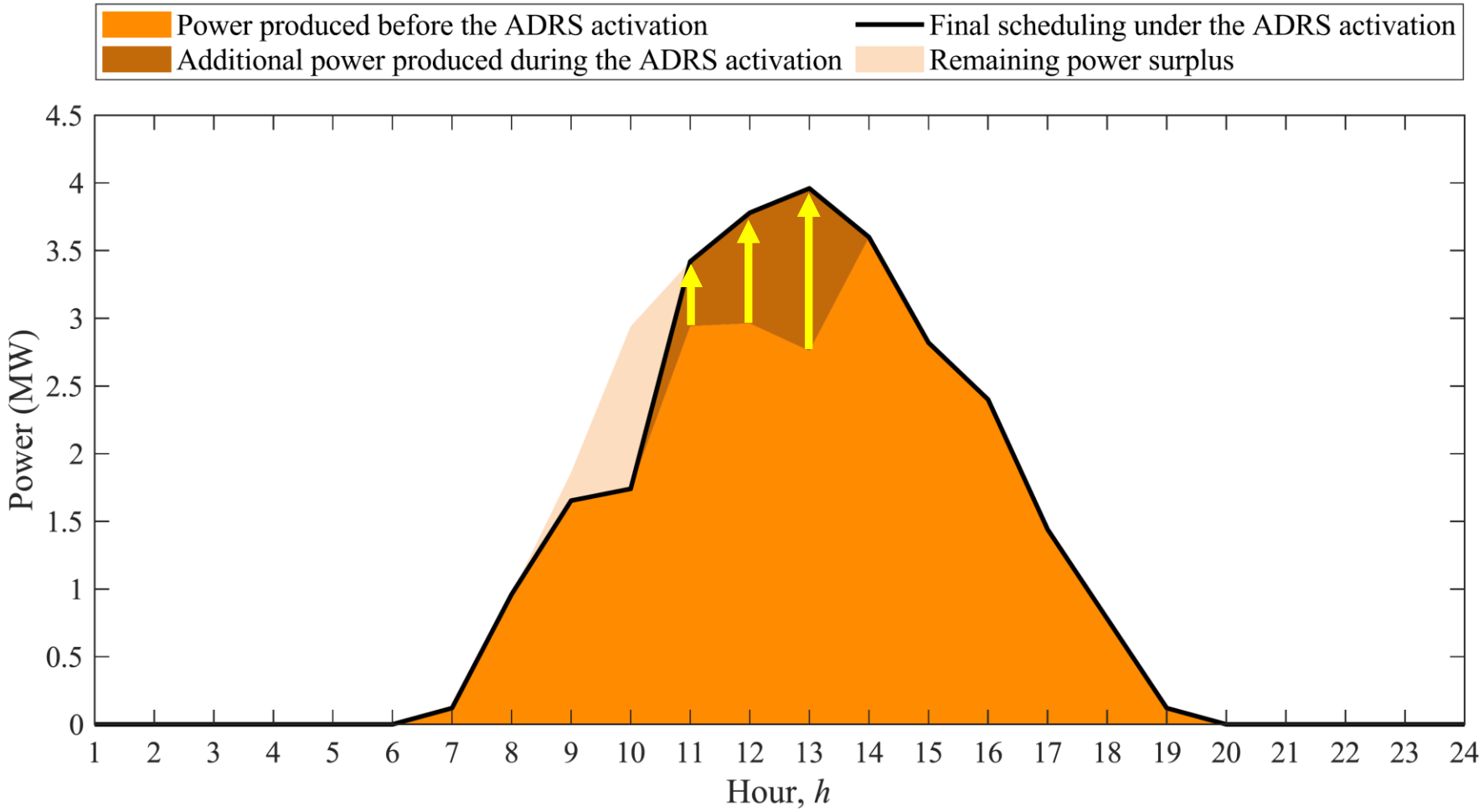
The reduction of the power consumed from the grid should be fulfilled by using the VPP's resources:

$$U_{h\xi}^{\mathrm{DR}} P^{\mathrm{DR}} = p_{h\omega\xi}^{\mathrm{H,DR}} + p_{h\omega\xi}^{\mathrm{D,F,DR}} + p_{h\omega\xi}^{\mathrm{PV,DR}} + p_{h\omega\xi}^{\mathrm{W,DR}}$$

$$+ p_{h\omega\xi}^{\mathrm{B,C,DR}} + p_{h\omega\xi}^{\mathrm{B,D,DR}} + p_{h\omega\xi}^{\mathrm{C,DR}}, \forall h, \forall \omega, \forall \xi$$

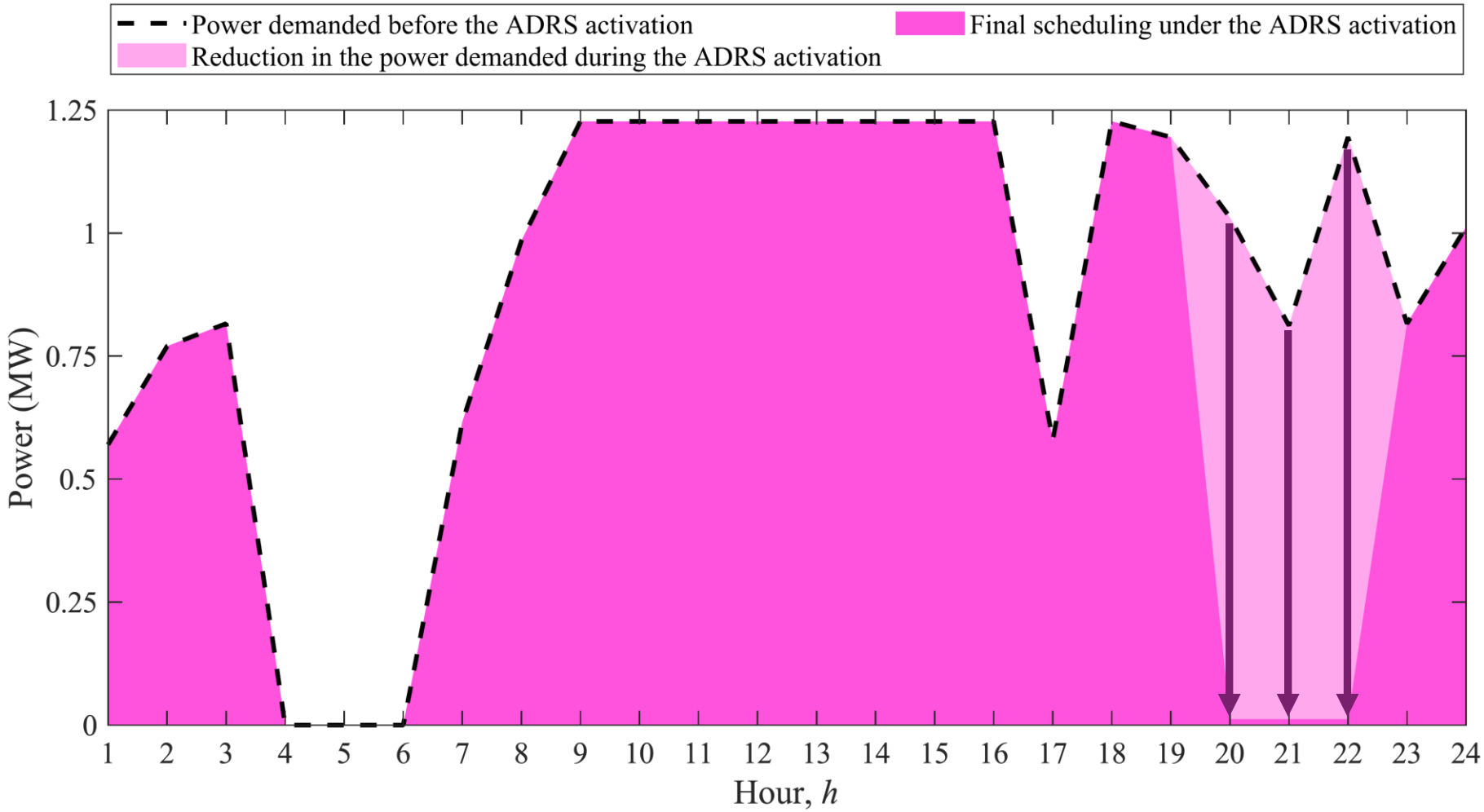
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Formulation – Rescheduling



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Formulation – Rescheduling



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Formulation – Rescheduling

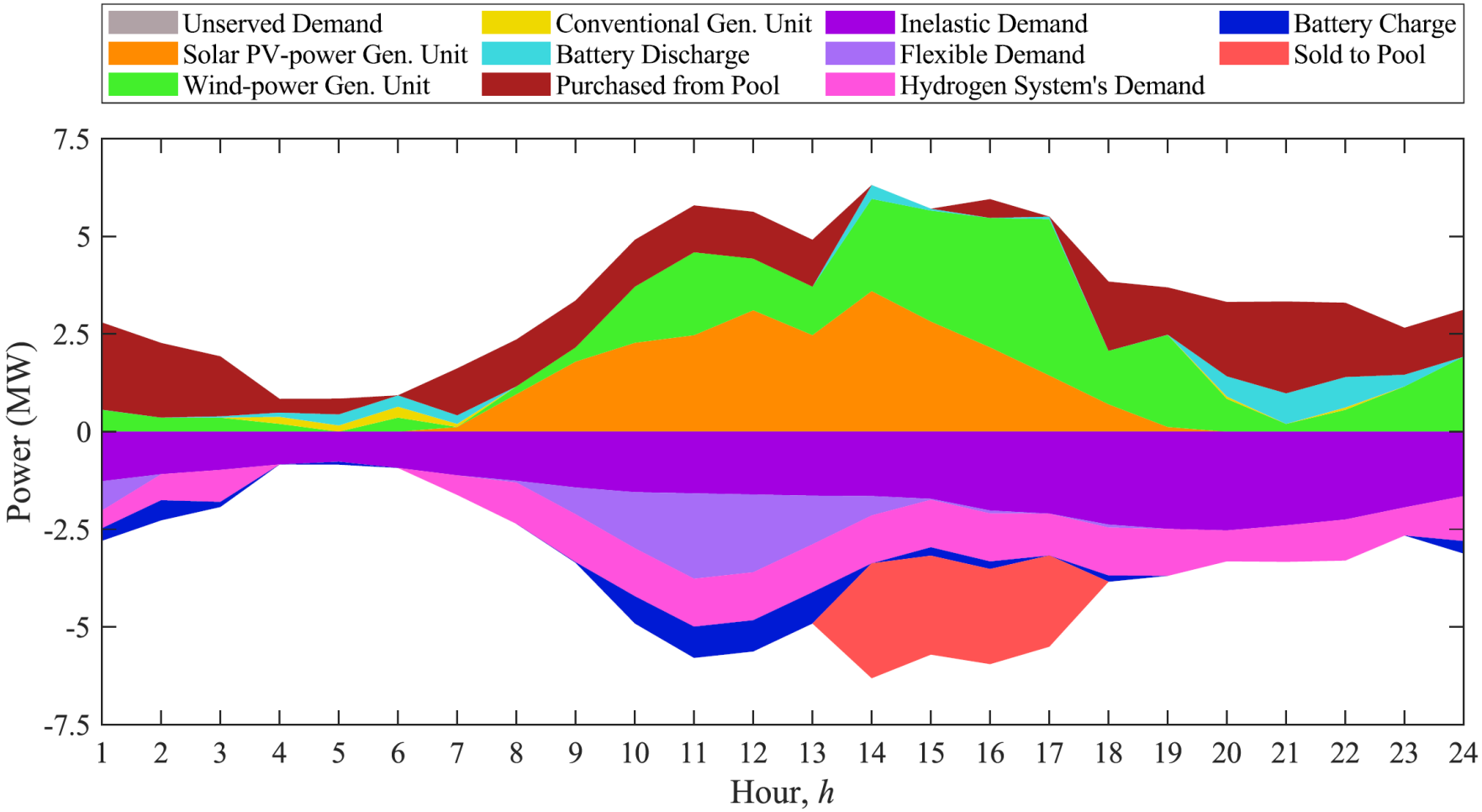
Recovery of energy reduced:

$$e_{\omega\xi}^{\mathrm{DA,R}} = \sum_{h \in \Omega^{\mathrm{H,DR}}} p_{h\omega\xi}^{\mathrm{H,DR}} + p_{h\omega\xi}^{\mathrm{D,F,DR}}, \forall \omega, \forall \xi$$

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

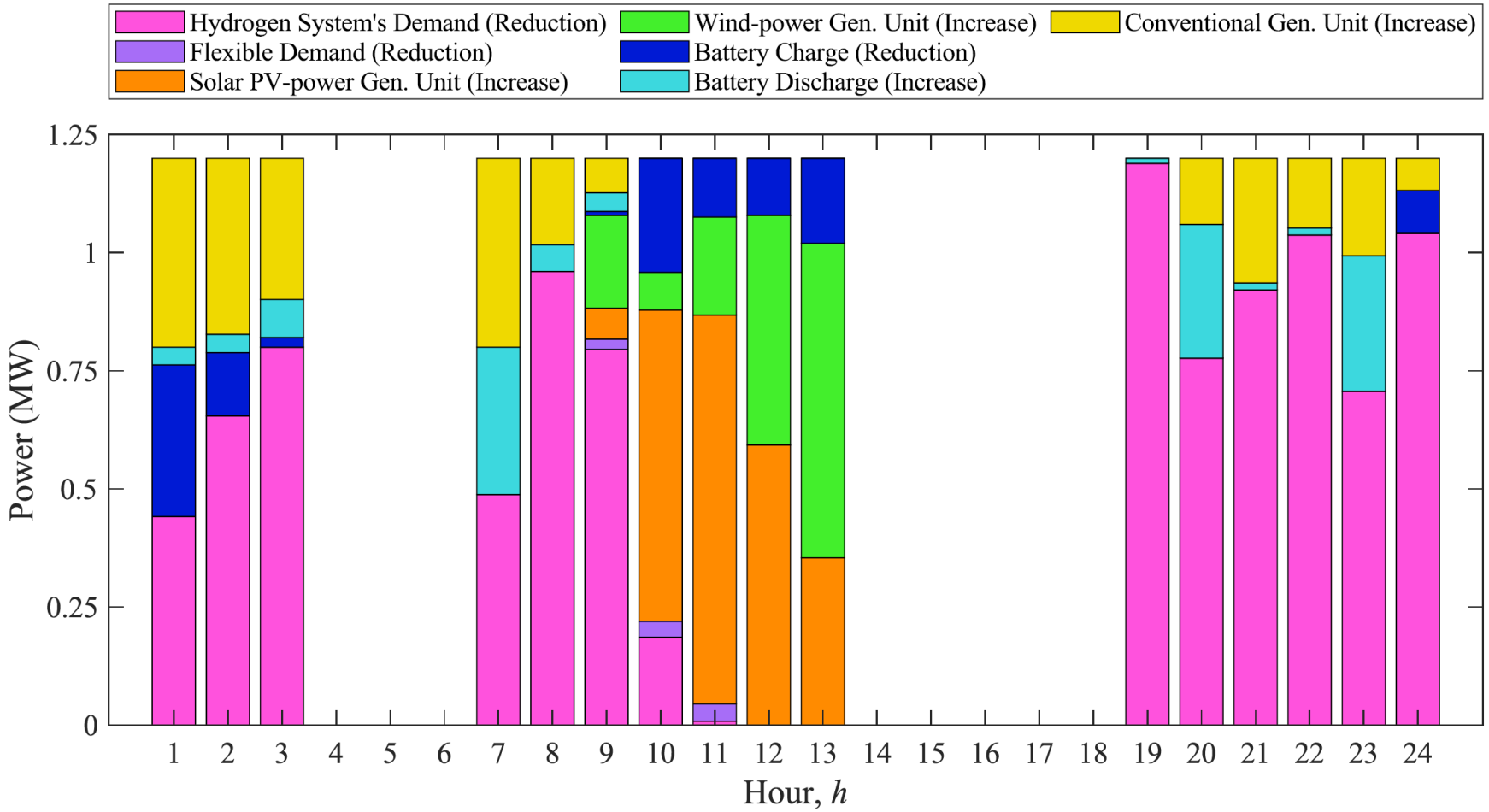
1.3. Case Study

Expected day-ahead scheduling of the VPP



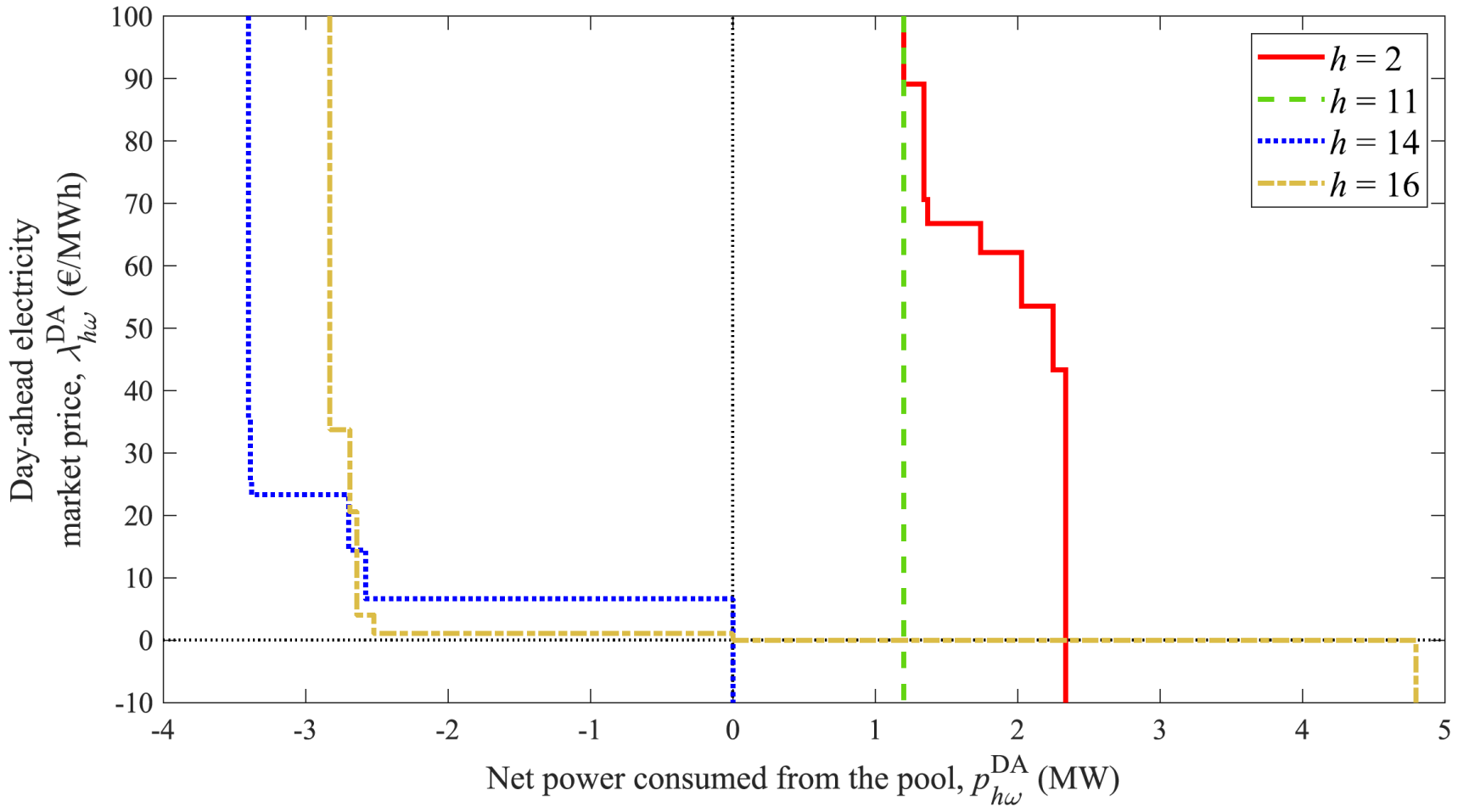
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Expected rescheduling of the VPP



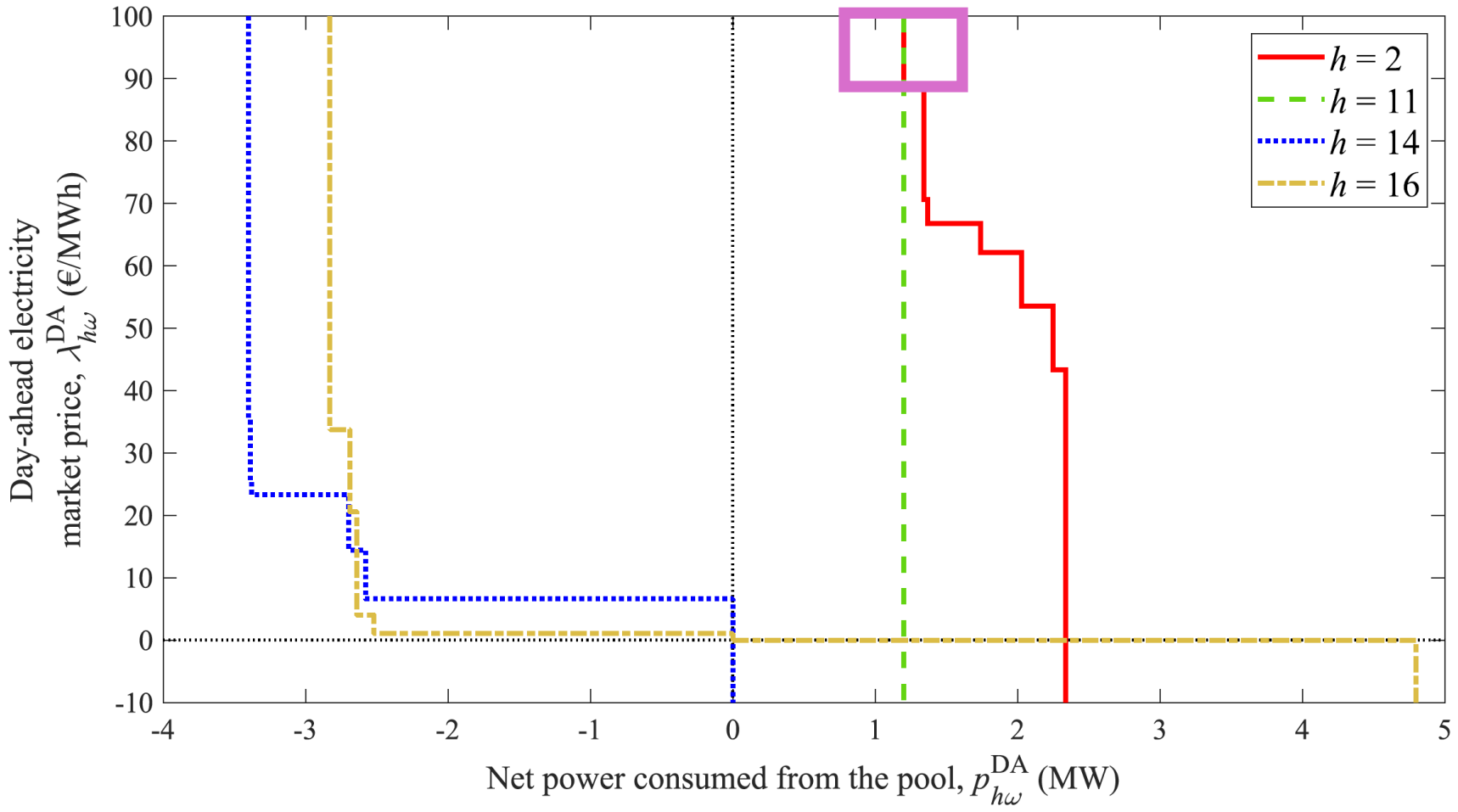
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DA bidding and offering curve



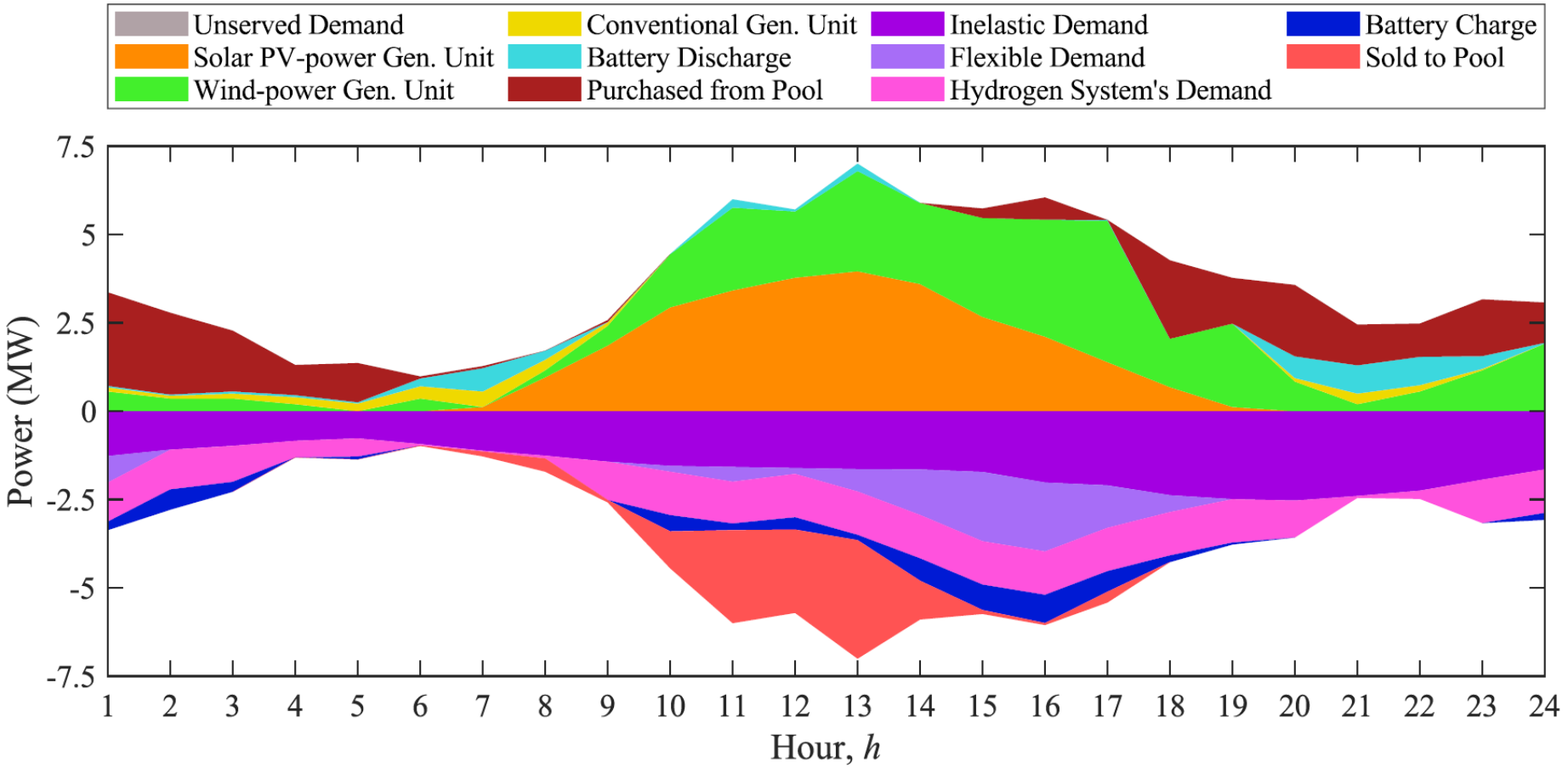
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DA bidding and offering curve



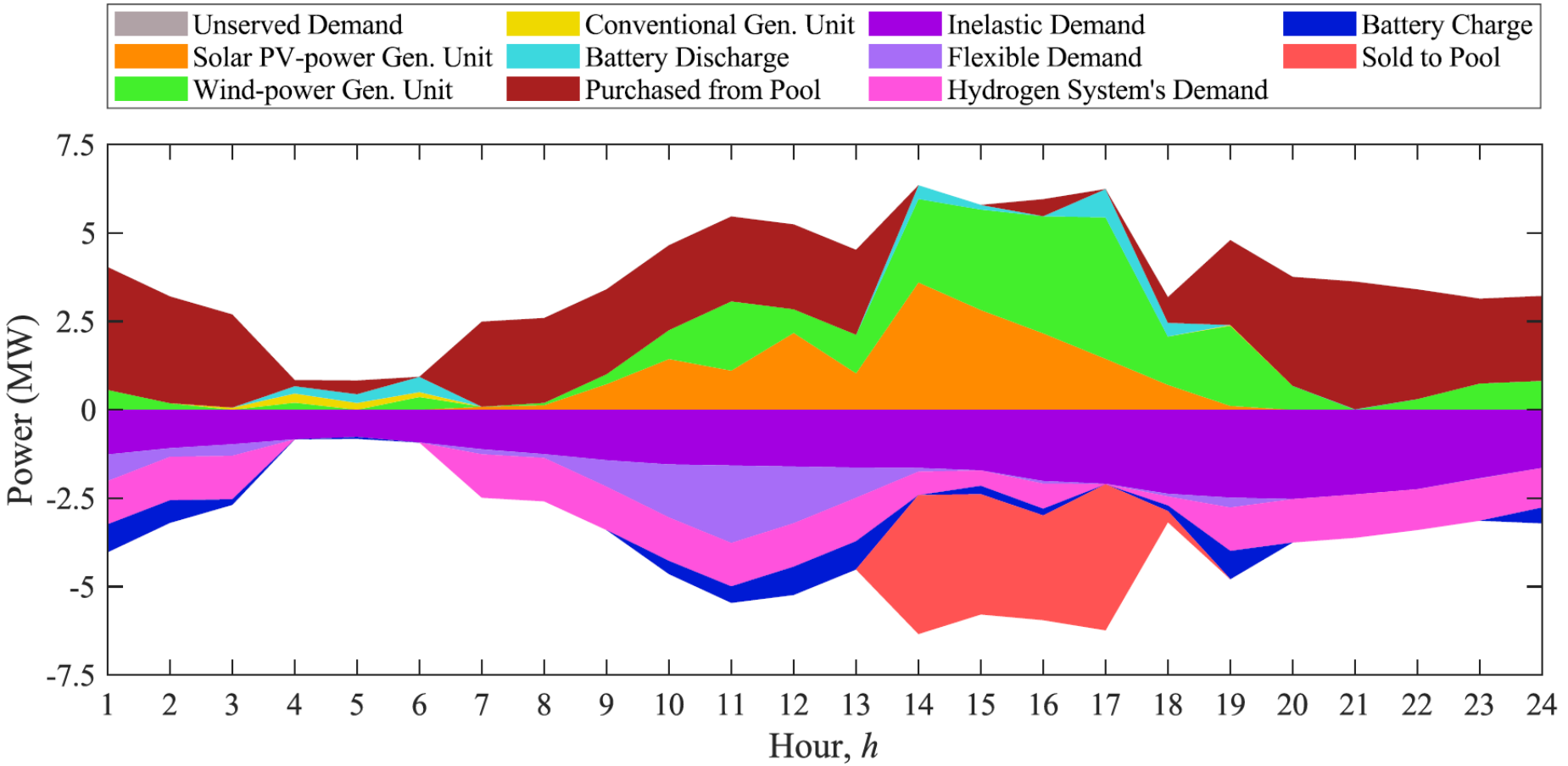
A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Analysis of the DR power provided: $P^{\mathrm{DR}} = 0$ MW



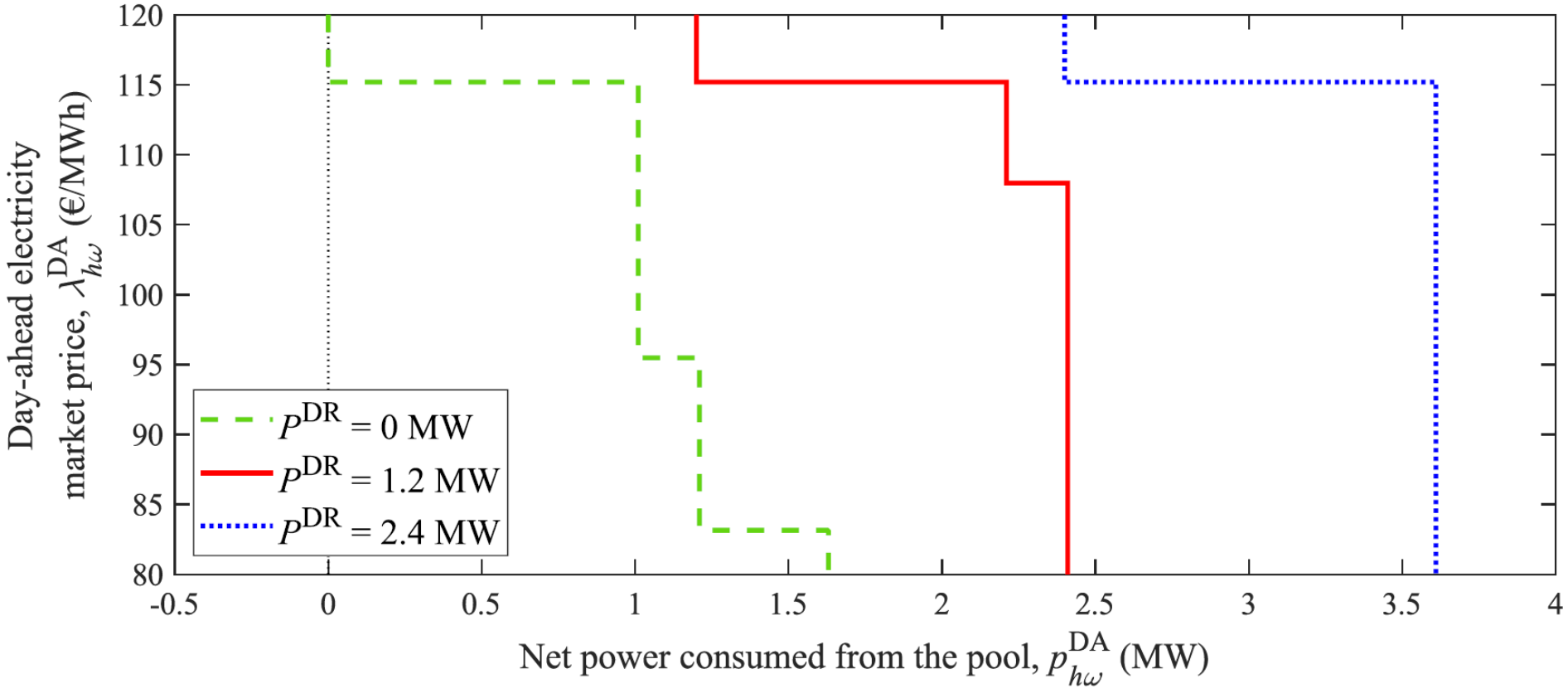
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Analysis of the DR power provided: $P^{\mathrm{DR}} = 2.4$ MW



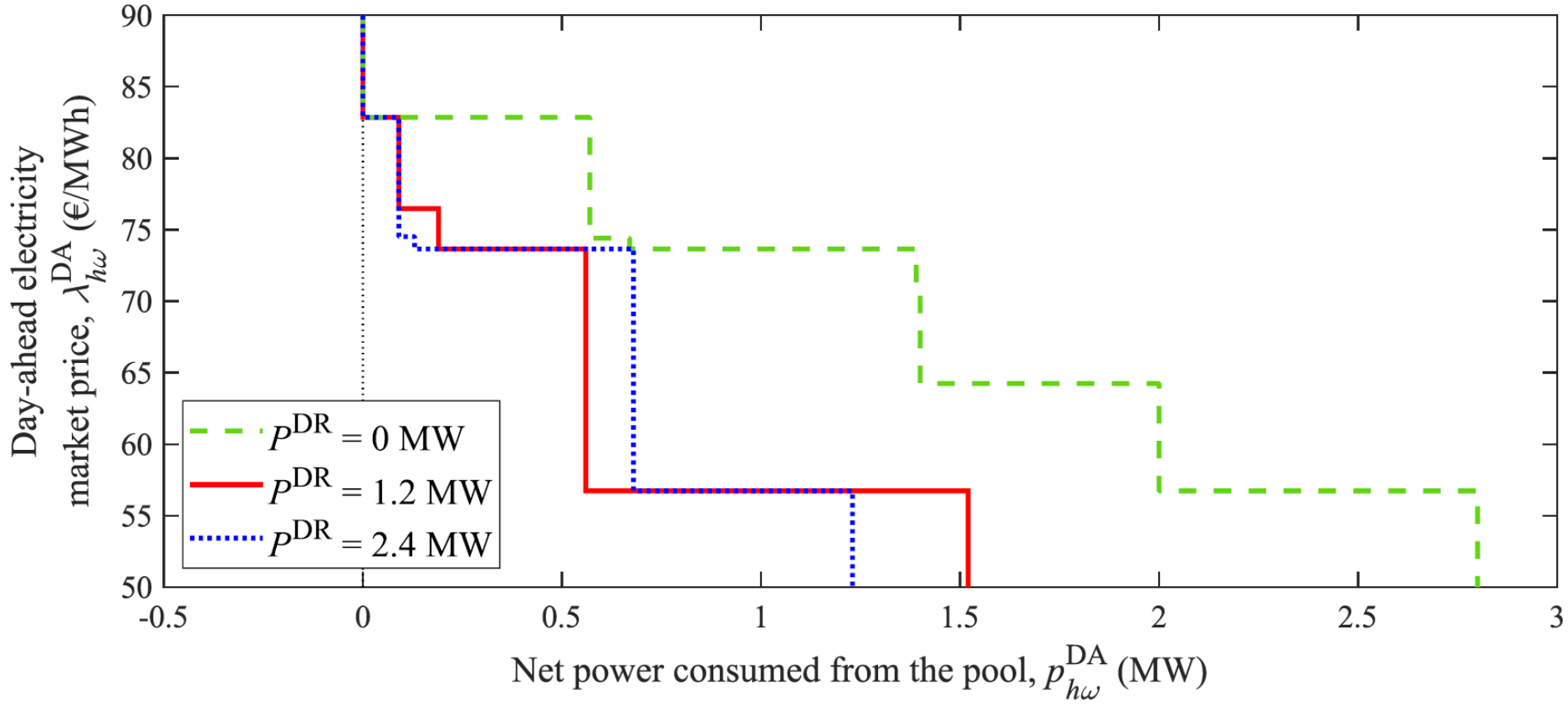
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Analysis of the DR power provided: $h = 21$



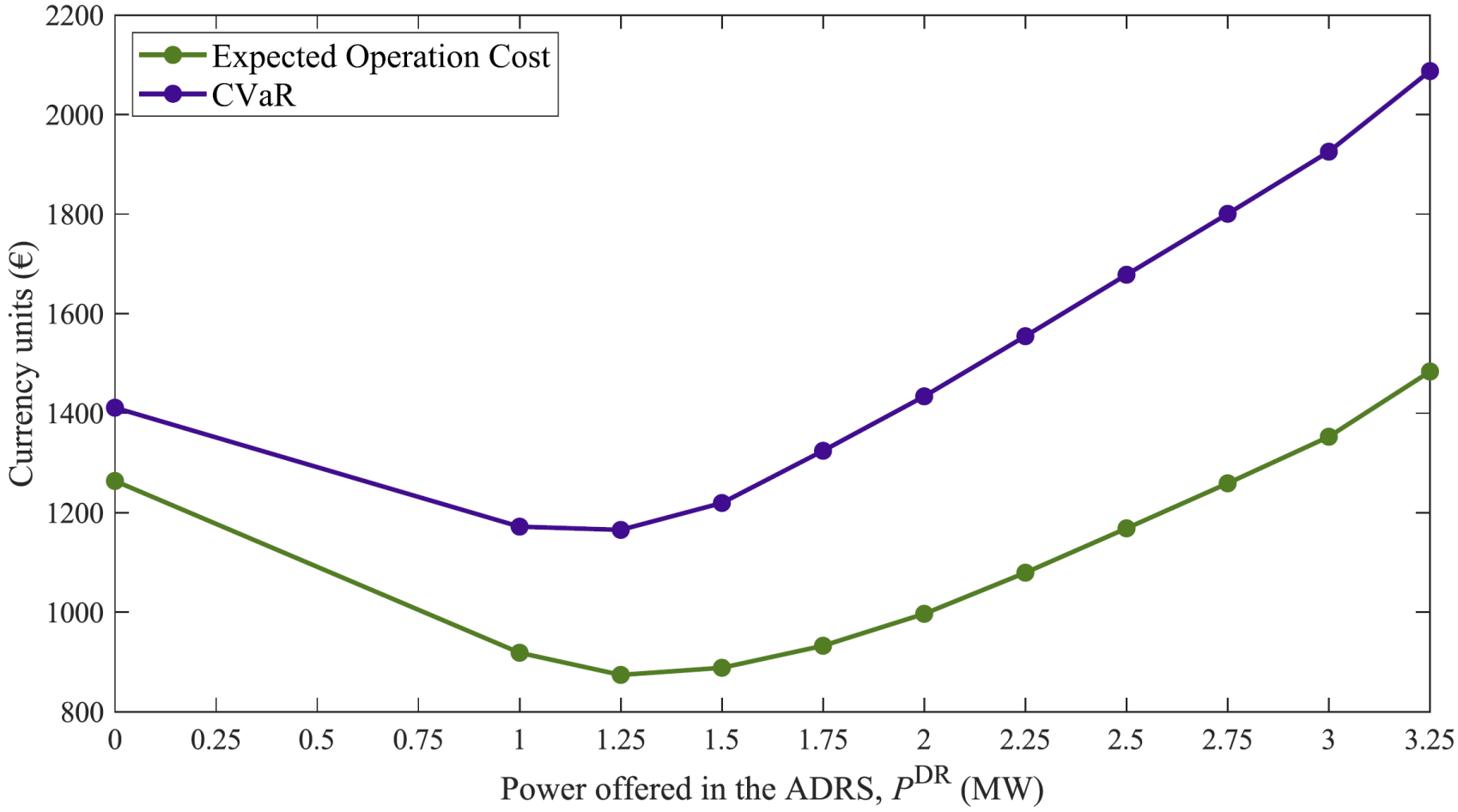
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Analysis of the DR power provided: $h = 5$



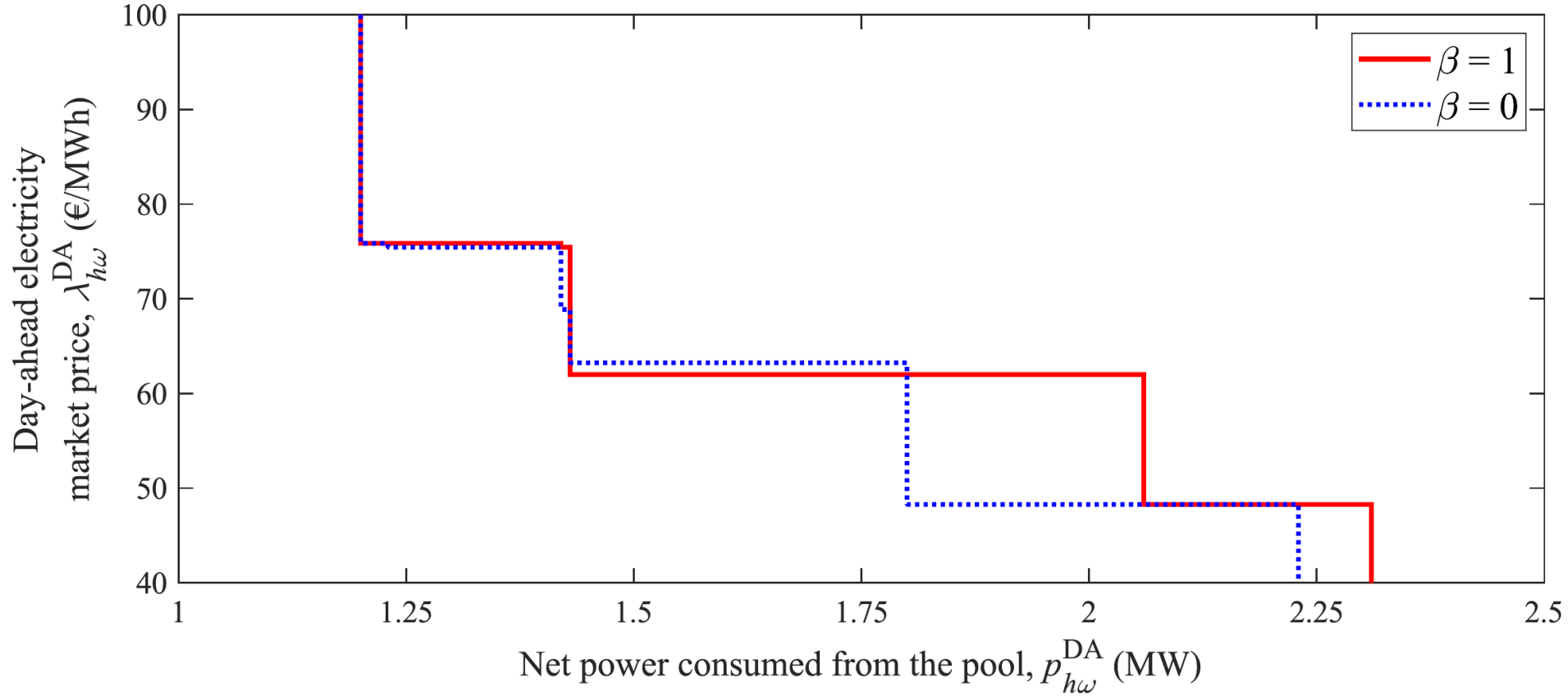
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Analysis of the DR power provided



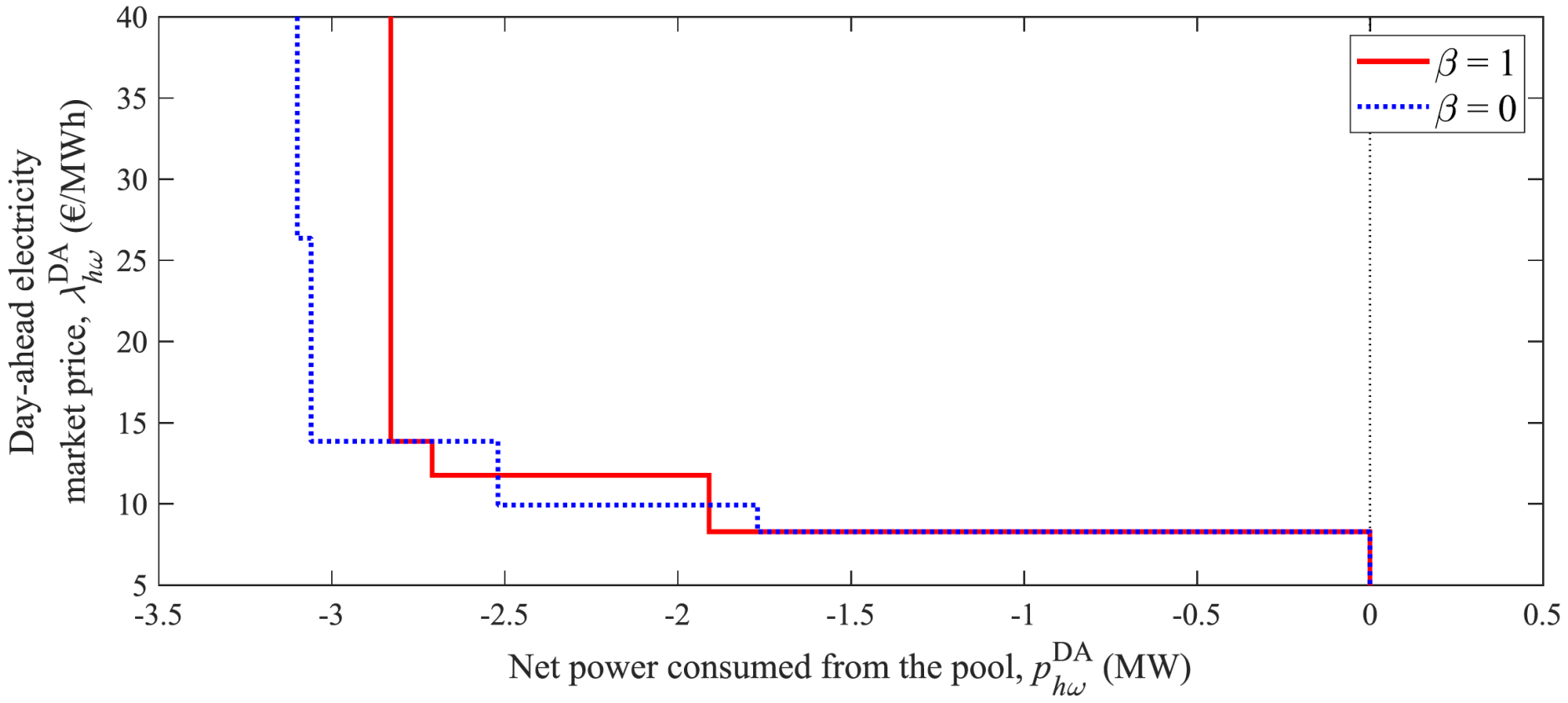
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Analysis of the risk: $h = 3$



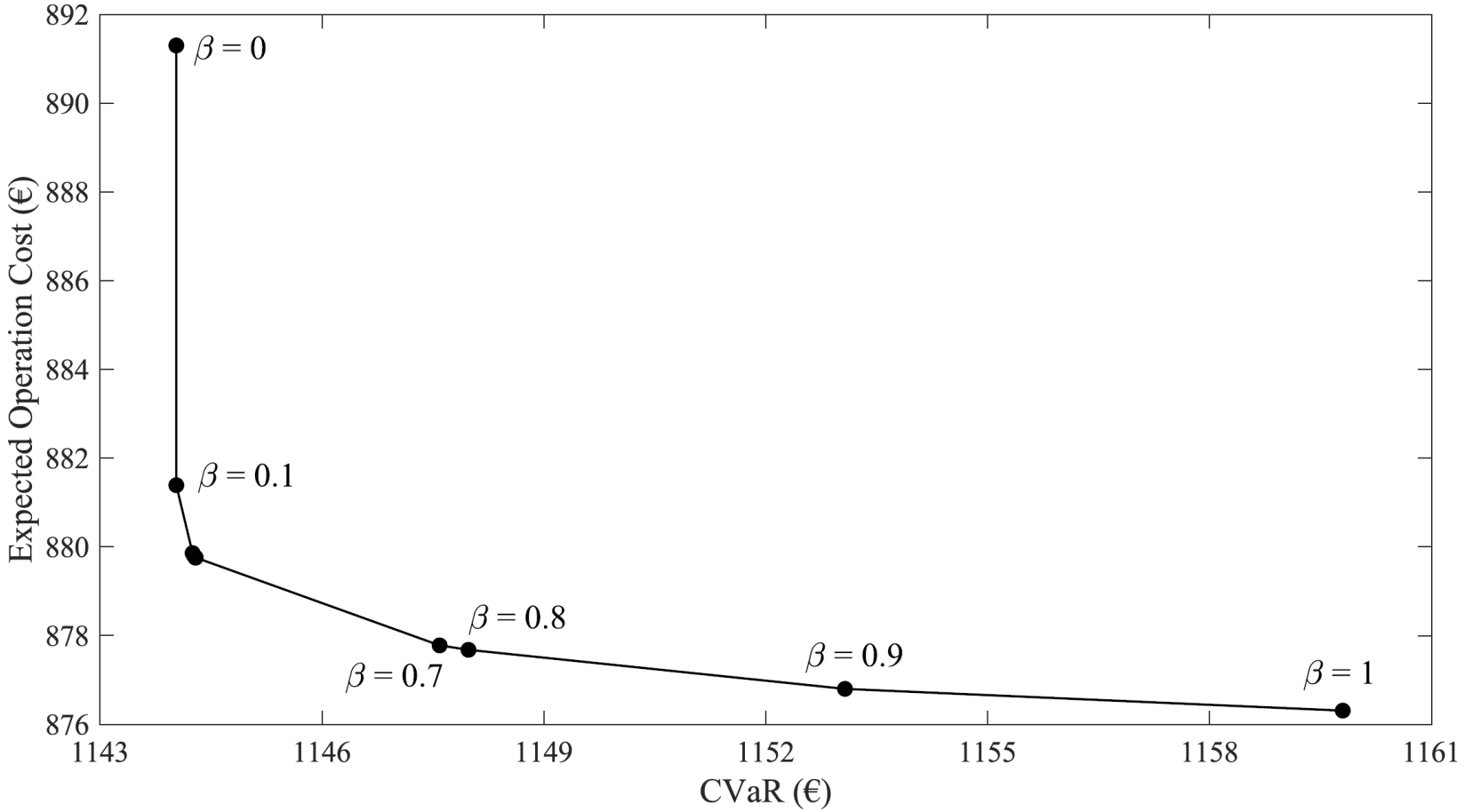
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Analysis of the risk: $h = 15$



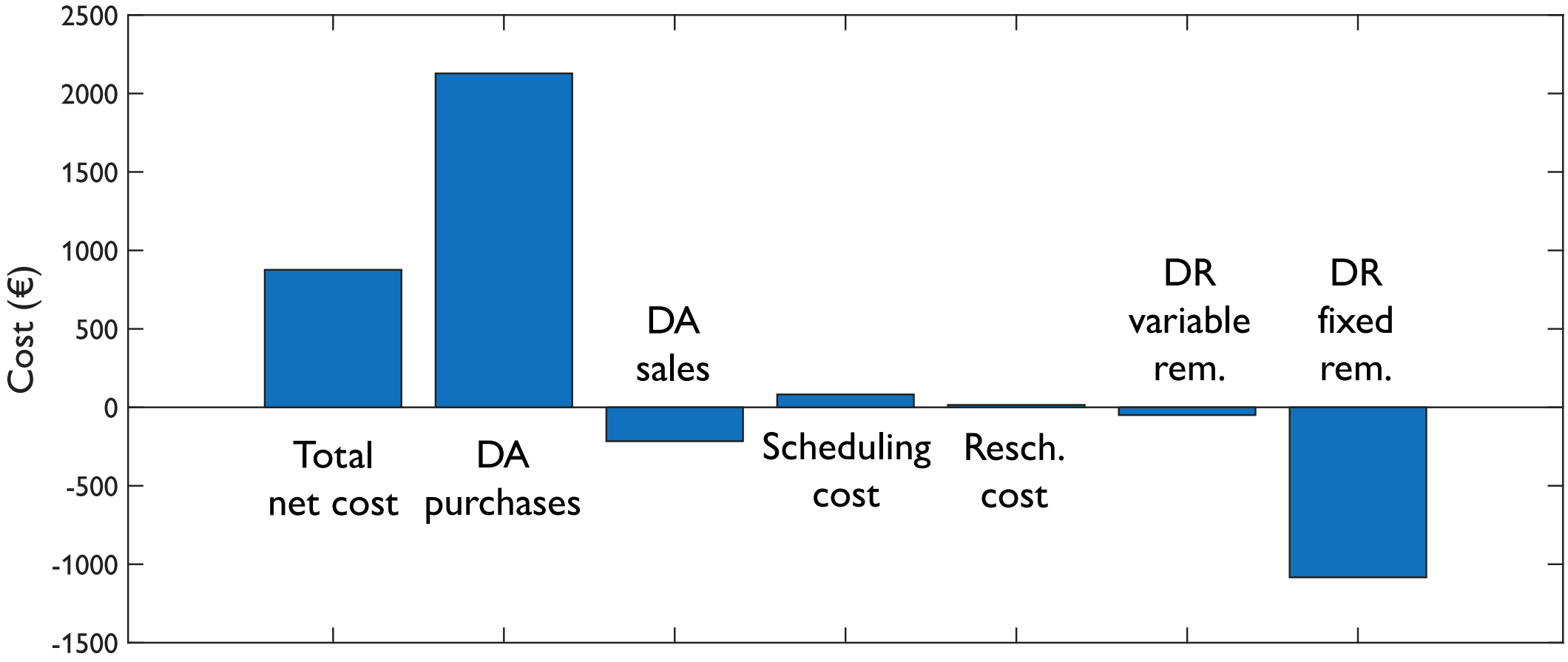
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Analysis of the risk: efficient frontier



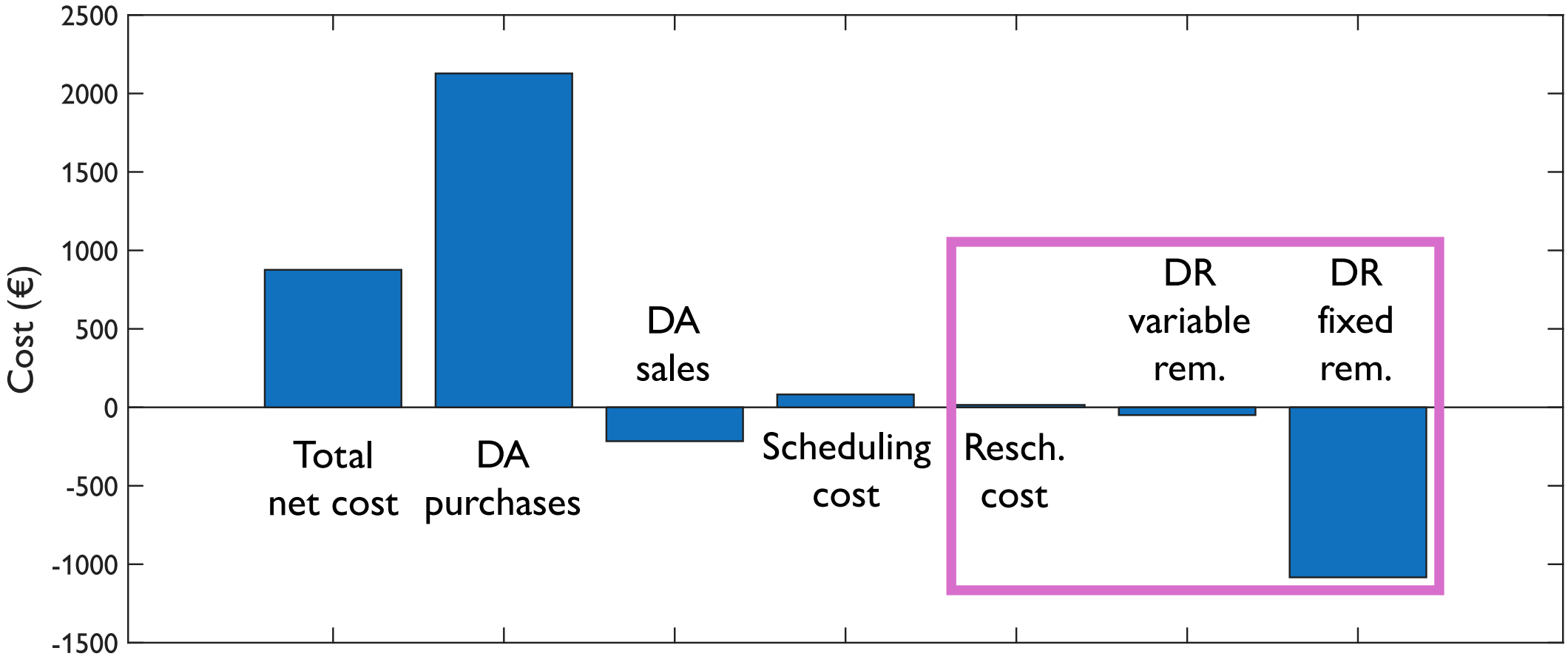
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Disaggregation of the total net cost



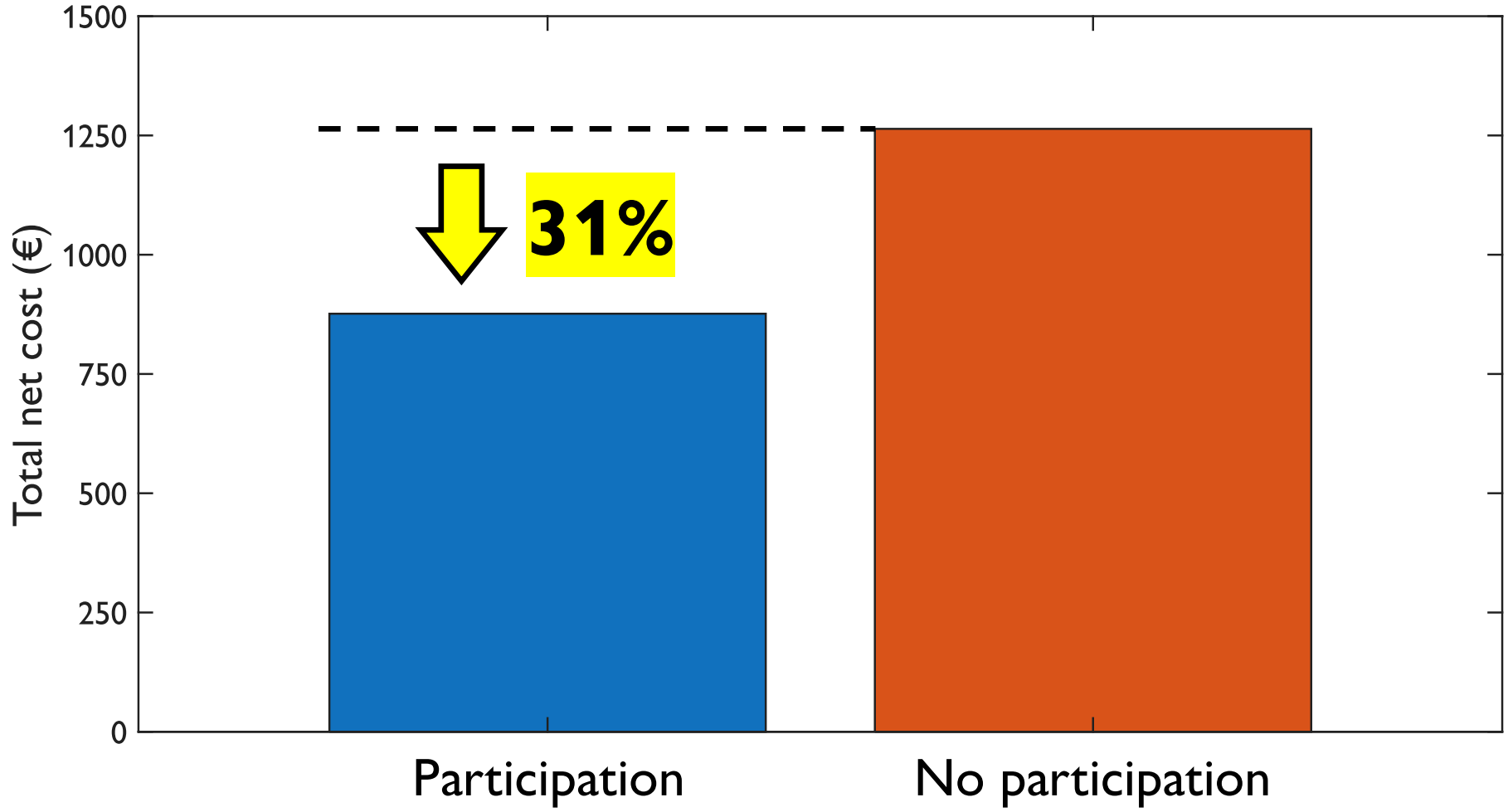
A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Disaggregation of the total net cost



A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Impact of the DR service's participation



A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Computational performance

CPLEX solver, relative optimality gap = 10^{-7}.

Time	Constraints	Variables	
		Continuous	Binary
< 1 h	113,391	51,392	6640

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

1.4. Conclusions

Conclusions

1. The VPP's participation in the mFRR-DR significantly affects the optimal DA bidding and offering curves, as well as the resulting net cost.

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Conclusions

1. The VPP's participation in the mFRR-DR significantly affects the optimal DA bidding and offering curves, as well as the resulting net cost.

2. The VPP reduces its grid consumption primarily by increasing renewable energy generation during midday and by decreasing the hydrogen system's energy use at the beginning and end of the day.

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Conclusions

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2. The VPP reduces its grid consumption primarily by increasing renewable energy generation during midday and by decreasing the hydrogen system's energy use at the beginning and end of the day.

3. The participation in the mFRR-DR leads to low rescheduling cost and high fixed remuneration.

A. Pleite-Paniagua, Á. García-Cerezo, and M. Carrión, "Building day-ahead market bidding and offering curves of a virtual power plant participating in a manual frequency restoration reserve-based demand response program", *Sustainable Energy, Grids and Networks*, vol. 47, art. no. 102426, Sep. 2026, DOI: 10.1016/j.segan.2026.102426.

Acknowledgments

Grant BDNS 837625 (25CO1/004439), funded by:



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Grants 101160614 (EU-DREAM Project), 101168796 (MCSA Doctoral Network FITNESS), and 101230578 (INNO-TREC project), funded by:



Grant 2025-GRIN-38264, funded by:



Paper available online



2. Single imbalance pricing scheme via stochastic robust optimization

Single imbalance pricing scheme via SP-RO

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

2.1. Context

Imbalance mechanism

	Imbalance pricing scheme	
	Single	Dual
Surplus vs deficit	Same price	Different price

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

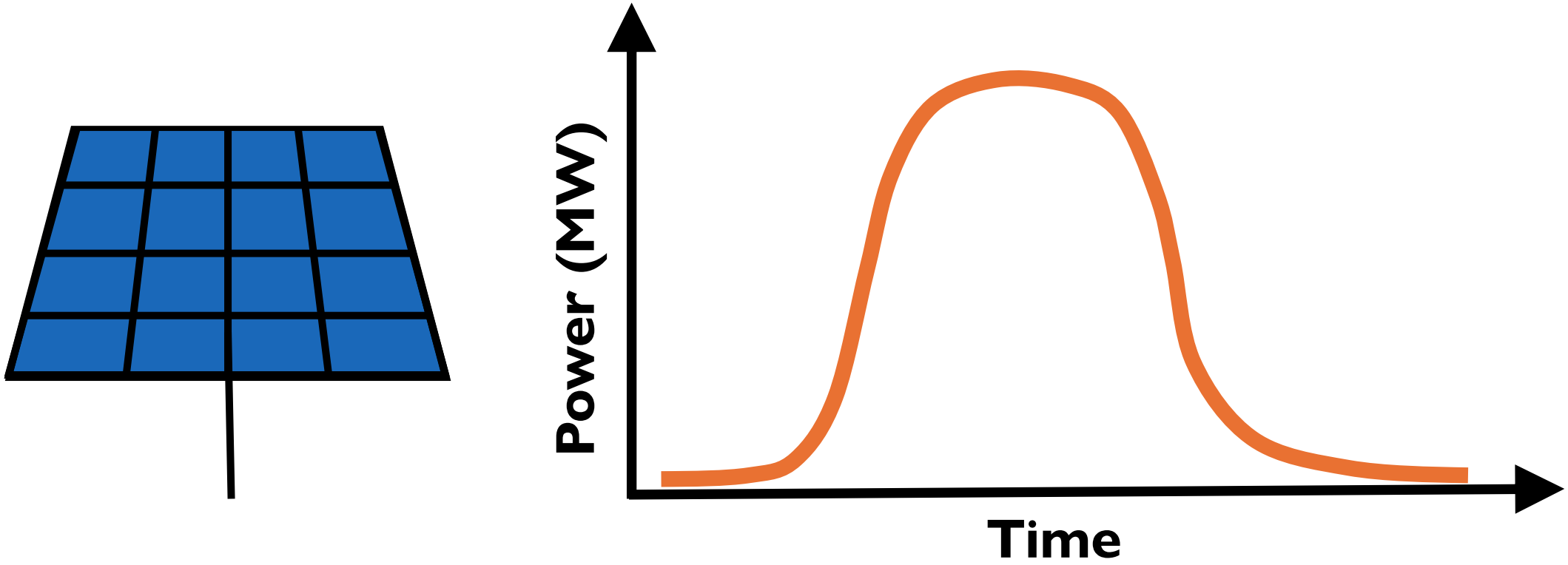
Imbalance mechanism

	Imbalance pricing scheme	
	Single	Dual
Surplus vs deficit	Same price	Different price

European Transmission System Operators

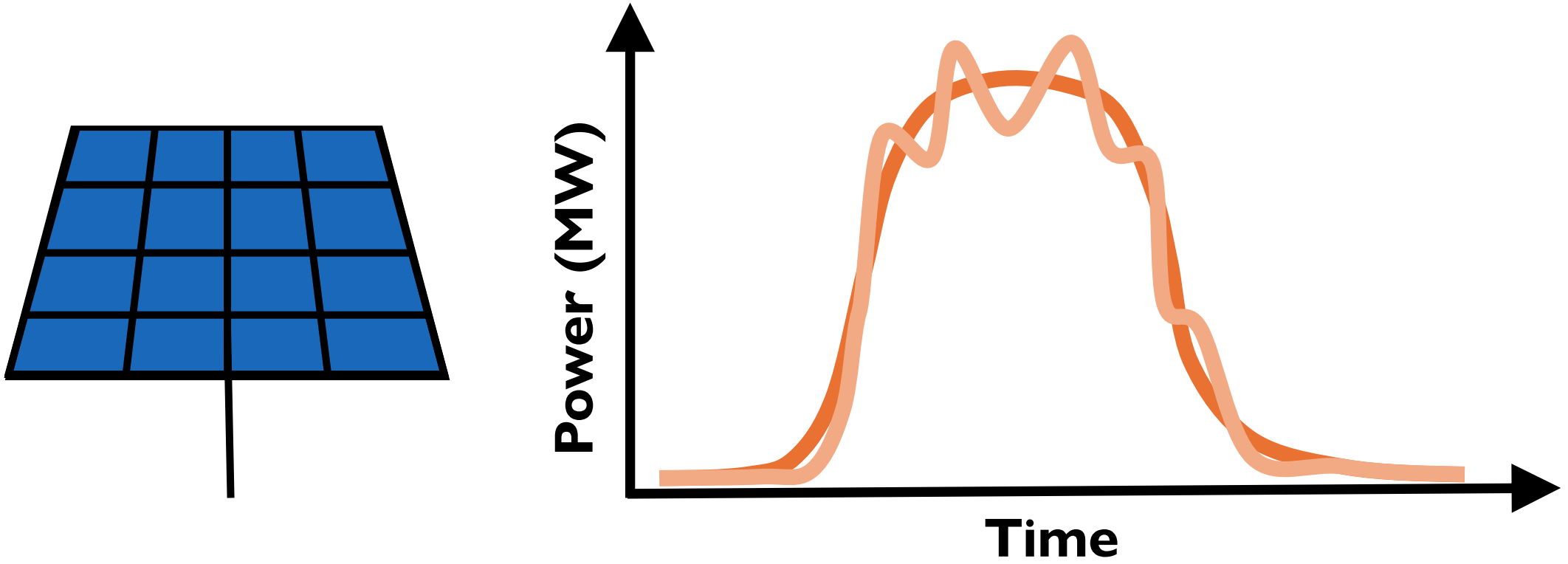
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Renewable Energy Sources (RES)



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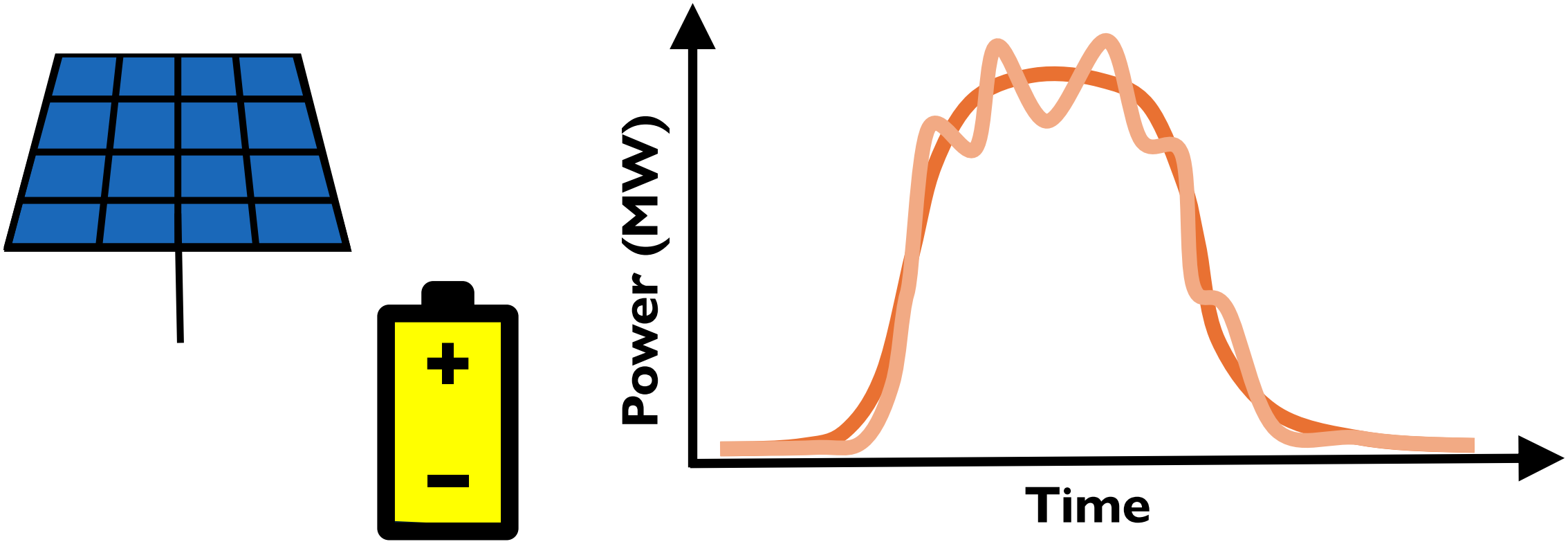
Renewable Energy Sources (RES)



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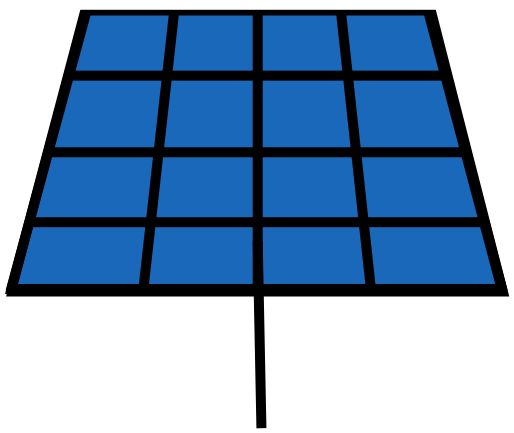
Balance Responsible Party (BRP)

Financial entity responsible for the imbalances.

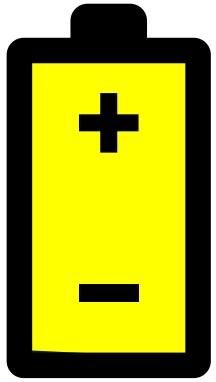


Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

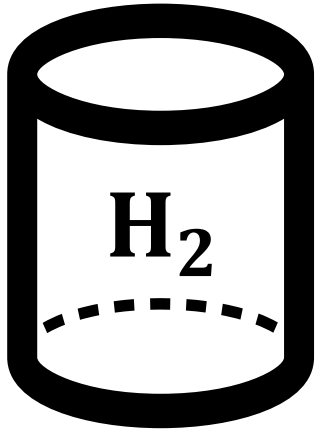
Price-taker Virtual Power Plant (VPP)



PV unit



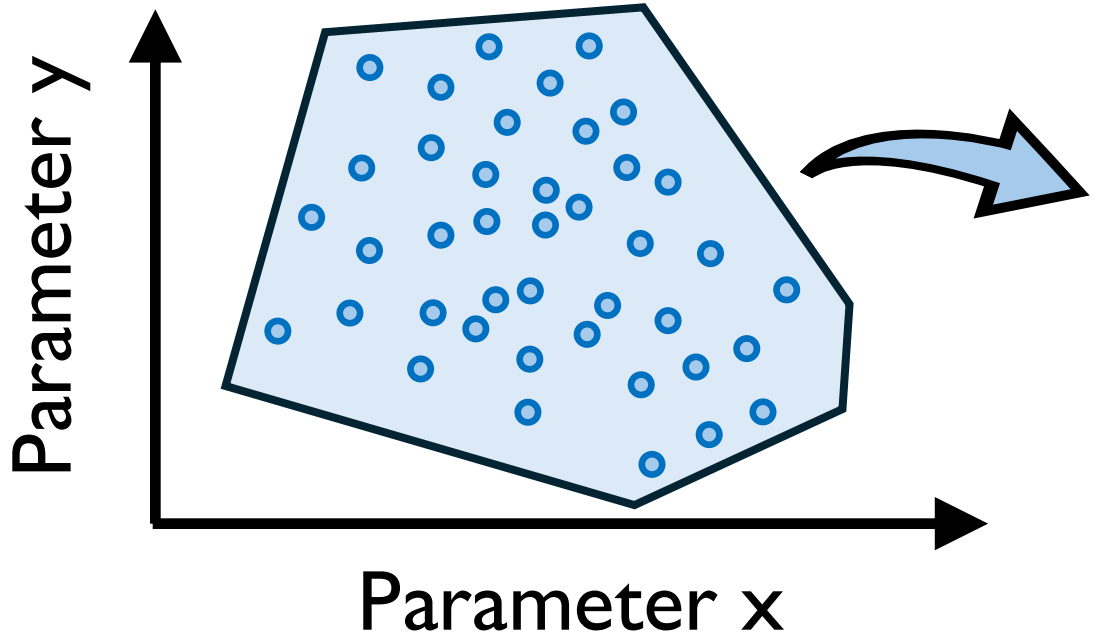
Battery



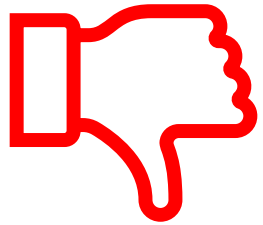
Electrolyzer

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Stochastic programming



Discrete set of scenarios



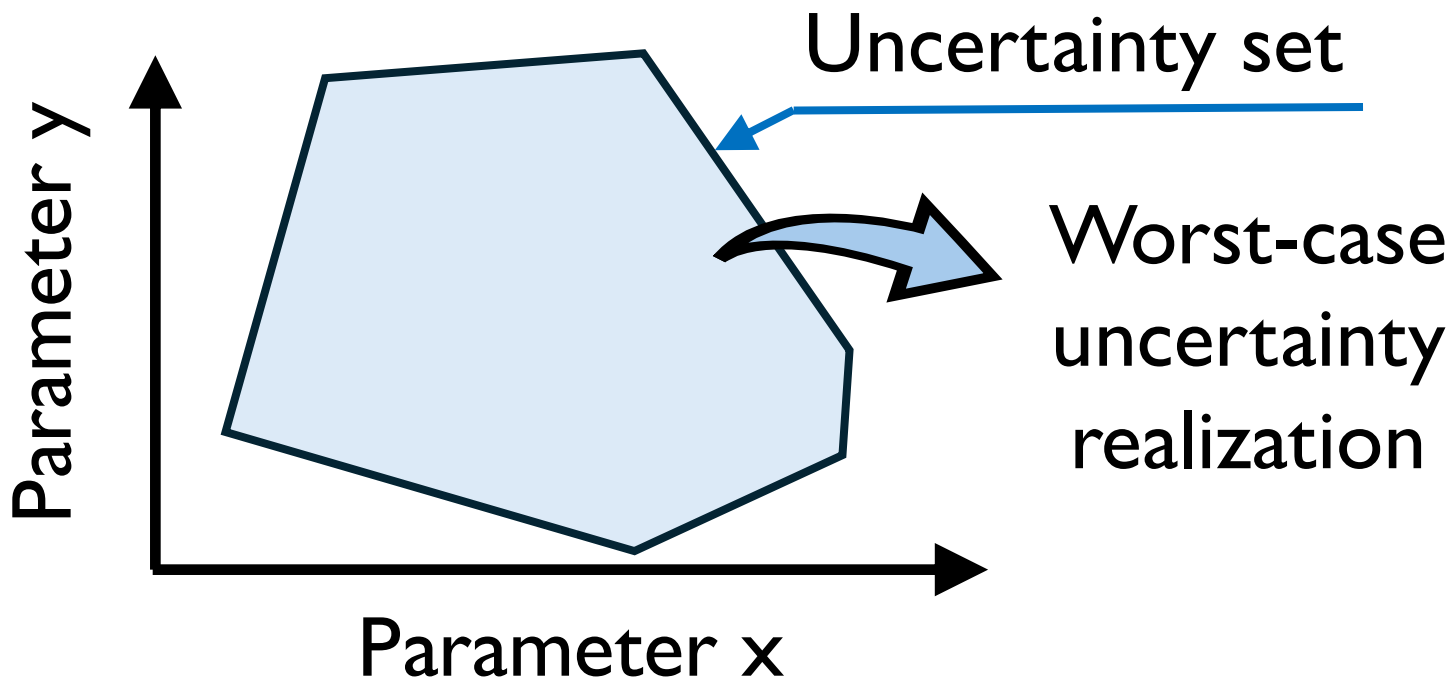
Lack of information about the probability density function

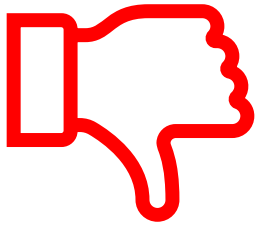
Risk-neutral framework → Expected value

Risk-averse framework → Conditional value-at-risk

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Robust optimization





Too conservative solution

Robust optimization without recourse

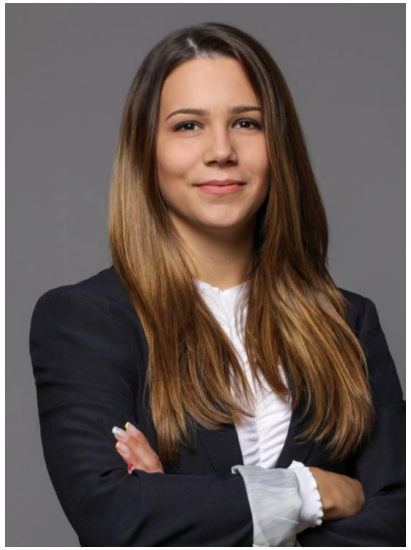
Robust optimization with recourse

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

2.2. Previous Optimization Model

Previous optimization model

A hybrid renewable-battery-electrolyzer facility under the single imbalance pricing scheme, *CSEE Journal of Power and Energy Systems*, paper accepted to be published (early access) (2025).



Petra Miljan Drašković



Ivan Pavić



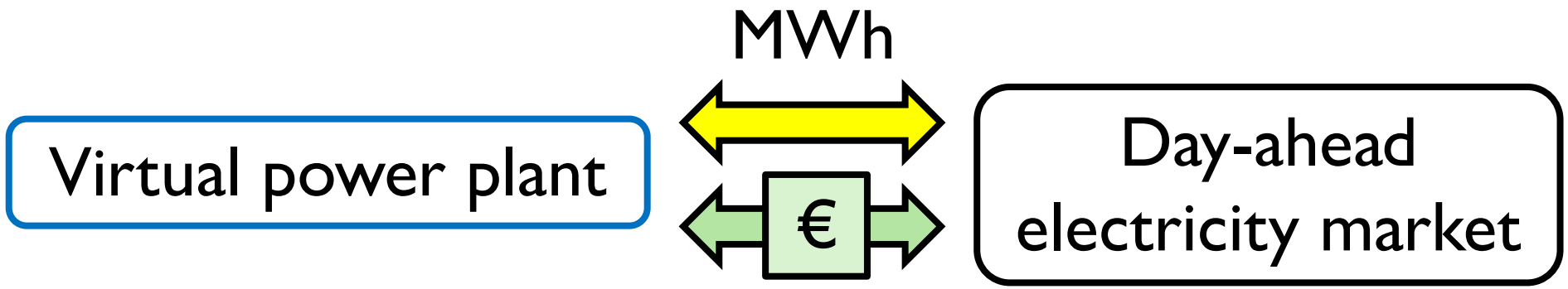
Karlo Šepetanc



Hrvoje Pandžić

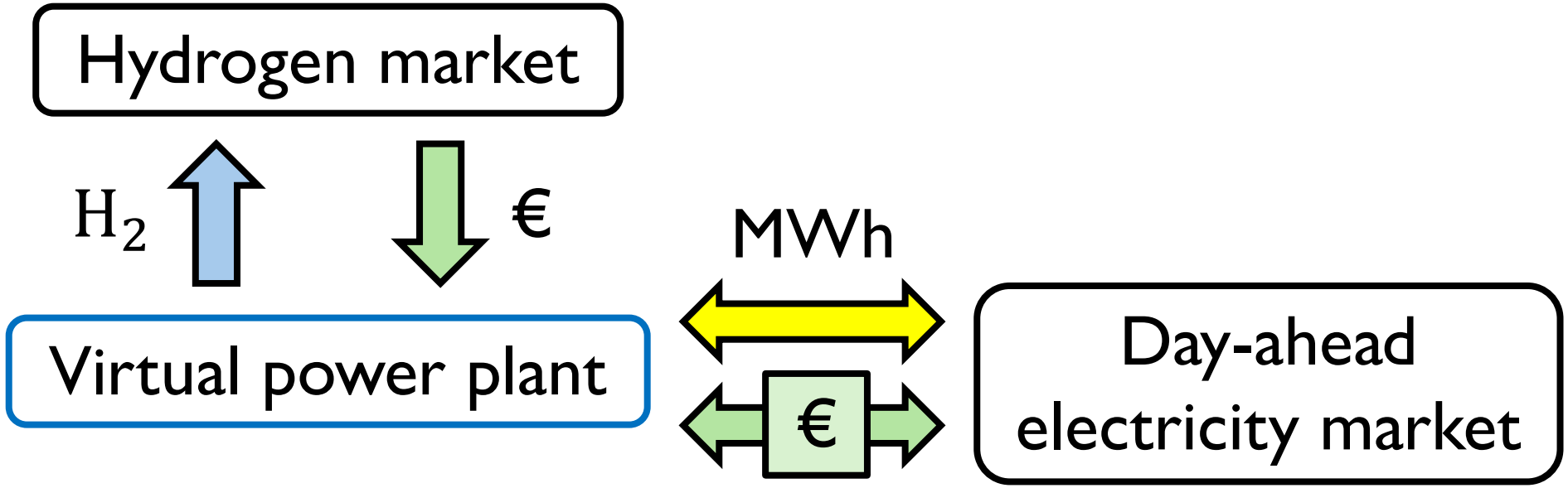
UNIVERSITY OF ZAGREB
Faculty of Electrical Engineering and Computing

Previous optimization model



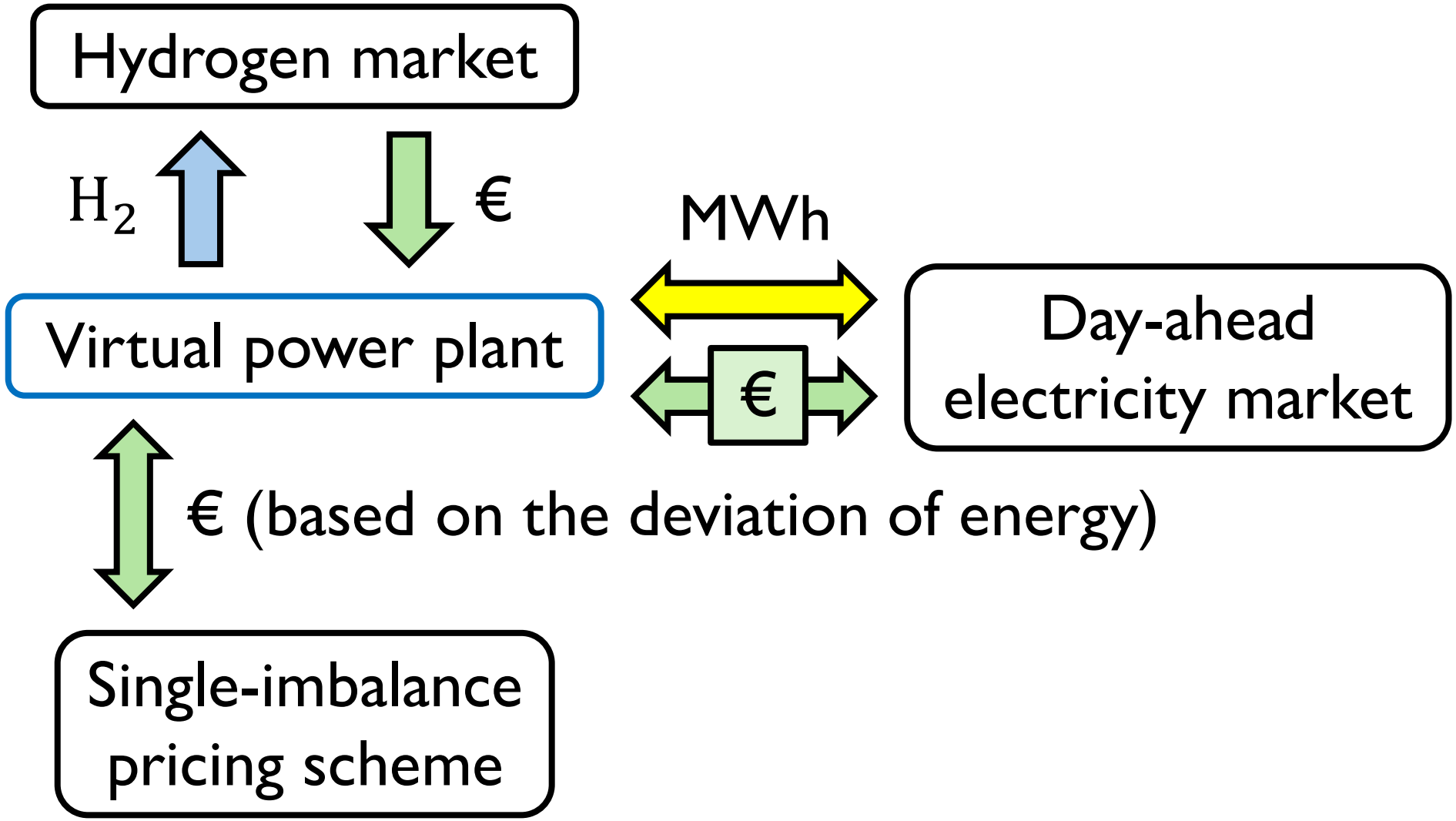
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Previous optimization model



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Previous optimization model



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Previous optimization model

Objective function:

$$max \quad \dots + \sum_{t \in \Omega^{T}} \sum_{s \in \Omega^{S}} \pi_{s}^{S} \left(\dots + \boldsymbol{p}_{\boldsymbol{t,s}}^{\mathbf{IB,WC}} \right)$$

Renewable production availability (RPA) scenarios s

Worst-case imbalance (IB) profit

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Imbalance mechanism

How is the profit of the imbalance mechanism computed?

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Imbalance mechanism

How is the profit of the imbalance mechanism computed?
Croatian power system:

$$p_{t,s}^{\mathrm{IB}} = \lambda_t^{\mathrm{DA}}\left(1 \pm K^{\mathrm{IB}}\right) d_{t,s}^{\mathrm{IB}}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$\pm \rightarrow$ Imbalance Price Deviation (IPD)

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Surplus of energy in the power system $\rightarrow -K^{\mathrm{IB}}$.

Deficit of energy in the power system $\rightarrow +K^{\mathrm{IB}}$.

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Worst-case situation

What is the worst-case realization of IPD for the VPP?

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 \pm K^{\mathrm{IB}}\right) d_{t,s}^{\mathrm{IB}}, \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$\pm$?

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Worst-case situation: surplus

If $\boldsymbol{d_{t,s}^{\mathrm{IB}} > 0}$, the VPP receives income for its **surplus** of energy.

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Worst-case situation: surplus

If $\boldsymbol{d_{t,s}^{\mathrm{IB}} > 0}$, the VPP receives income for its **surplus** of energy.

The worst-case situation for the VPP is that:

a) There is a surplus of energy in the power system:

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 - K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}.$$

b) There is a deficit of energy in the power system:

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 + K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}.$$

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Worst-case situation: surplus

If $\boldsymbol{d_{t,s}^{\mathrm{IB}} > 0}$, the VPP receives income for its **surplus** of energy.

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Worst-case situation: deficit

If $\boldsymbol{d_{t,s}^{\mathrm{IB}} < 0}$, the VPP pays for its **deficit** of energy.

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Worst-case situation: summary

VPP-surplus $\left(d_{t,s}^{\mathrm{IB}} > 0\right)$: $p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 - K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}$.

VPP-deficit $\left(d_{t,s}^{\mathrm{IB}} < 0\right)$: $p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 + K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}$.

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VPP-surplus $\left(d_{t,s}^{\mathrm{IB}} > 0\right)$: $p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 - K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}$.

VPP-deficit $\left(d_{t,s}^{\mathrm{IB}} < 0\right)$: $p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(1 + K^{\mathrm{IB}}\right)d_{t,s}^{\mathrm{IB}}$.

Hence, **independently of the sign of $d_{t,s}^{\mathrm{IB}}$**, the worst-case situation is:

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(d_{t,s}^{\mathrm{IB}} - K^{\mathrm{IB}}\left|d_{t,s}^{\mathrm{IB}}\right|\right), \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

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Previous optimization model

Objective function:

$$max \quad \ldots + \sum_{t \in \Omega^{\mathrm{T}}} \sum_{s \in \Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \left(\ldots + \lambda_t^{\mathrm{DA}} \left(d_{t,s}^{\mathrm{IB}} - K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \right) \right)$$

What is the drawback of this approach?

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Objective function:

$$max \quad \ldots + \sum_{t \in \Omega^{\mathrm{T}}} \sum_{s \in \Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \left(\ldots + \lambda_t^{\mathrm{DA}} \left(d_{t,s}^{\mathrm{IB}} - K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \right) \right)$$

What is the drawback of this approach?

It is too conservative.

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Robust optimization

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(\dots + \lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

$$- \sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \max_{b_{t,s}^{\mathrm{IB}}} \left\{ \sum_{t\in\Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \left(2b_{t,s}^{\mathrm{IB}} - 1\right) \right.$$

subject to:

$$\left. \begin{aligned} & 0 \le b_{t,s}^{\mathrm{IB}} \le 1\,, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}} \\ & \sum_{t\in\Omega^{\mathrm{T}}} b_{t,s}^{\mathrm{IB}} \le \boldsymbol{\Gamma^{\mathrm{IB}}}\,, \quad \forall s \in \Omega^{\mathrm{S}} \end{aligned} \right\}$$

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Robust optimization

Optimistic case: $\Gamma^{\mathrm{IB}} = 0$

$$0 \leq b_{t,s}^{\mathrm{IB}} \leq 1, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$\sum_{t \in \Omega^{\mathrm{T}}} b_{t,s}^{\mathrm{IB}} \leq \mathbf{0}, \quad \forall s \in \Omega^{\mathrm{S}}$$

Hence, the IPD realizations are **beneficial** for the VPP:

$$b_{t,s}^{\mathrm{IB}} = \mathbf{0}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}} \left(d_{t,s}^{\mathrm{IB}} + K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \right), \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}.$$

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Robust optimization

Pessimistic case: $\Gamma^{\mathrm{IB}} = 24$

$$0 \leq b_{t,s}^{\mathrm{IB}} \leq 1, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$\sum_{t \in \Omega^{\mathrm{T}}} b_{t,s}^{\mathrm{IB}} \leq \mathbf{24}, \quad \forall s \in \Omega^{\mathrm{S}}$$

Hence, the IPD realizations are **harmful** for the VPP:

$$b_{t,s}^{\mathrm{IB}} = \mathbf{1}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}}\left(d_{t,s}^{\mathrm{IB}} - K^{\mathrm{IB}}\left|d_{t,s}^{\mathrm{IB}}\right|\right), \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}.$$

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Robust optimization

Intermediate case: $\Gamma^{\mathrm{IB}} = 13$

$$\mathbf{0} \leq b_{t,s}^{\mathrm{IB}} \leq \mathbf{1}, \qquad \sum_{t \in \Omega^{\mathrm{T}}} b_{t,s}^{\mathrm{IB}} \leq \mathbf{13}$$

Imbalance profit in 13 time periods t:

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}} \left(d_{t,s}^{\mathrm{IB}} - K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \right) \rightarrow \text{Harmful for the VPP}$$

Imbalance profit in $24 - 13 = 11$ time periods t:

$$p_{t,s}^{\mathrm{IB,WC}} = \lambda_t^{\mathrm{DA}} \left(d_{t,s}^{\mathrm{IB}} + K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \right) \rightarrow \text{Beneficial for the VPP}$$

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Robust optimization

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(\dots + \lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

$$- \sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \max_{b_{t,s}^{\mathrm{IB}}} \left\{ \sum_{t\in\Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right| \left(2b_{t,s}^{\mathrm{IB}} - 1\right) \right.$$

subject to:

$$0 \le b_{t,s}^{\mathrm{IB}} \le 1 : z_{t,s}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$\left. \sum_{t\in\Omega^{\mathrm{T}}} b_{t,s}^{\mathrm{IB}} \le \Gamma^{\mathrm{IB}} : \delta_s, \quad \forall s \in \Omega^{\mathrm{S}} \right\}$$

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Robust optimization – Dual problem

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(\dots + \lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

$$-\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \underset{\delta_s, z_{t,s}}{min} \left\{ \sum_{t\in\Omega^{\mathrm{T}}} \left(z_{t,s} - \lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right|\right) + \Gamma^{\mathrm{IB}} \delta_s \right.$$

subject to:

$$\delta_s + z_{t,s} \geq 2\lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \left| d_{t,s}^{\mathrm{IB}} \right|, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$\delta_s \geq 0, \quad \forall s \in \Omega^{\mathrm{S}}$$

$$\left. z_{t,s} \geq 0, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}} \right\}$$

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subject to:

$$\delta_s + z_{t,s} \geq 2\lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \left|d_{t,s}^{\mathrm{IB}}\right|, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

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Linearization of the absolute value

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(\dots + \lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

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Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

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Linearization of the absolute value

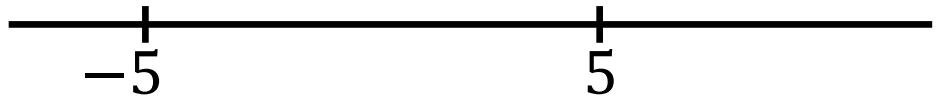
$$\min_{y_{t,s}} y_{t,s}$$

subject to:

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Example 1:

$d_{t,s}^{\mathrm{IB}} = 5 \rightarrow$



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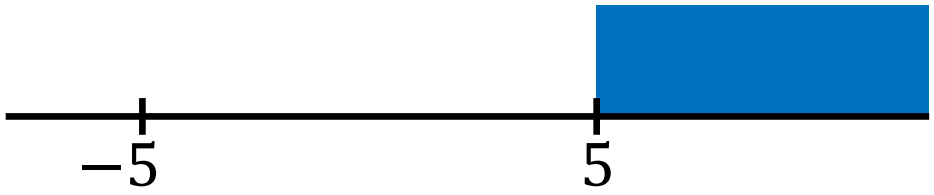
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$d_{t,s}^{\mathrm{IB}} = 5 \rightarrow$



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Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

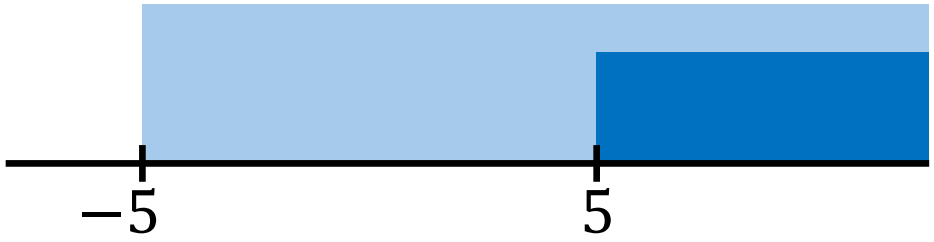
subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 1:

$$d_{t,s}^{\mathrm{IB}} = 5 \rightarrow$$



Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

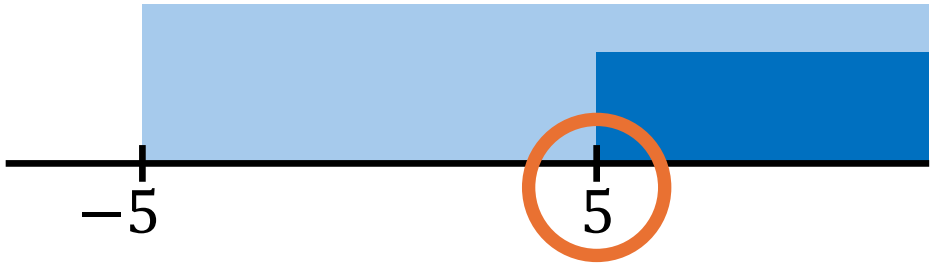
subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 1:

$d_{t,s}^{\mathrm{IB}} = 5 \rightarrow$



$\rightarrow y_{t,s} = 5 = \left|d_{t,s}^{\mathrm{IB}}\right|$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

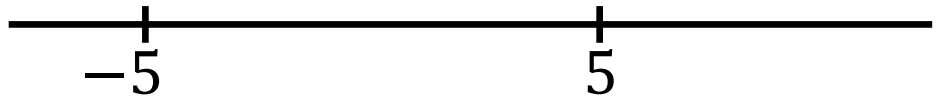
$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 2:

$d_{t,s}^{\mathrm{IB}} = -5 \rightarrow$



Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 2:

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Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 2:

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Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Example 2:

$d_{t,s}^{\mathrm{IB}} = -5 \rightarrow$



$\rightarrow y_{t,s} = 5 = \left|d_{t,s}^{\mathrm{IB}}\right|$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

$$\min_{y_{t,s}} y_{t,s}$$

subject to:

$$y_{t,s} \geq d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$y_{t,s} \geq -d_{t,s}^{\mathrm{IB}} : \mu_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Karush–Kuhn–Tucker (KKT) conditions:

$$1 - \mu_{t,s}^{+} - \mu_{t,s}^{-} = 0, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$0 \leq \left(y_{t,s} - d_{t,s}^{\mathrm{IB}}\right) \perp \mu_{t,s}^{+} \geq 0\,, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$0 \leq \left(y_{t,s} + d_{t,s}^{\mathrm{IB}}\right) \perp \mu_{t,s}^{-} \geq 0\,, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

Disjunctive transformation:

$$0 \leq y_{t,s} - d_{t,s}^{\mathrm{IB}} \leq M\left(1 - x_{t,s}^{+}\right), \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$0 \leq \mu_{t,s}^{+} \leq M x_{t,s}^{+}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$0 \leq y_{t,s} + d_{t,s}^{\mathrm{IB}} \leq M\left(1 - x_{t,s}^{-}\right), \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$0 \leq \mu_{t,s}^{-} \leq M x_{t,s}^{-}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$
$$x_{t,s}^{+}, x_{t,s}^{-} \in \{0, 1\}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Linearization of the absolute value

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(\dots + \lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

$$+ \sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \left(\sum_{t\in\Omega^{\mathrm{T}}} \left(\lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \boldsymbol{y_{t,s}} - z_{t,s}\right) - \Gamma^{\mathrm{IB}}\delta_s \right)$$

subject to:

$$\delta_s + z_{t,s} \geq 2\lambda_t^{\mathrm{DA}} K^{\mathrm{IB}} \boldsymbol{y_{t,s}}, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

$$\delta_s \geq 0, \quad \forall s \in \Omega^{\mathrm{S}}$$

$$z_{t,s} \geq 0, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}$$

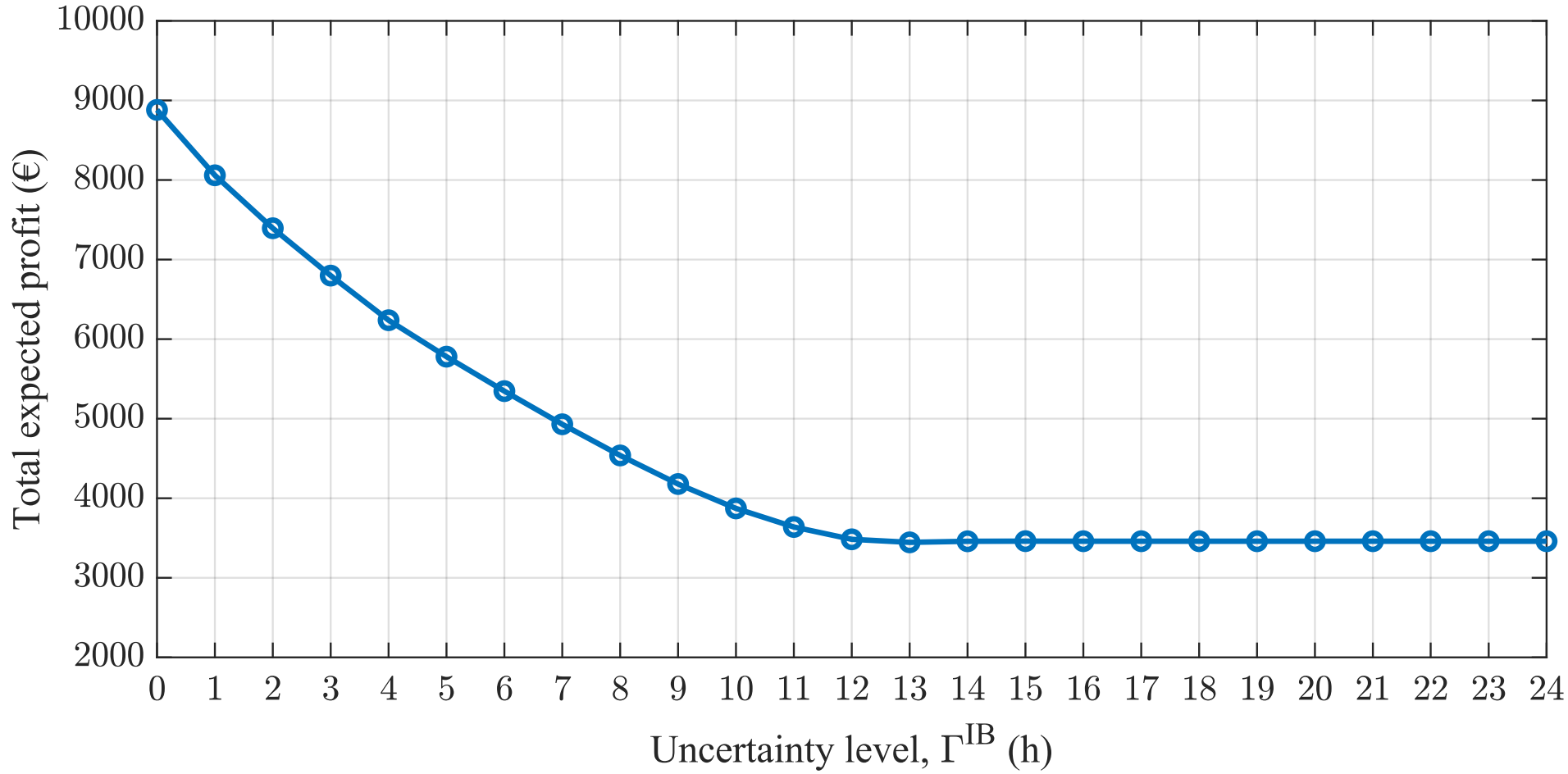
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Previous optimization model

How can we improve this model?

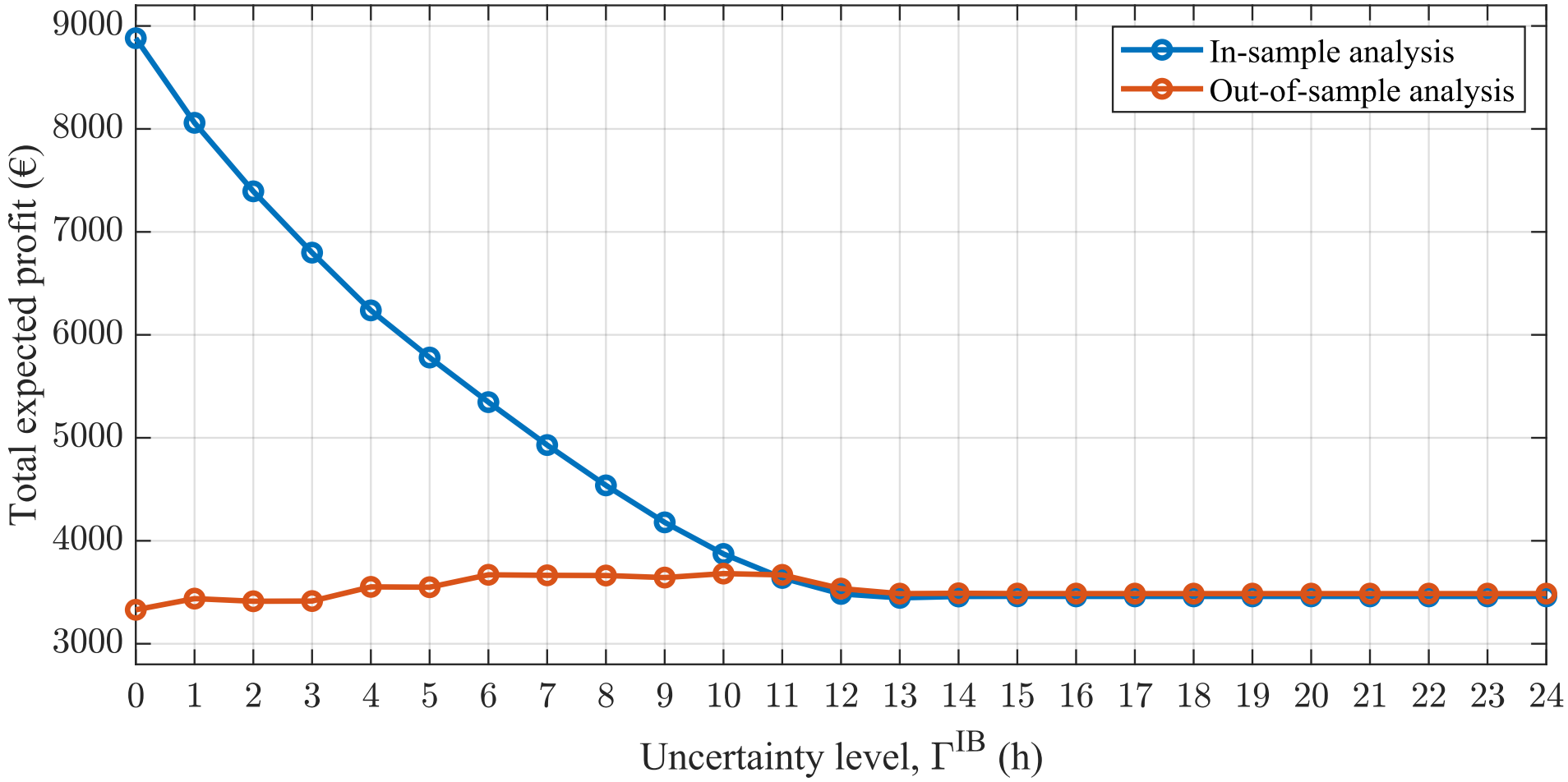
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Previous optimization model



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Previous optimization model



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Previous optimization model

Main issue of this model:

it is very optimistic for low values of Γ^{IB}.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

2.3. Proposed Optimization Model

Proposed optimization model



Petra Miljan Drašković



Hrvoje Pandžić

UNIVERSITY OF ZAGREB

Faculty of Electrical Engineering and Computing

Proposed optimization model

Previous model: only one value of Γ^{IB} is considered in the optimization problem.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

Previous model: only one value of Γ^{IB} is considered in the optimization problem.

Proposed model: multiple values of Γ^{IB} are **simultaneously** considered in the optimization problem.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

Stochastic Programming + Robust Optimization

for the same source of uncertainty

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

Stochastic Programming + Robust Optimization

for the same source of uncertainty

IPD scenarios ξ linked to different values of the uncertainty budget $\boldsymbol{\Gamma_{\xi}^{IB}}$:

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \ldots, 24]$$

$$\pi_{\xi}^{\mathrm{X}} = \left[\pi_{\xi_1}^{\mathrm{X}}, \pi_{\xi_2}^{\mathrm{X}}, \ldots, \pi_{\xi_{25}}^{\mathrm{X}}\right]$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

We use a scenario-based approach, but we do not directly model the realization of the imbalance price deviation.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

We use a scenario-based approach, but we do not directly model the realization of the imbalance price deviation.

We model **how optimistic or pessimistic each scenario ξ is** through the value of the uncertainty budget $\Gamma_{\xi}^{\mathrm{IB}}$.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

The values of the imbalance price deviations are identified by solving the inner problem (robust optimization).

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

$$max \quad \dots + \sum_{t\in\Omega^{\mathrm{T}}}\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}}\left(+\lambda_t^{\mathrm{DA}} d_{t,s}^{\mathrm{IB}}\right)$$

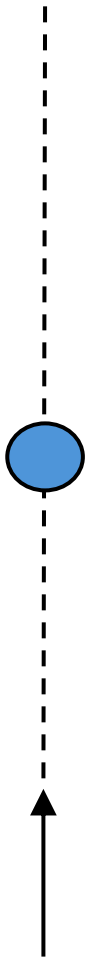
$$-\sum_{s\in\Omega^{\mathrm{S}}} \pi_s^{\mathrm{S}} \sum_{\boldsymbol{\xi}\in\boldsymbol{\Omega}^{\mathbf{X}}} \boldsymbol{\pi}_{\boldsymbol{\xi}}^{\mathbf{X}} \max_{b_{t,s,\boldsymbol{\xi}}^{\mathrm{IB}}} \left\{ \sum_{t\in\Omega^{\mathrm{T}}} \lambda_t^{\mathrm{DA}} K^{IB} \left| d_{t,s}^{IB} \right| \left(2 b_{t,s,\boldsymbol{\xi}}^{\mathrm{IB}} - 1\right) \right.$$

subject to:

$$0 \le b_{t,s,\boldsymbol{\xi}}^{\mathrm{IB}} \le 1\,, \quad \forall t \in \Omega^{\mathrm{T}}, \forall s \in \Omega^{\mathrm{S}}, \forall \boldsymbol{\xi} \in \boldsymbol{\Omega}^{\mathbf{X}}$$

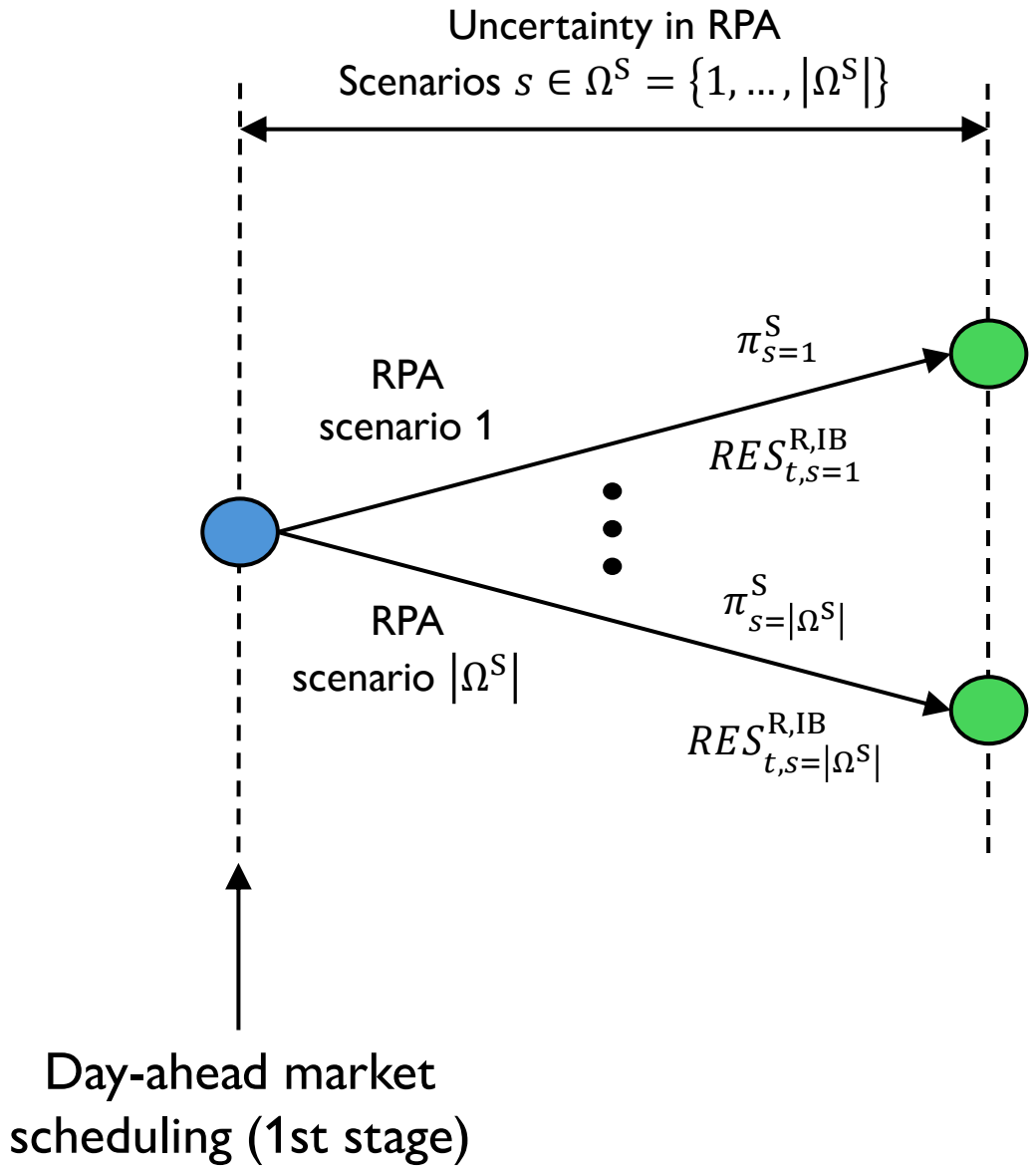
$$\left. \sum_{t\in\Omega^{\mathrm{T}}} b_{t,s,\boldsymbol{\xi}}^{\mathrm{IB}} \le \boldsymbol{\Gamma}_{\boldsymbol{\xi}}^{\mathbf{IB}}, \quad \forall s \in \Omega^{\mathrm{S}}, \forall \boldsymbol{\xi} \in \boldsymbol{\Omega}^{\mathbf{X}} \right\}$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

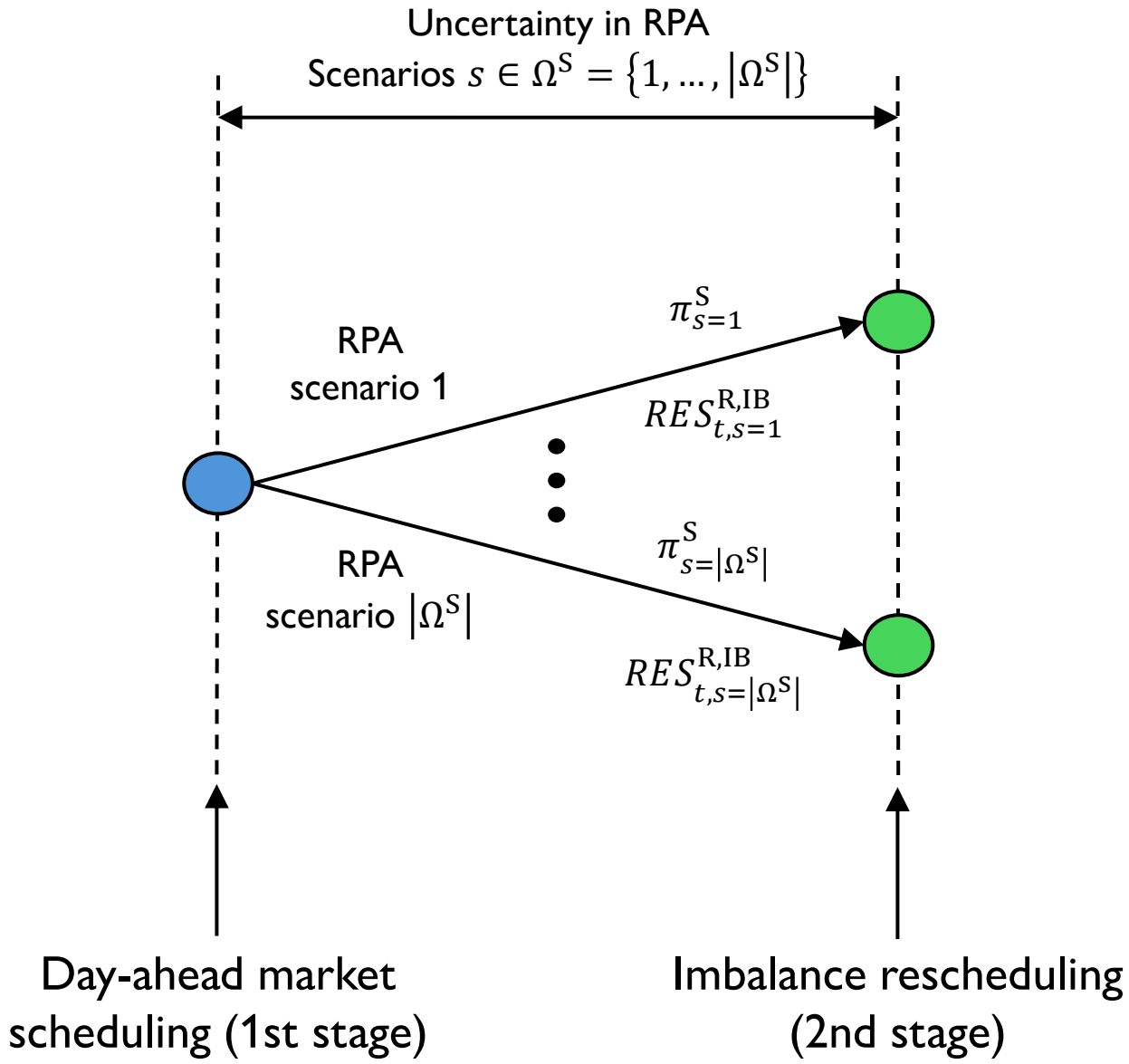


Day-ahead market scheduling (1st stage)

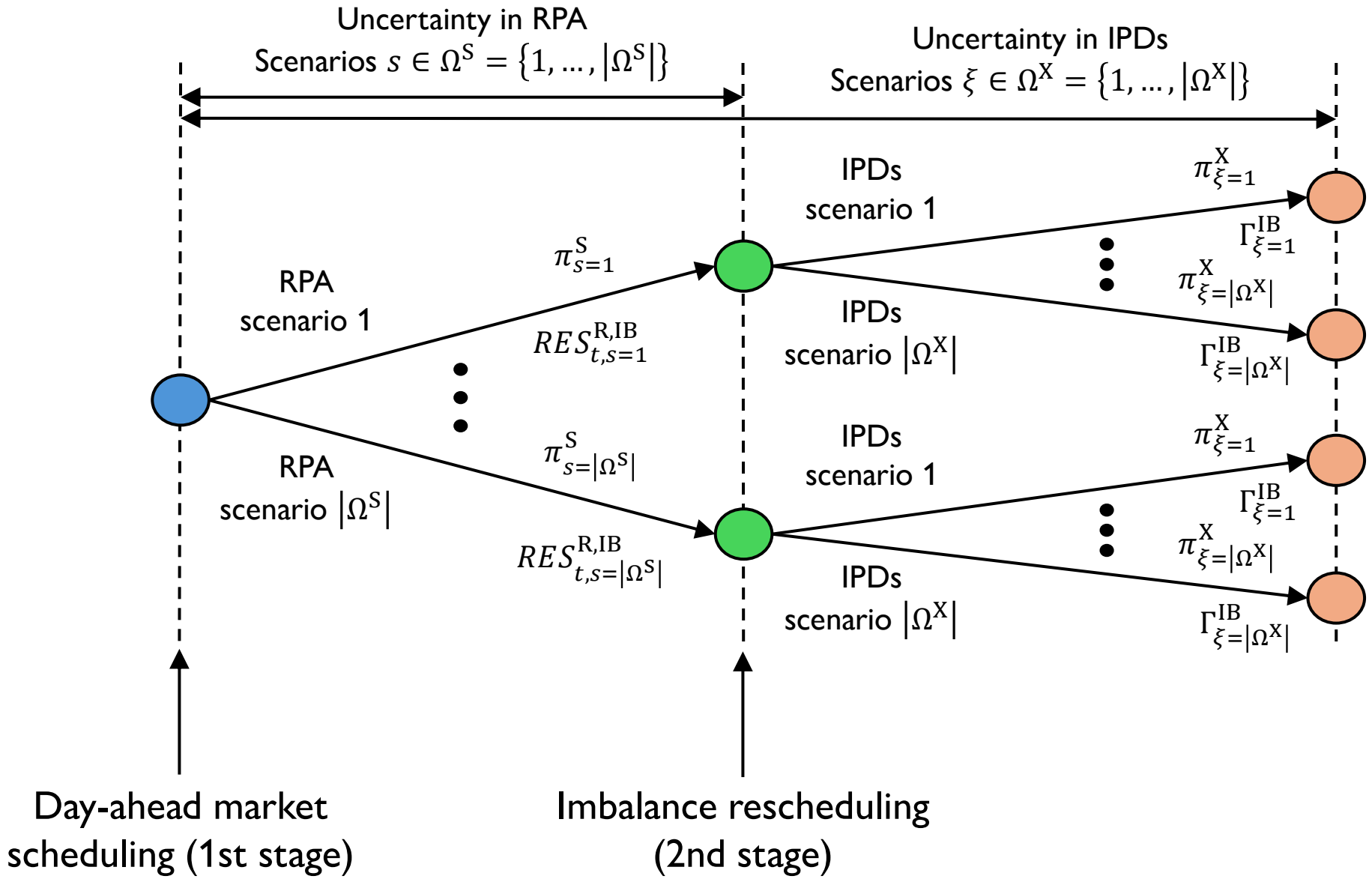
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.



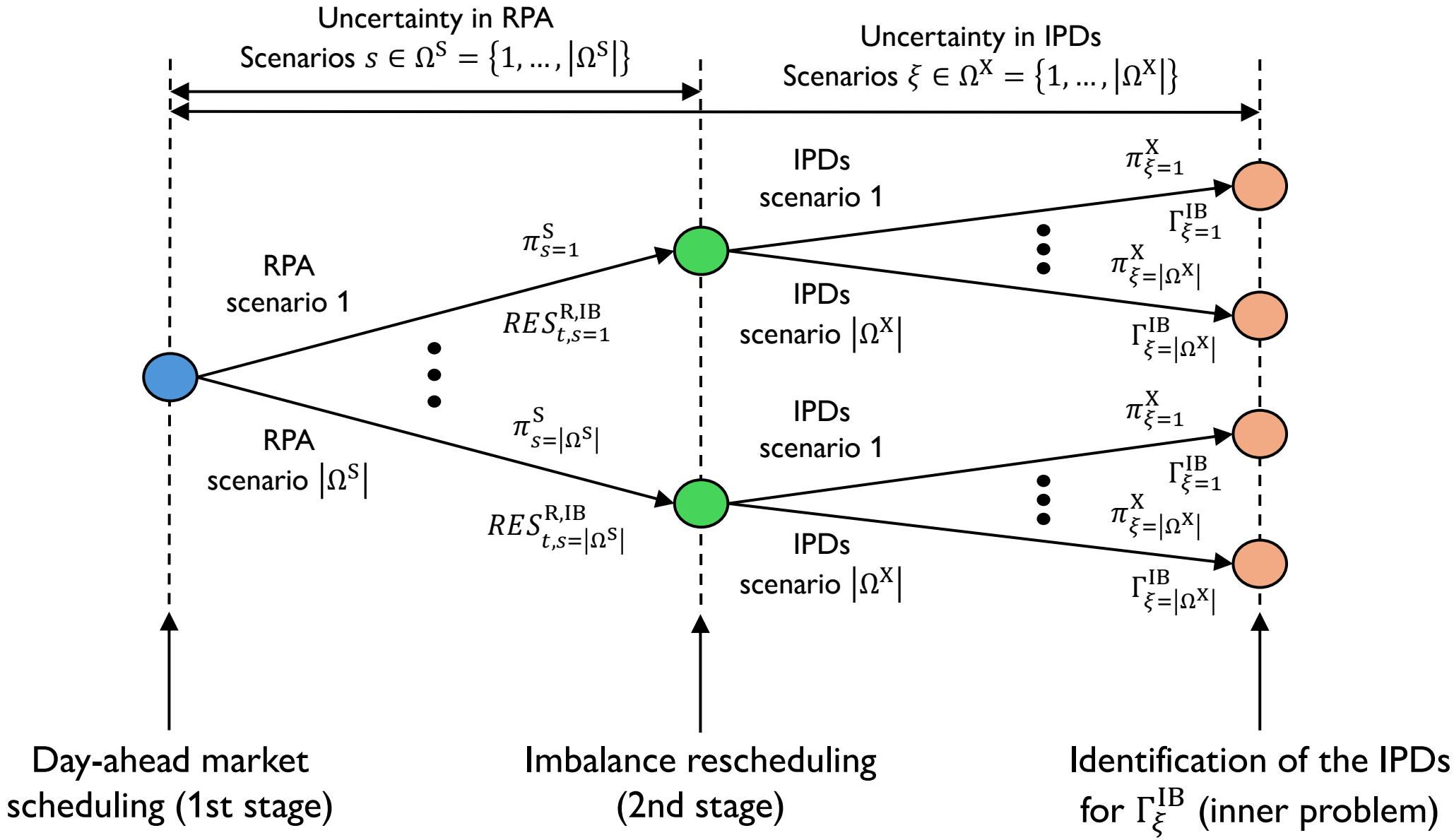
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Risk aversion

Previous model → Risk aversion is controlled through the uncertainty budget Γ^{IB}.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Risk aversion

Previous model → Risk aversion is controlled through the uncertainty budget Γ^{IB}.

Proposed model → How can we control the risk aversion?

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Risk aversion

We could modify the values of the uncertainty budget evaluated:

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \ldots, 24]$$

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \ldots, 12]$$

$$\Gamma_{\xi}^{\mathrm{IB}} = [12, 13, 14, \ldots, 24]$$

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 5, 7, 18, 23, 24]$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Risk aversion

We could modify the values of the weights:

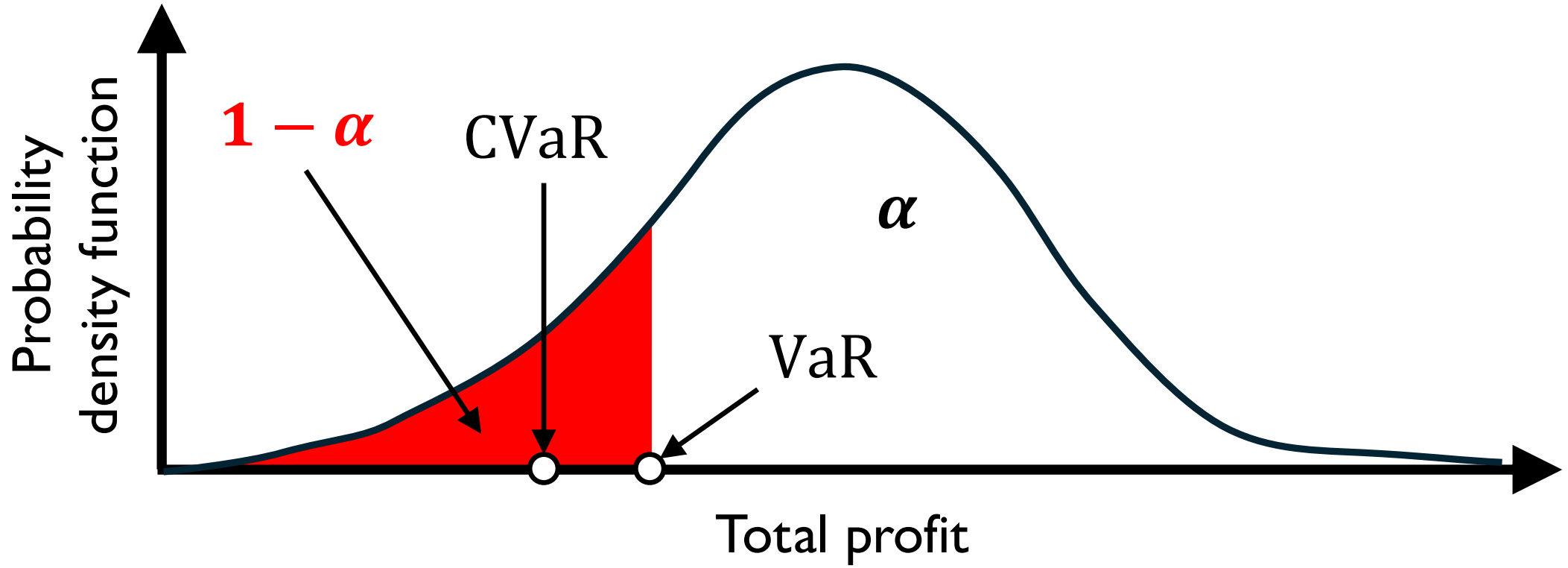
$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \ldots, 24]$$

$$\pi_{\xi}^{\mathrm{X}} = \left[\pi_{\xi_1}^{\mathrm{X}}, \pi_{\xi_2}^{\mathrm{X}}, \ldots, \pi_{\xi_{25}}^{\mathrm{X}}\right]$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Risk aversion

We decided to model the risk aversion through the conditional value-at-risk (CVaR):



Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

Model	RPA	IPD	Risk	Risk-aversion parameter
Previous	Stochastic Programming	Stochastic Programming	IPD	Uncertainty budget Γ^{IB}
Proposed	Stochastic Programming	Stochastic Programming + Robust Optimization	RPA and IPD	Confidence level α

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

Does the proposed optimization model enhance the results of the previous approach?

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Proposed optimization model

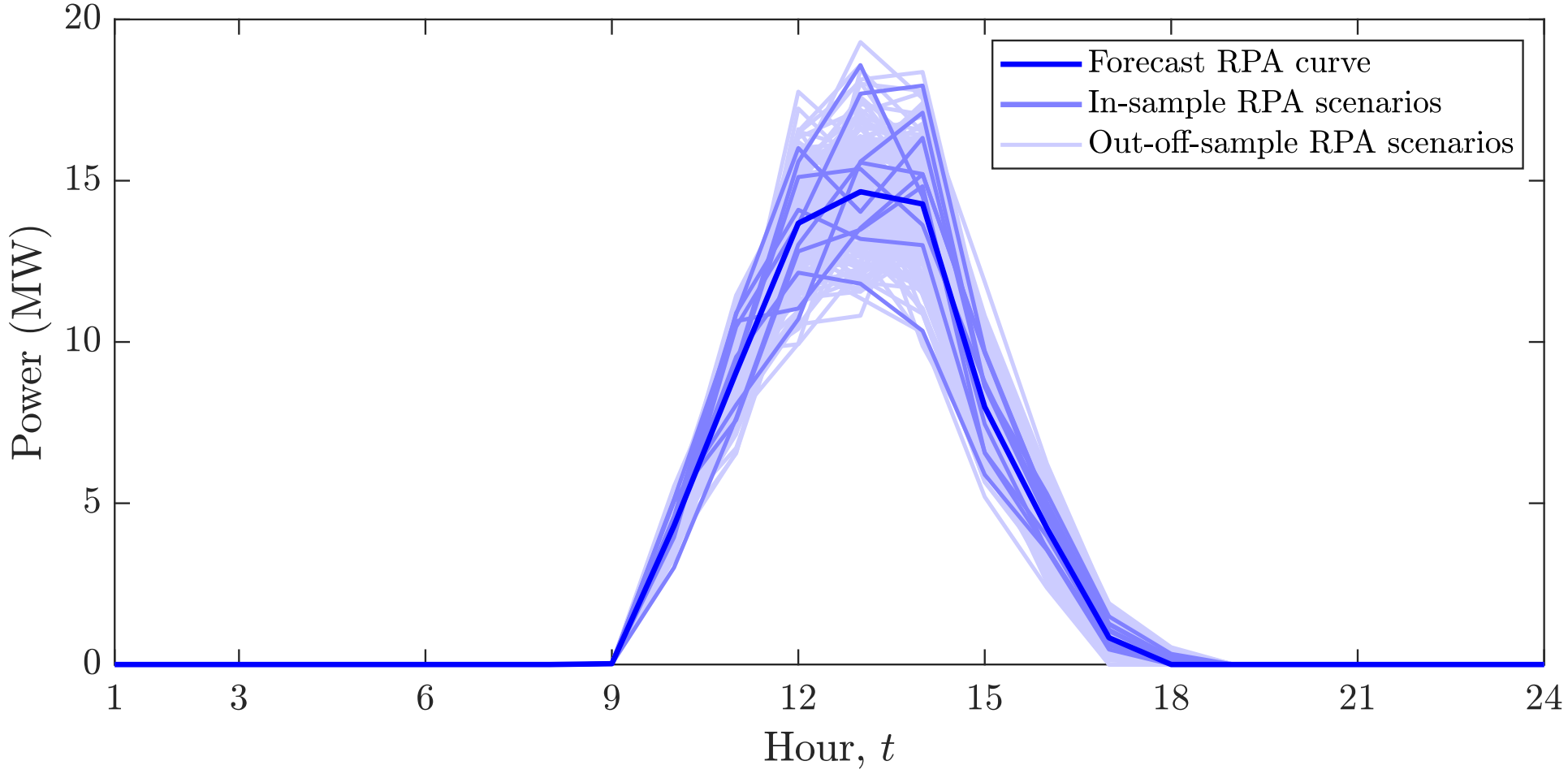
Does the proposed optimization model enhance the results of the previous approach?

Spoiler: Yes

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

2.4. Case Study

RPA scenarios



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IPD scenarios

In-sample analyses

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \dots, 24]$$

$$\pi_{\xi}^{\mathrm{X}} = [1/25, 1/25, \dots, 1/25]$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

IPD scenarios

In-sample analyses

$$\Gamma_{\xi}^{\mathrm{IB}} = [0, 1, 2, \dots, 24]$$

$$\pi_{\xi}^{\mathrm{X}} = [1/25, 1/25, \dots, 1/25]$$

Out-of-sample analyses

366 historical daily curves of the power system IPD

$$\lambda_t^{\mathrm{IB}} = \lambda_t^{\mathrm{DA}}\left(1 \pm K^{\mathrm{IB}}\right)$$

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

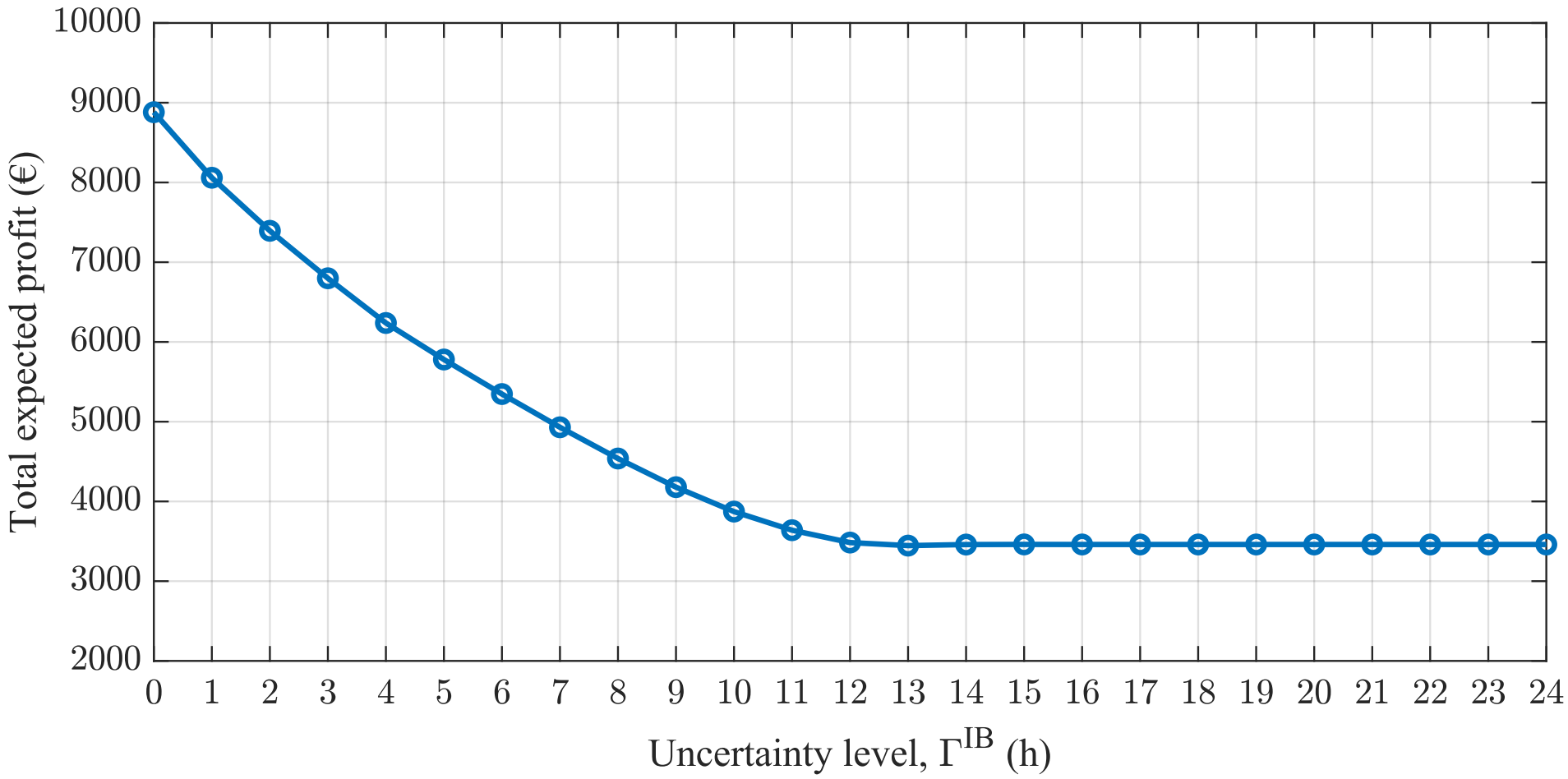
Research questions

RQ1: Does the proposed model reduce the arbitrage of the VPP between the day-ahead market and the imbalance settlement in comparison with the previous model?

RQ2: How do the market position and the operation of the VPP depend on the risk-aversion level of the decision maker?

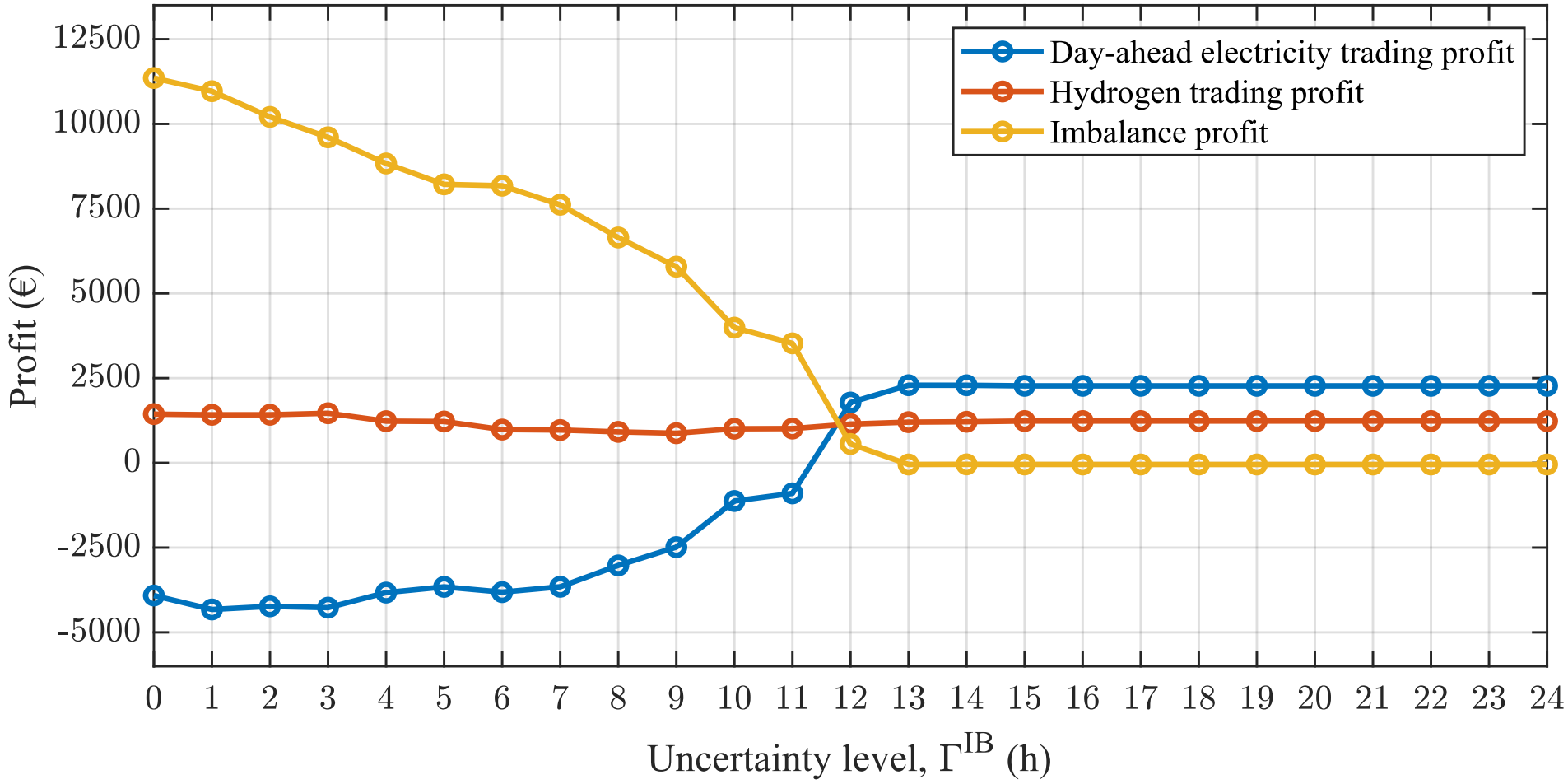
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Analysis of the profit: previous



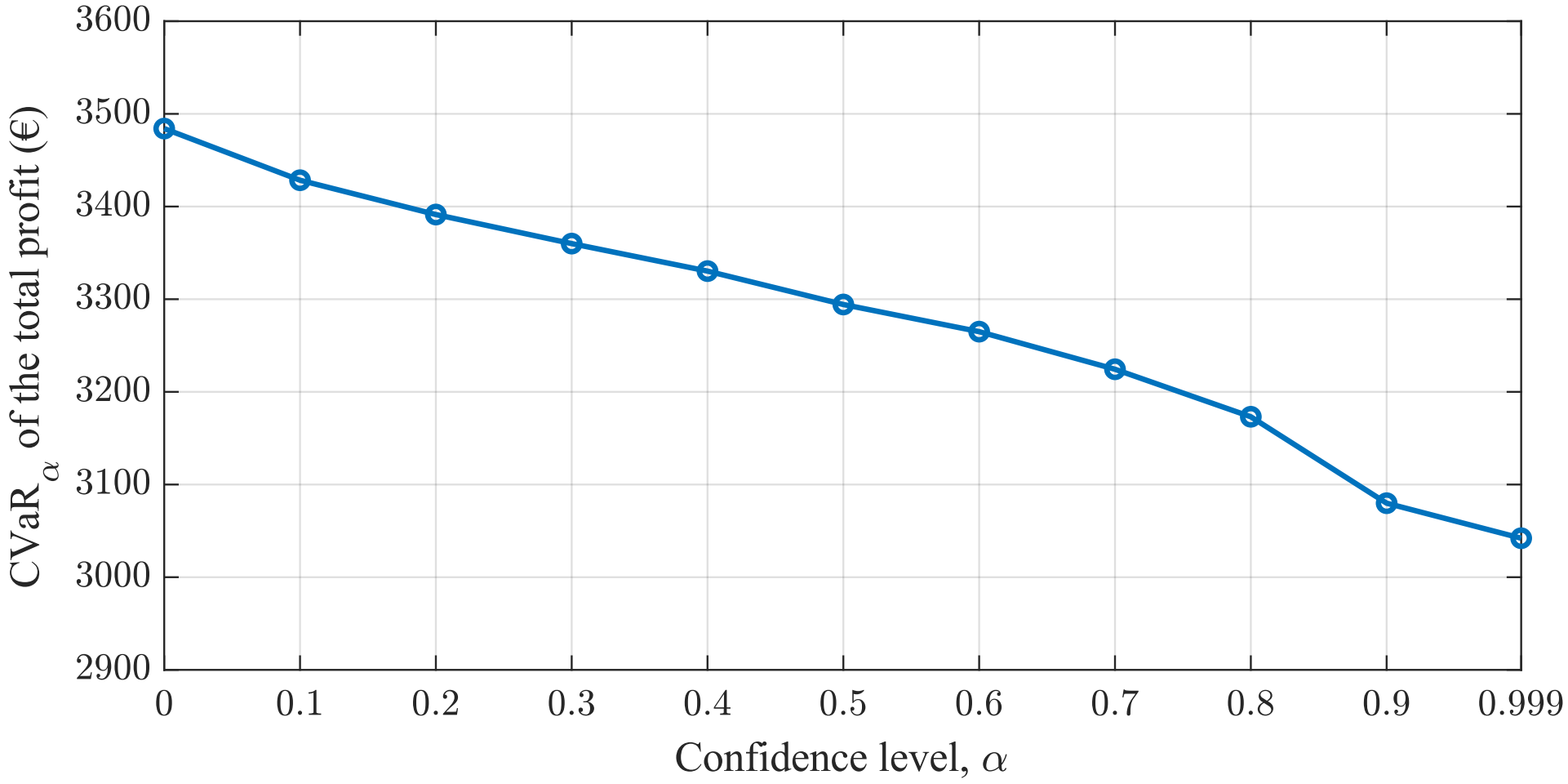
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Analysis of the profit: previous



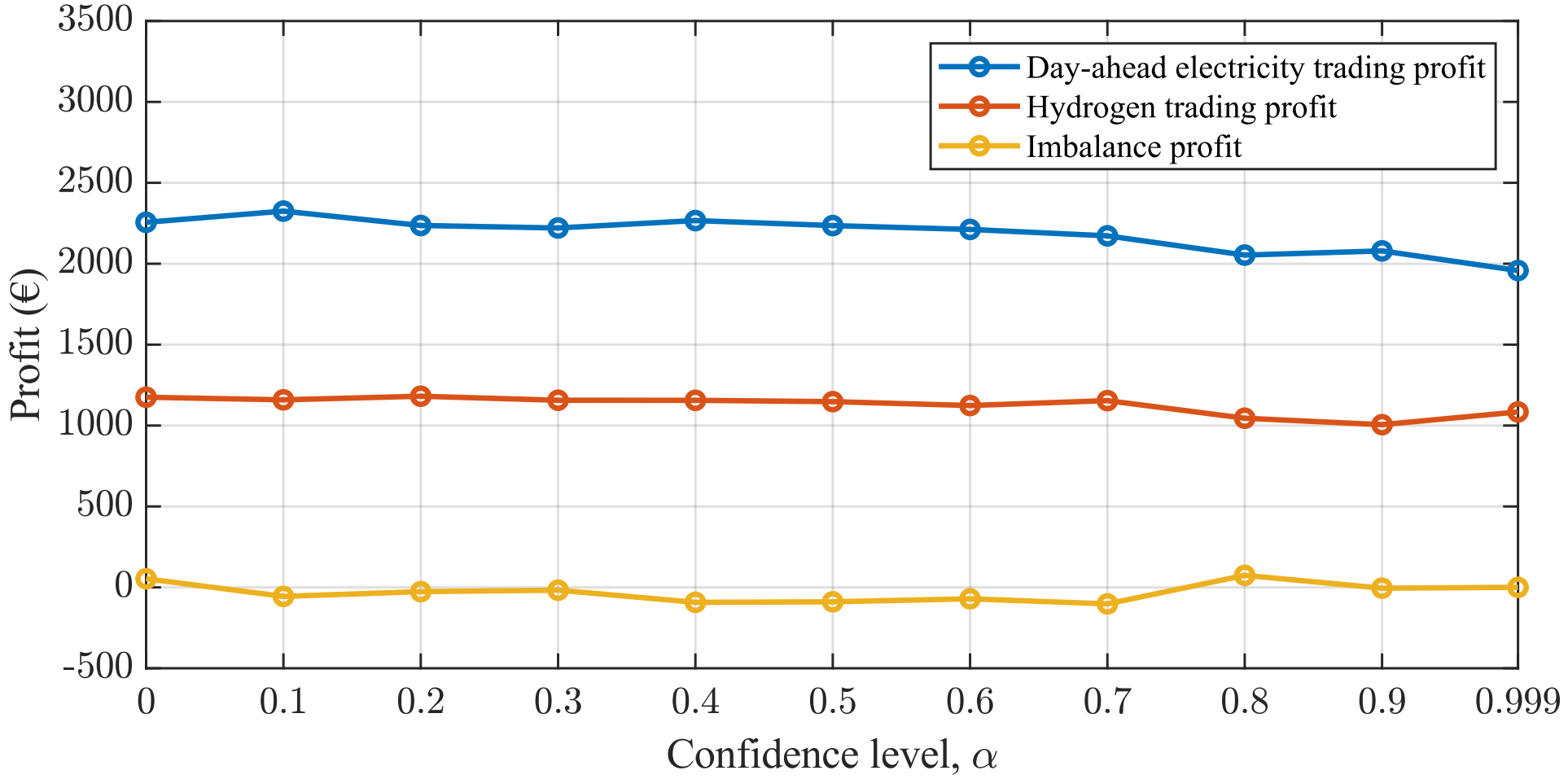
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Analysis of the profit: proposed



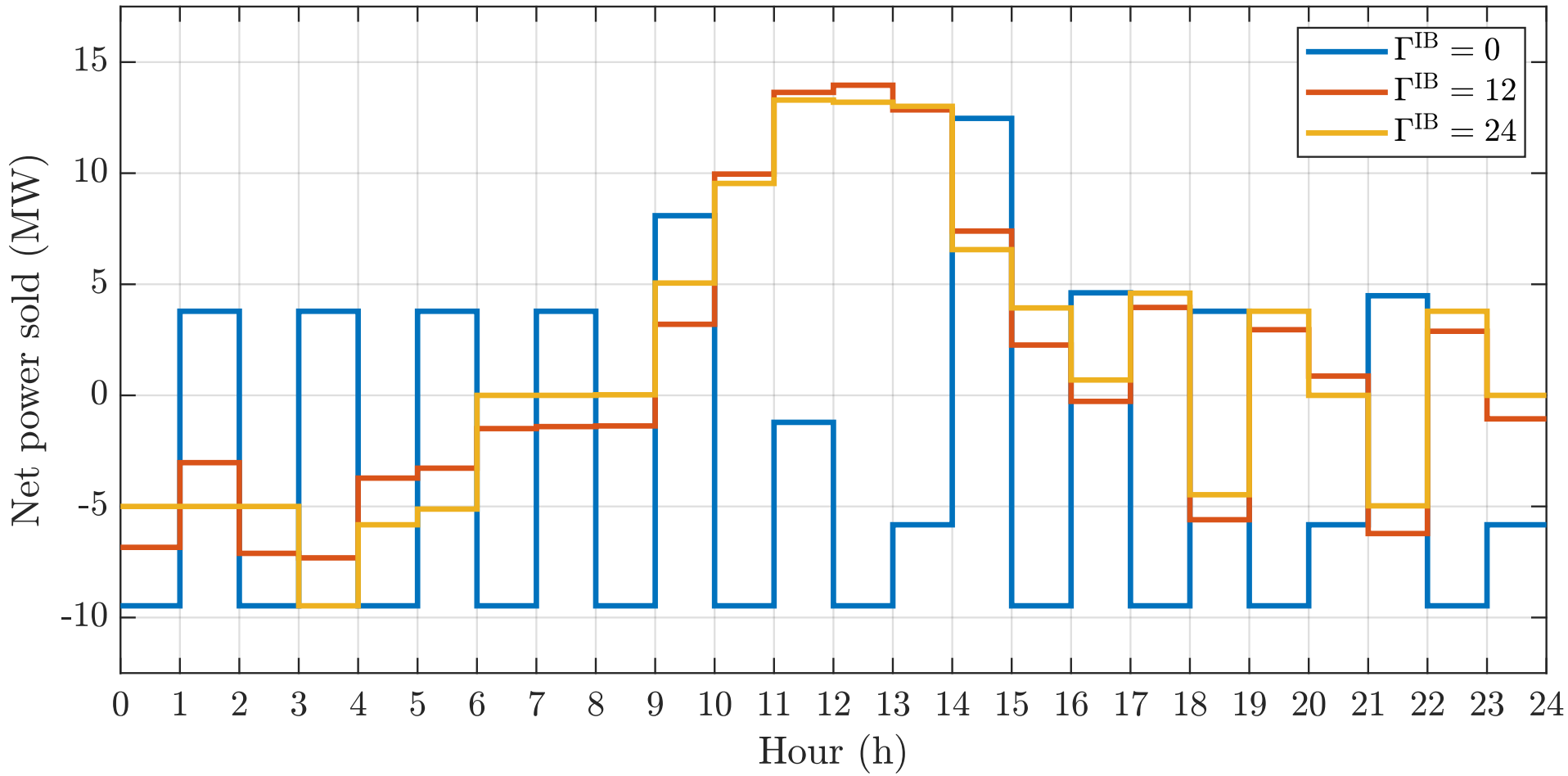
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Analysis of the profit: proposed



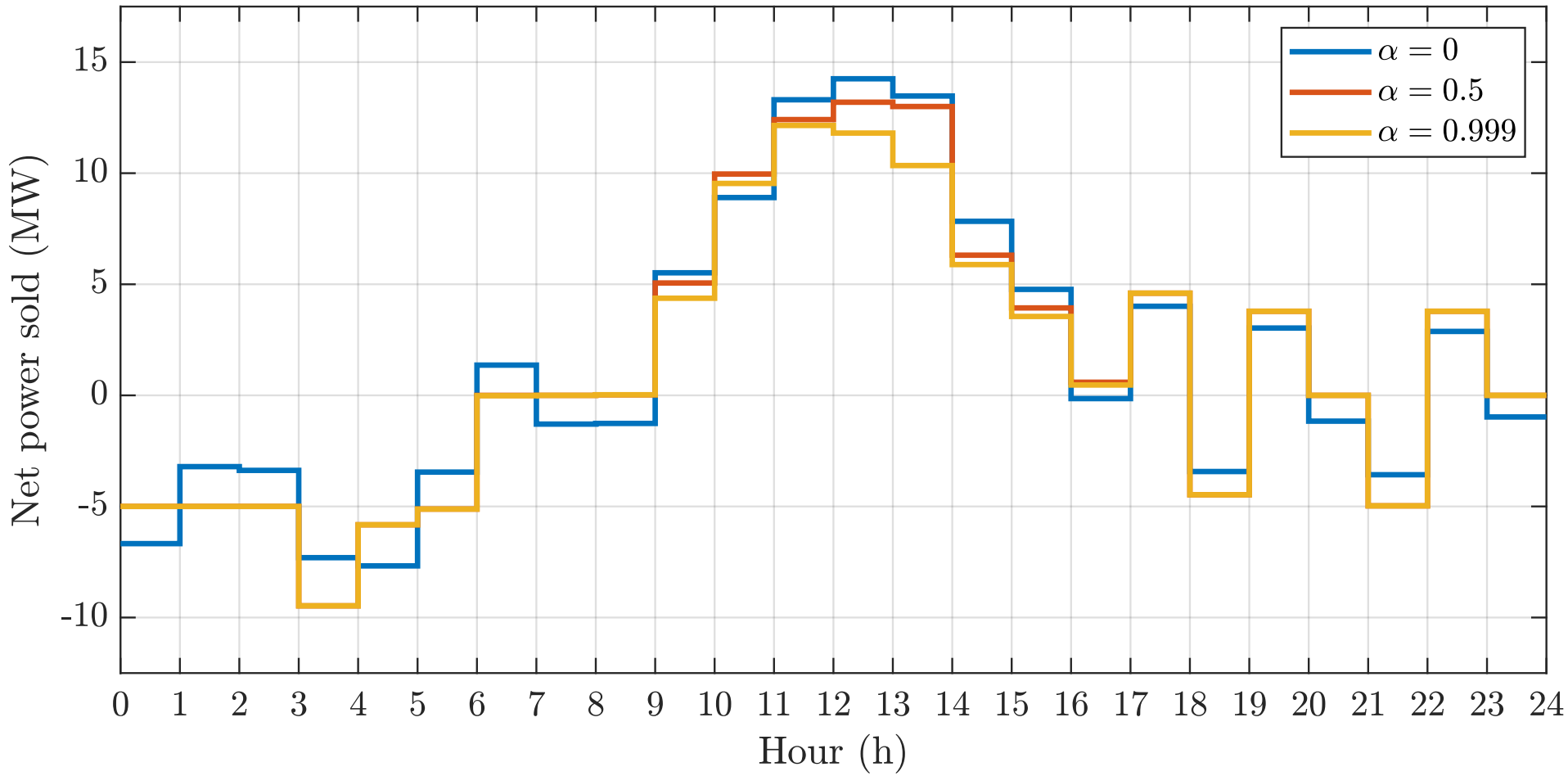
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Market position of the VPP: previous



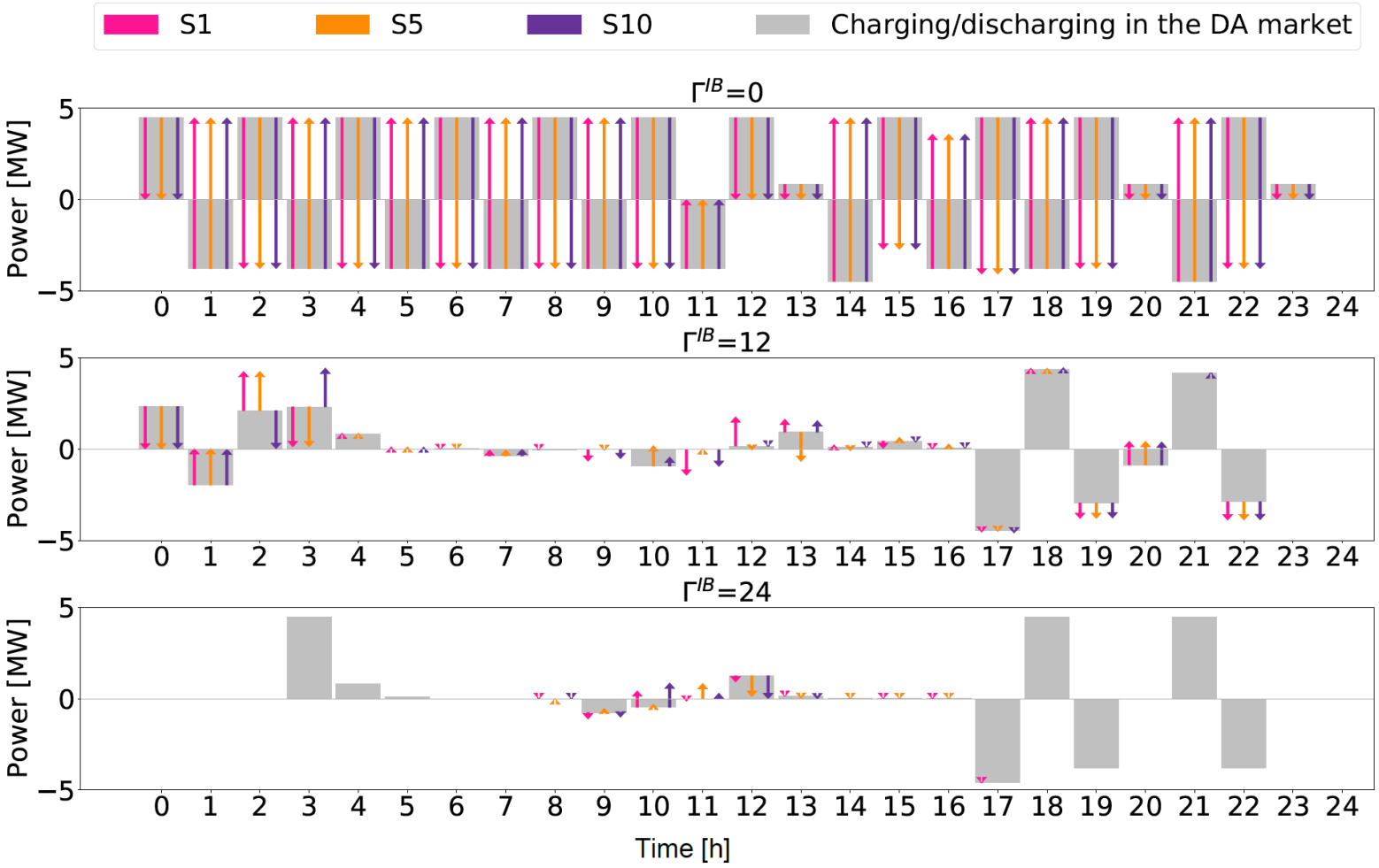
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Market position of the VPP: proposed



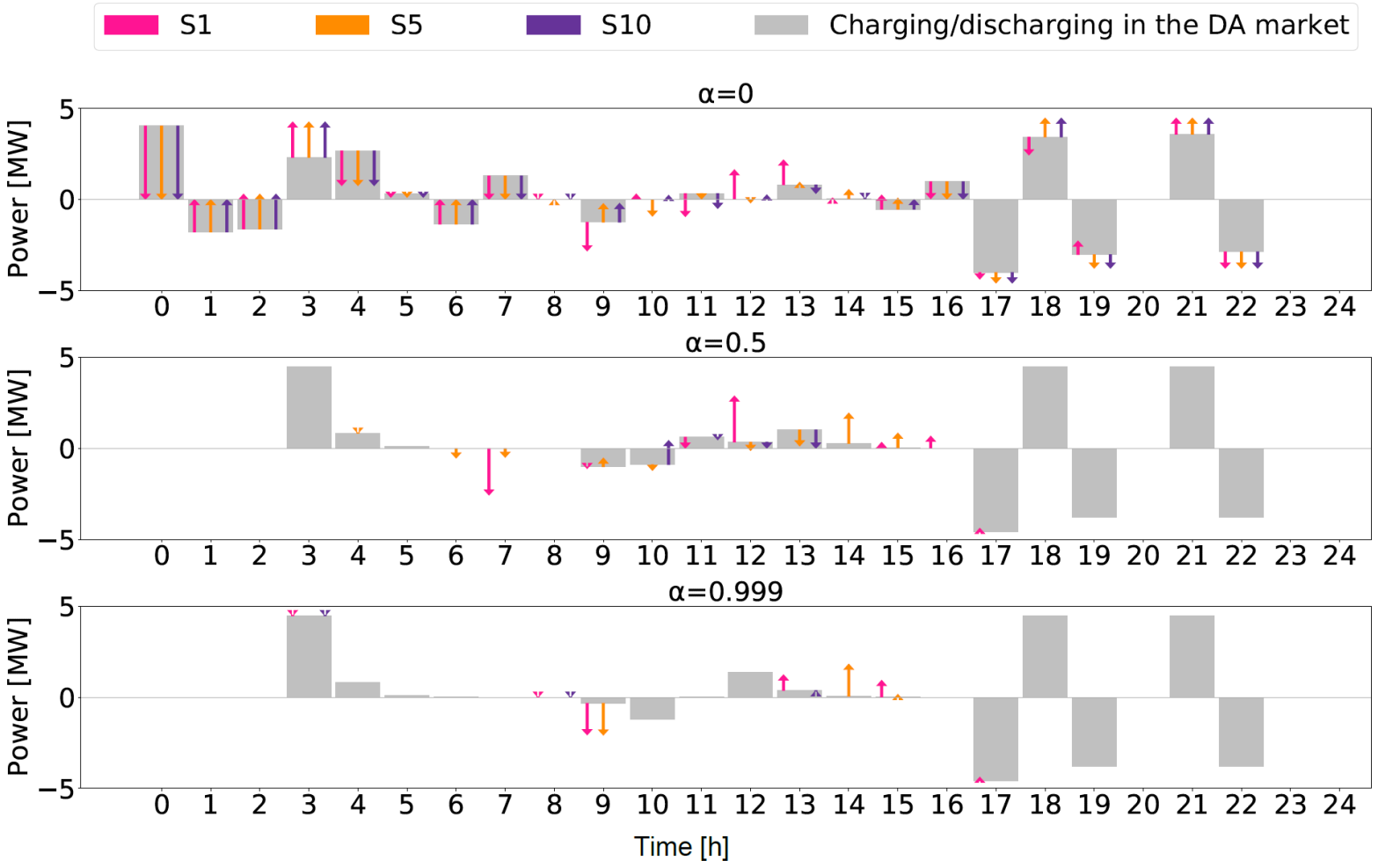
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Market position of the battery: previous



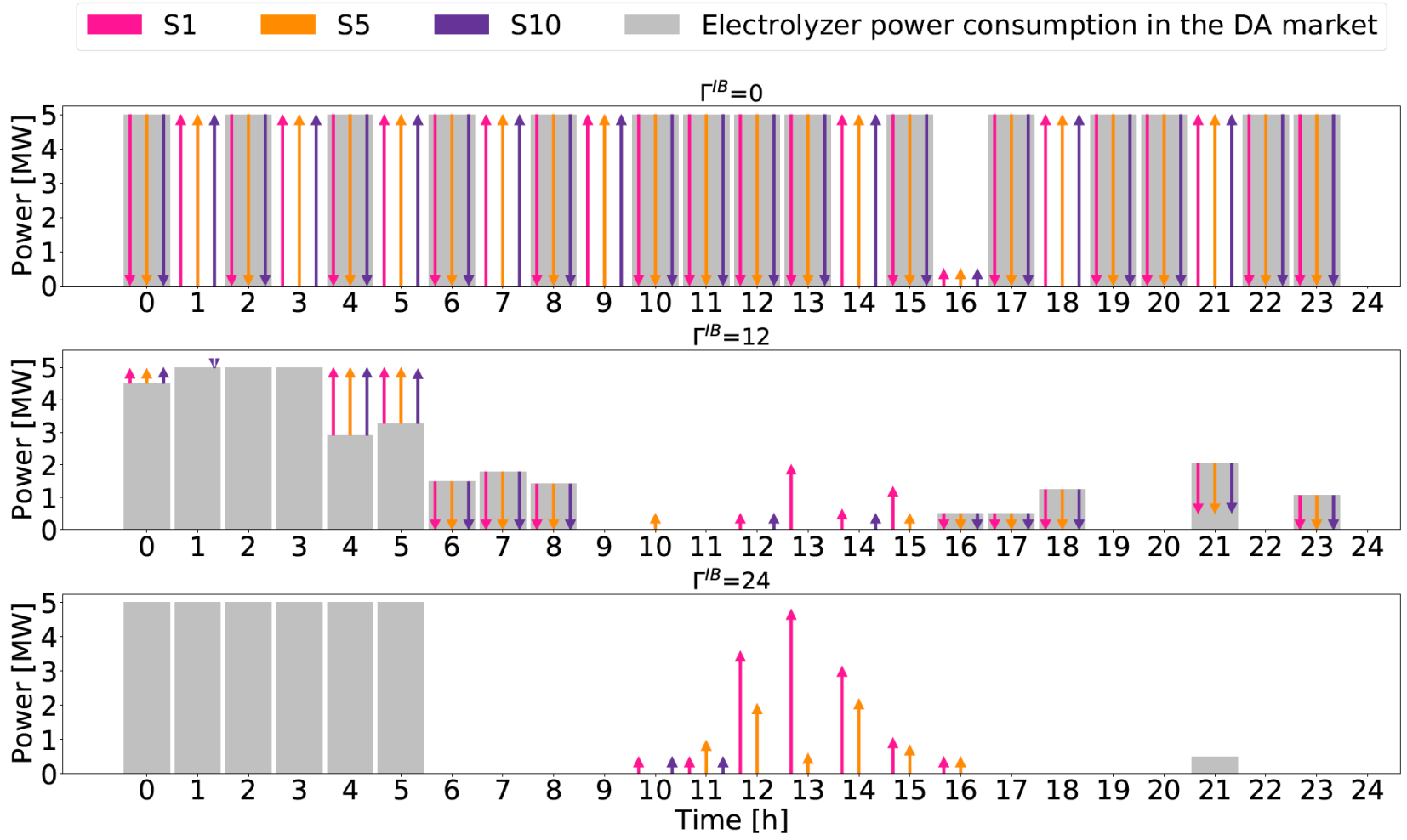
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Market position of the battery: proposed



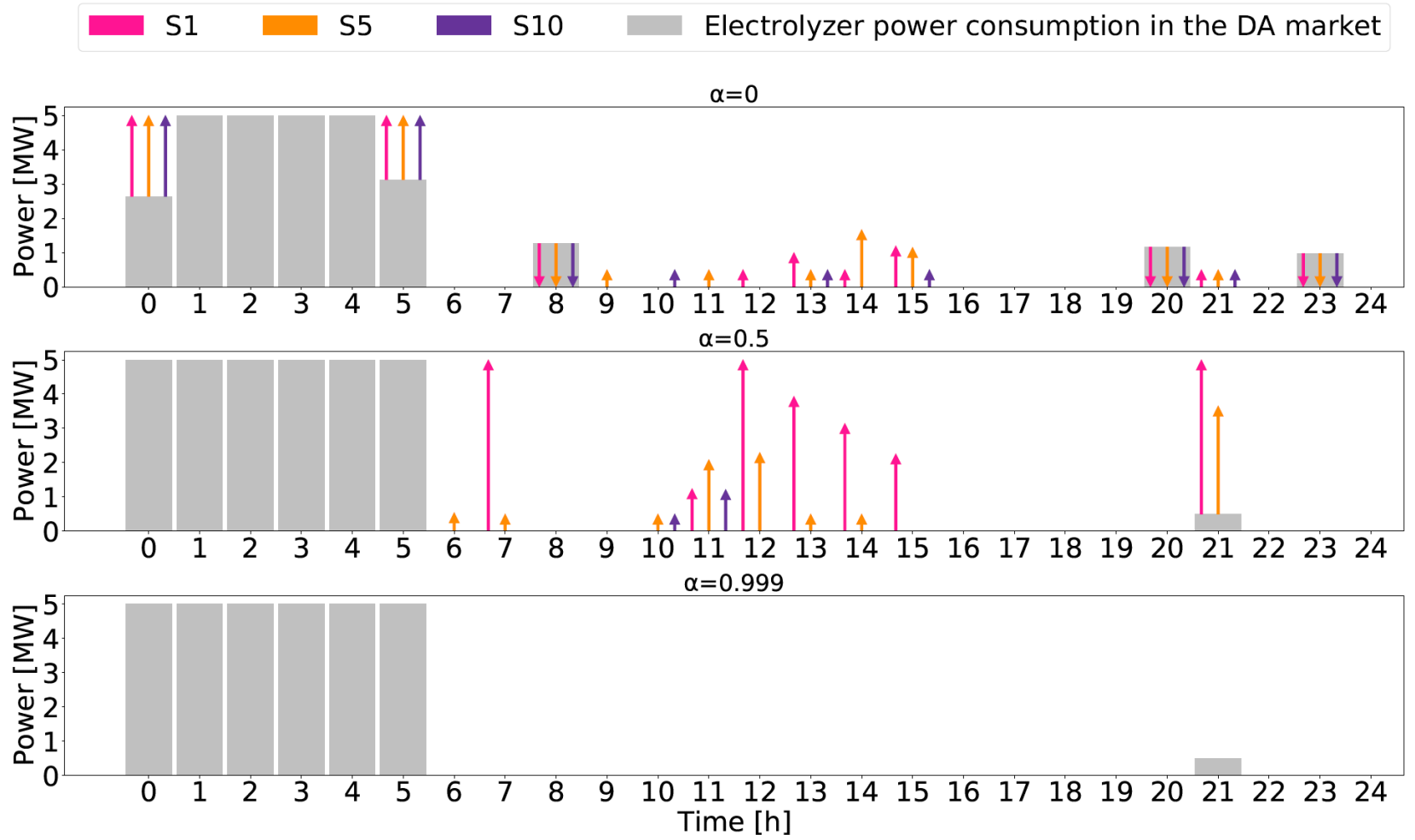
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Market position of the electrolyzer: previous



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Market position of the electrolyzer: proposed



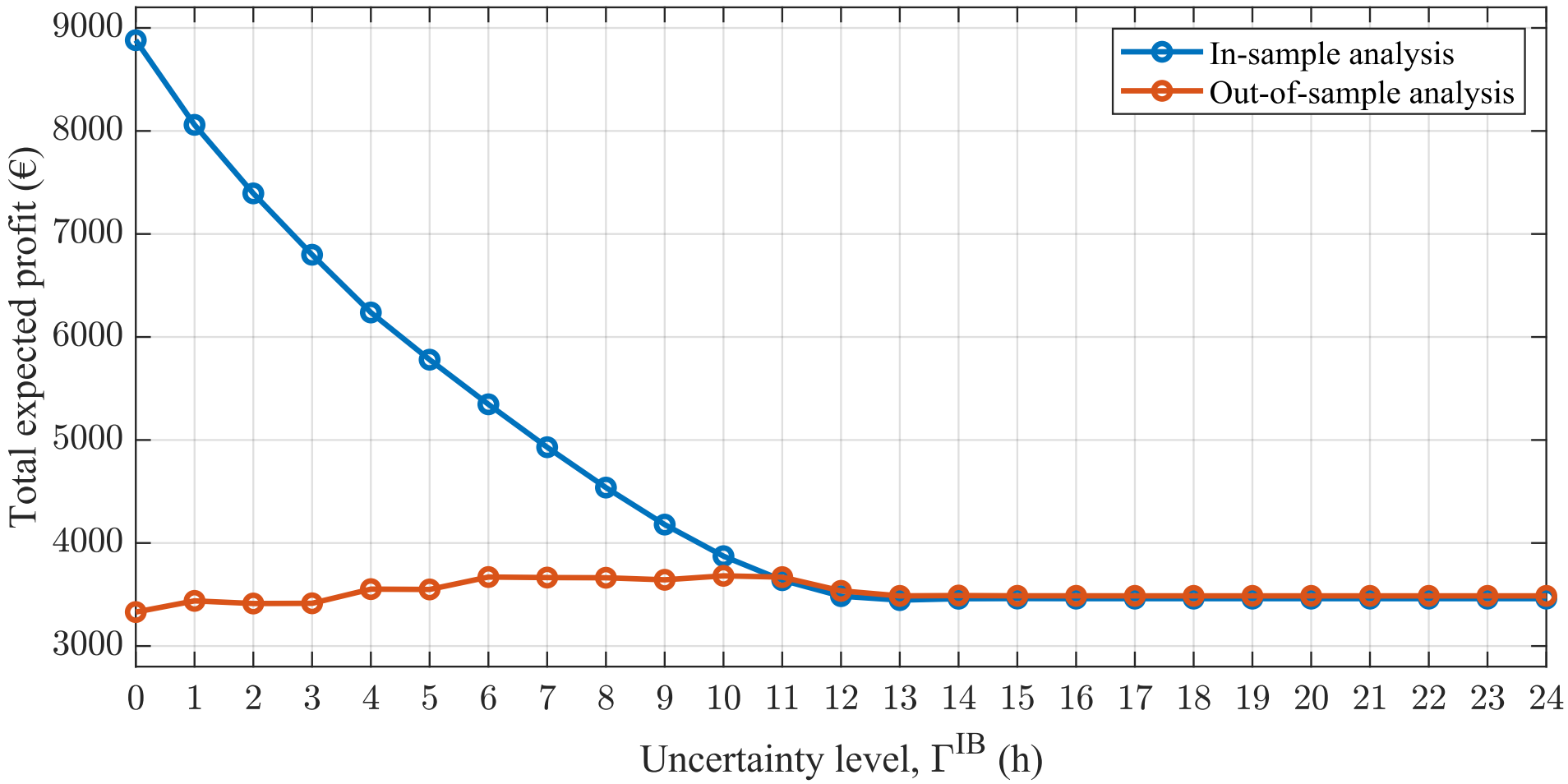
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Research questions

RQ4: Do the day-ahead bidding and offering decisions determined by the proposed formulation provide similar results between in-sample and out-of-sample analyses?

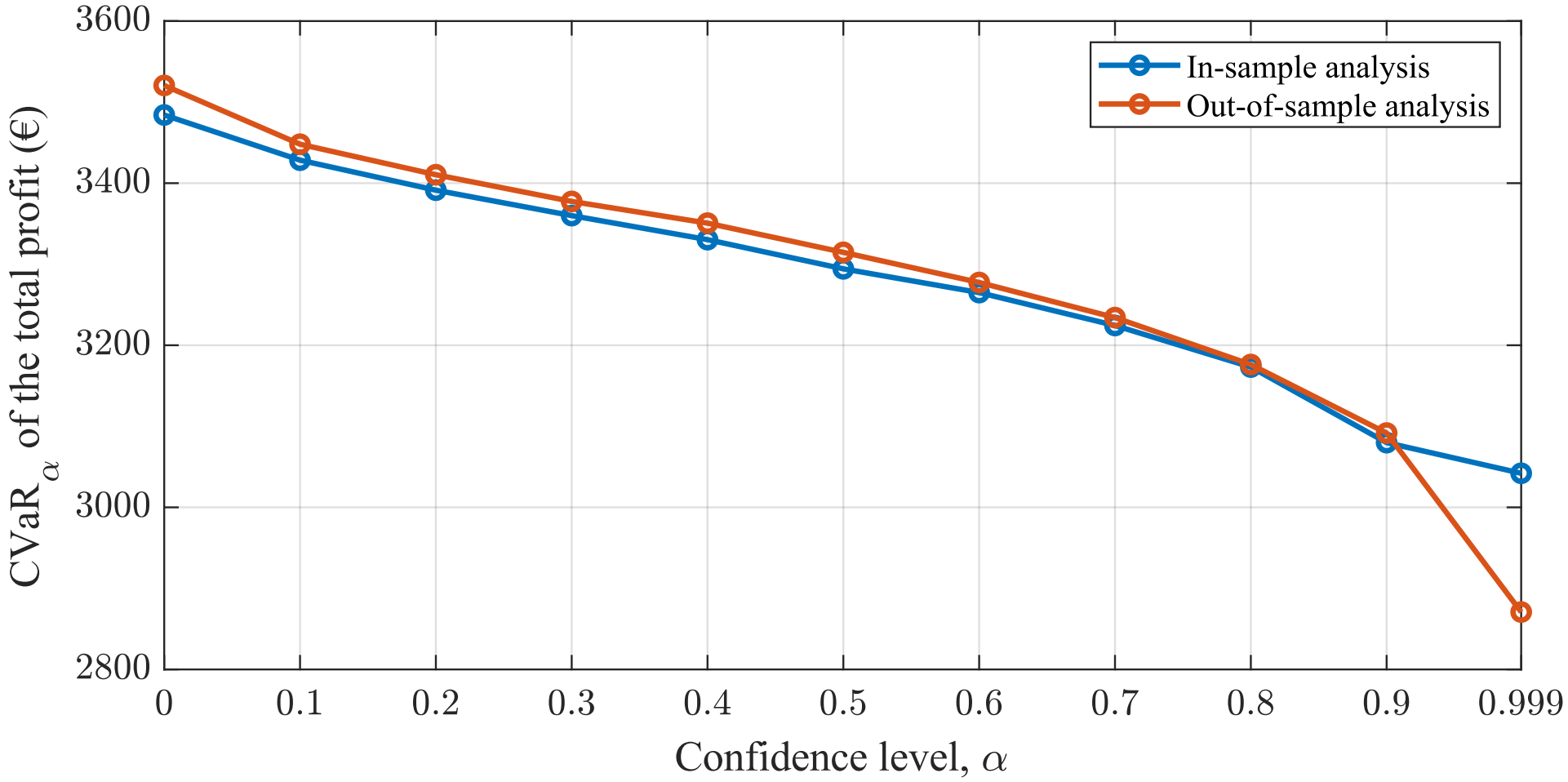
Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

In-sample vs out-of-sample: previous



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In-sample vs out-of-sample: proposed



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Computational performance

Gurobi solver, relative optimality gap = 1%.

Time	Constraints	Variables	
		Continuous	Binary
< 3 min	12,372	14,337	1,008

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

2.5. Conclusions

Conclusions

1. Proposed model → Stochastic programming + robust optimization (scenarios of the uncertainty budget).

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Conclusions

1. Proposed model → Stochastic programming + robust optimization (scenarios of the uncertainty budget).

2. Impact of the risk aversion on the market position.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Conclusions

1. Proposed model → Stochastic programming + robust optimization (scenarios of the uncertainty budget).

2. Impact of the risk aversion on the market position.

3. Proposed model → Better out-of-sample results than the previous model.

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Acknowledgments

Grant BDNS ID 795356, funded by: UCLM Universidad Castilla-La Mancha

European Union's NextGenerationEU programme, as part of the institutional project "Resilient Self-Healing Future Power Systems – RePowerFER" (University of Zagreb, Faculty of Electrical Engineering and Computing).

Grants PID2021-126566OB-I00, PID2021-122579OB-I00, and PID2025-171604OB-I00, funded by: MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES; Cofinanciado por la Unión Europea; AGENCIA ESTATAL DE INVESTIGACIÓN

Grants 101160614 (EU-DREAM Project), 101168796 (MCSA Doctoral Network FITNESS), and 101230578 (INNO-TREC project) funded by: Funded by the European Union

Grant 2025-GRIN-38264, funded by: Cofinanciado por la Unión Europea; Ministerio de Hacienda; Fondos Europeos; Castilla-La Mancha; UCLM Universidad de Castilla-La Mancha

Á. García-Cerezo, P. Miljan, and H. Pandžić, "Day-ahead market scheduling strategy of a virtual power plant under the single imbalance pricing scheme via stochastic robust optimization", *Energy Economics*, vol. 163, art. no. 109627, Nov. 2026, DOI: 10.1016/j.eneco.2026.109627.

Paper available online



3. Students' Projects Supervised



Figures in Spanish

Students' Projects Supervised

Author: Roberto Pos Arcos.

Project's Title: Generation expansion planning of power systems considering inertia constraints.

Supervisors: Álvaro García-Cerezo and Miguel Carrión.

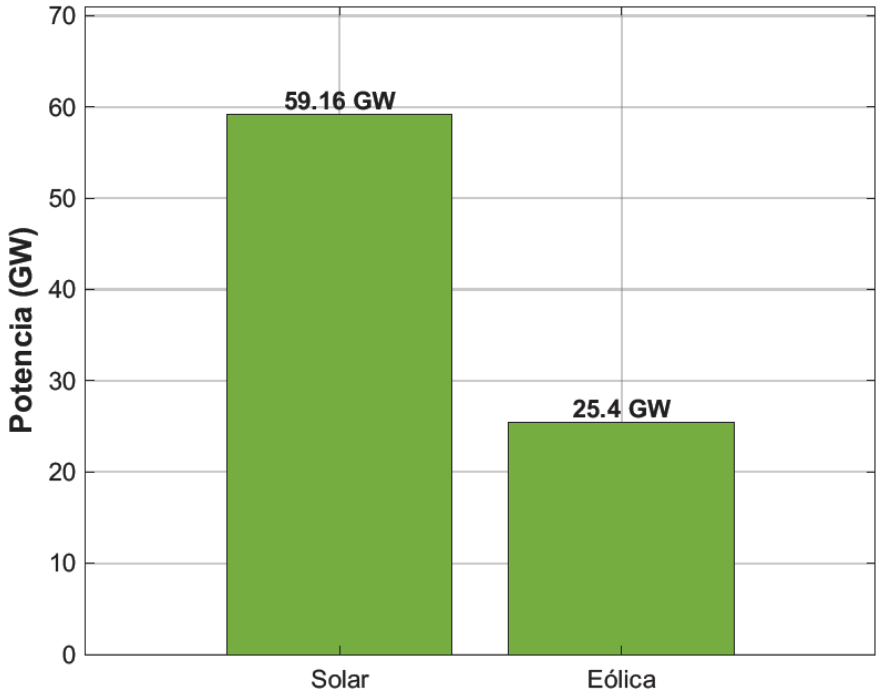
Bachelor's Final Project, Electrical Engineering Degree, Universidad de Castilla-La Mancha.

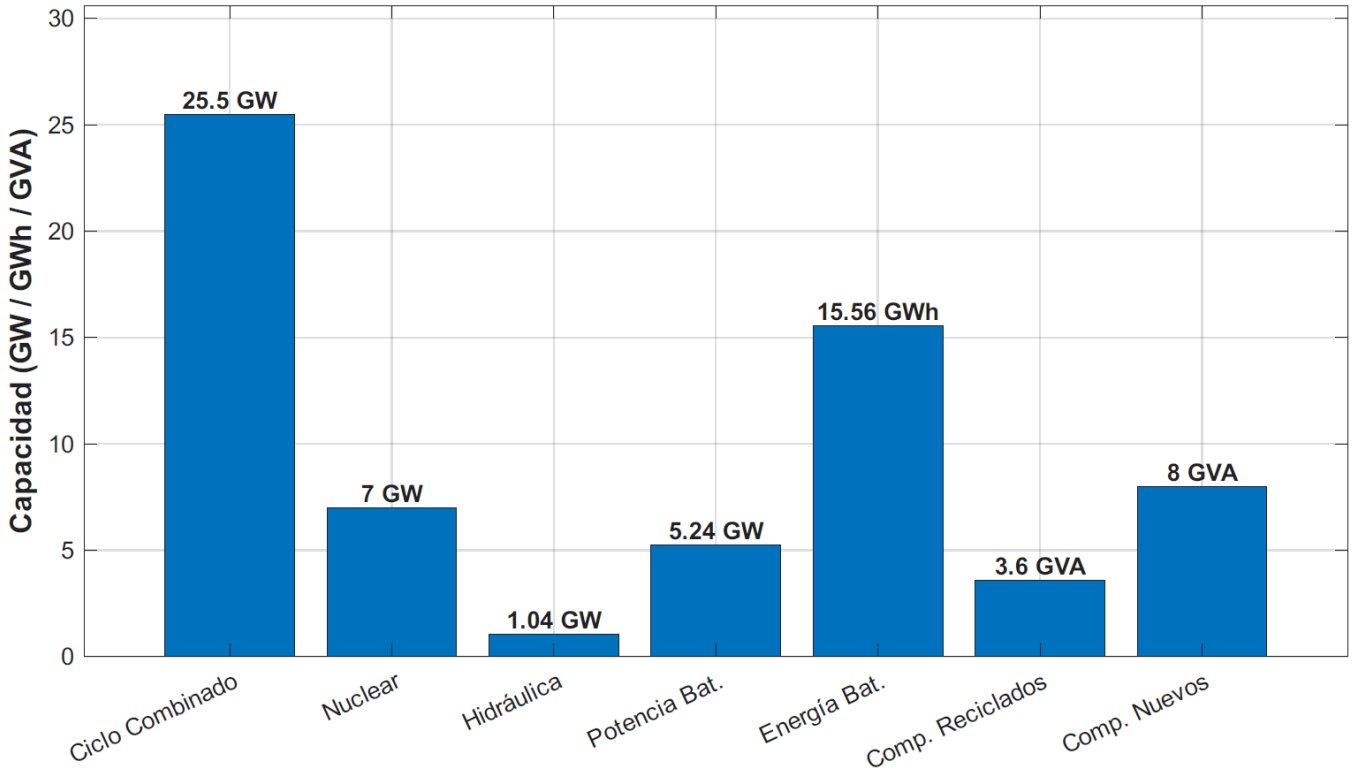
Date of defense: September 2026.

Students' Projects Supervised

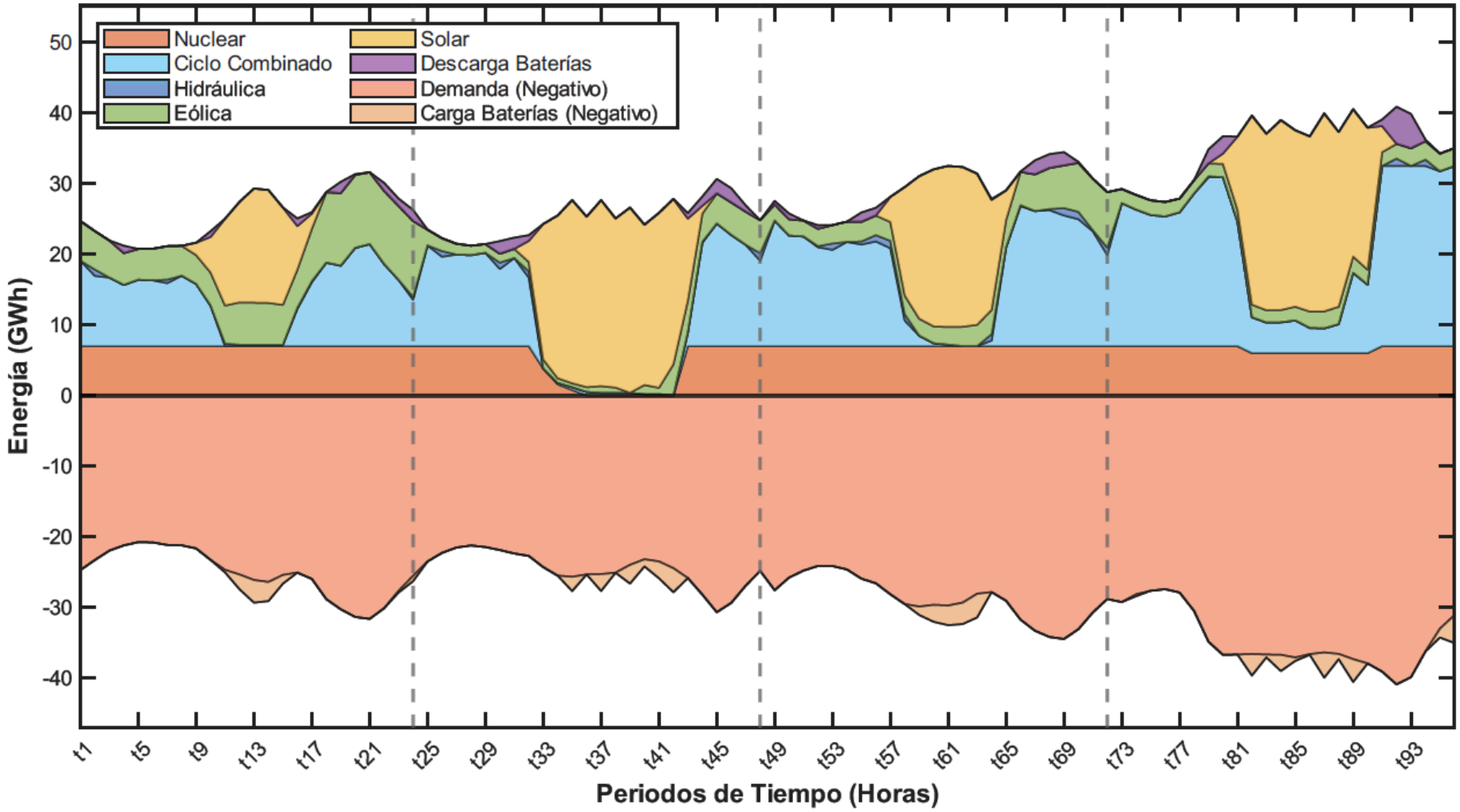


Students' Projects Supervised

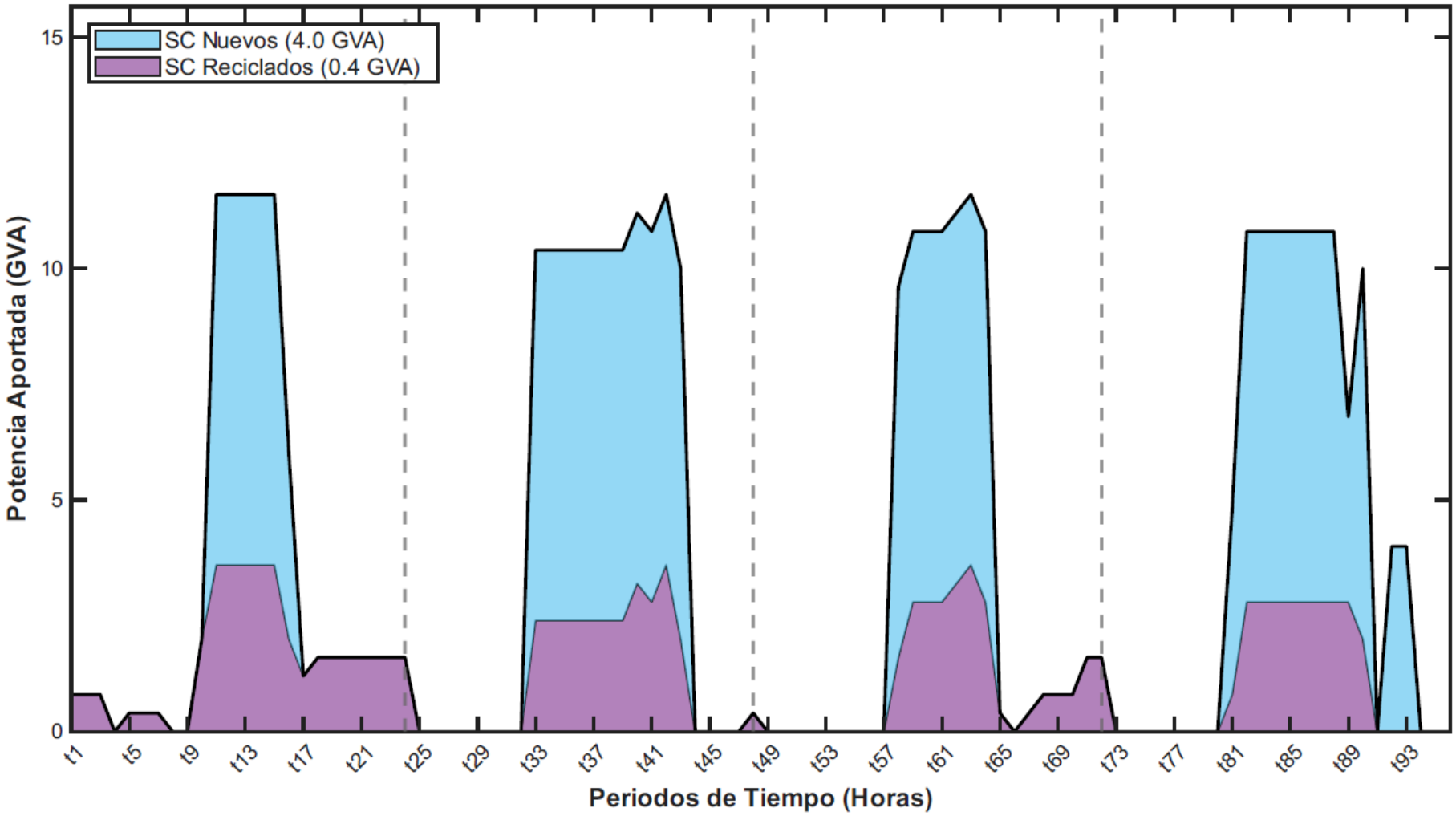




Students' Projects Supervised



Students' Projects Supervised



Students' Projects Supervised

Author: Jorge Muñoz Pedrosa.

Project's Title: Planning and operational decision-making for an industrial plant with self-consumption.

Supervisors: Álvaro García-Cerezo and Miguel Cañas-Carretón.

Bachelor's Final Project, Electrical Engineering Degree, Universidad de Castilla-La Mancha.

Date of defense: July 2026.

Students' Projects Supervised

GOBIERNO DE ESPAÑA
VICEPRESIDENCIA PRIMERA DEL GOBIERNO
MINISTERIO DE HACIENDA
SECRETARÍA DE ESTADO DE HACIENDA
DIRECCIÓN GENERAL DEL CATASTRO

CONSULTA DESCRIPTIVA Y GRÁFICA
DE DATOS CATASTRALES DE BIEN INMUEBLE

Referencia catastral: 0655004VK2105F0001TS

DATOS DESCRIPTIVOS DEL INMUEBLE

Localización:
CL RIO JARAMA 117
45007 TOLEDO [TOLEDO]

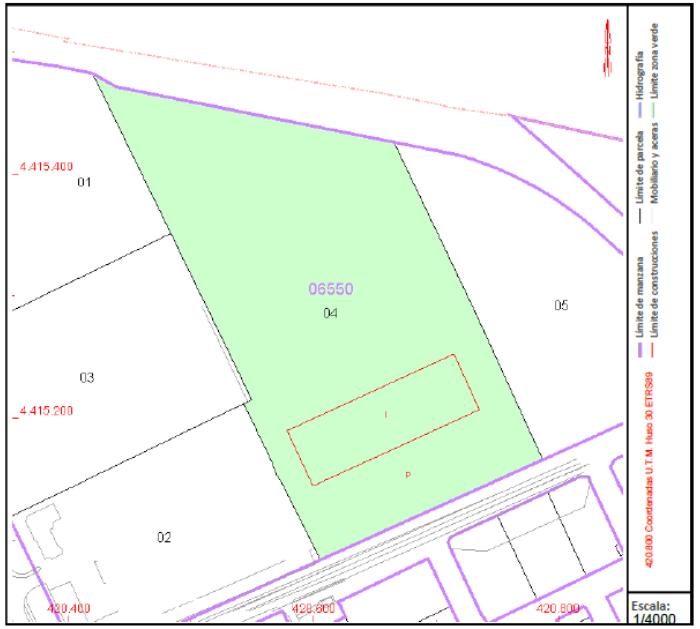
Clase: URBANO
Uso principal: Industrial
Superficie construida: 7.500 m2
Año construcción: 2012

CONSTRUCCIÓN

Destino	Escalera/Planta/Puerta	Superficie m²
INDUSTRIAL	1/00/01	7.500

PARCELA

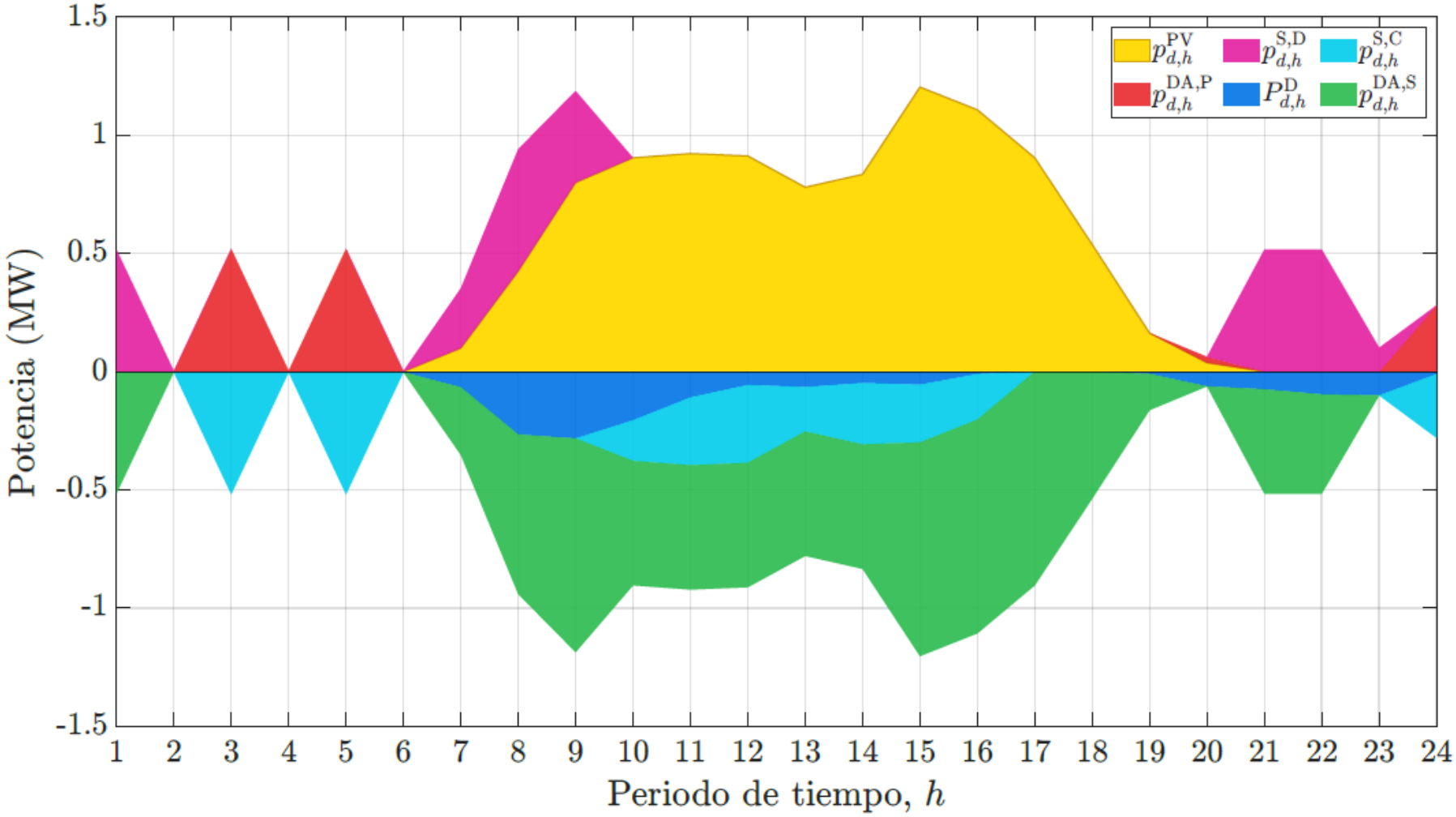
Superficie gráfica: 71.381 m2
Participación del inmueble: 100,00 %
Tipo: Parcela construida sin división horizontal



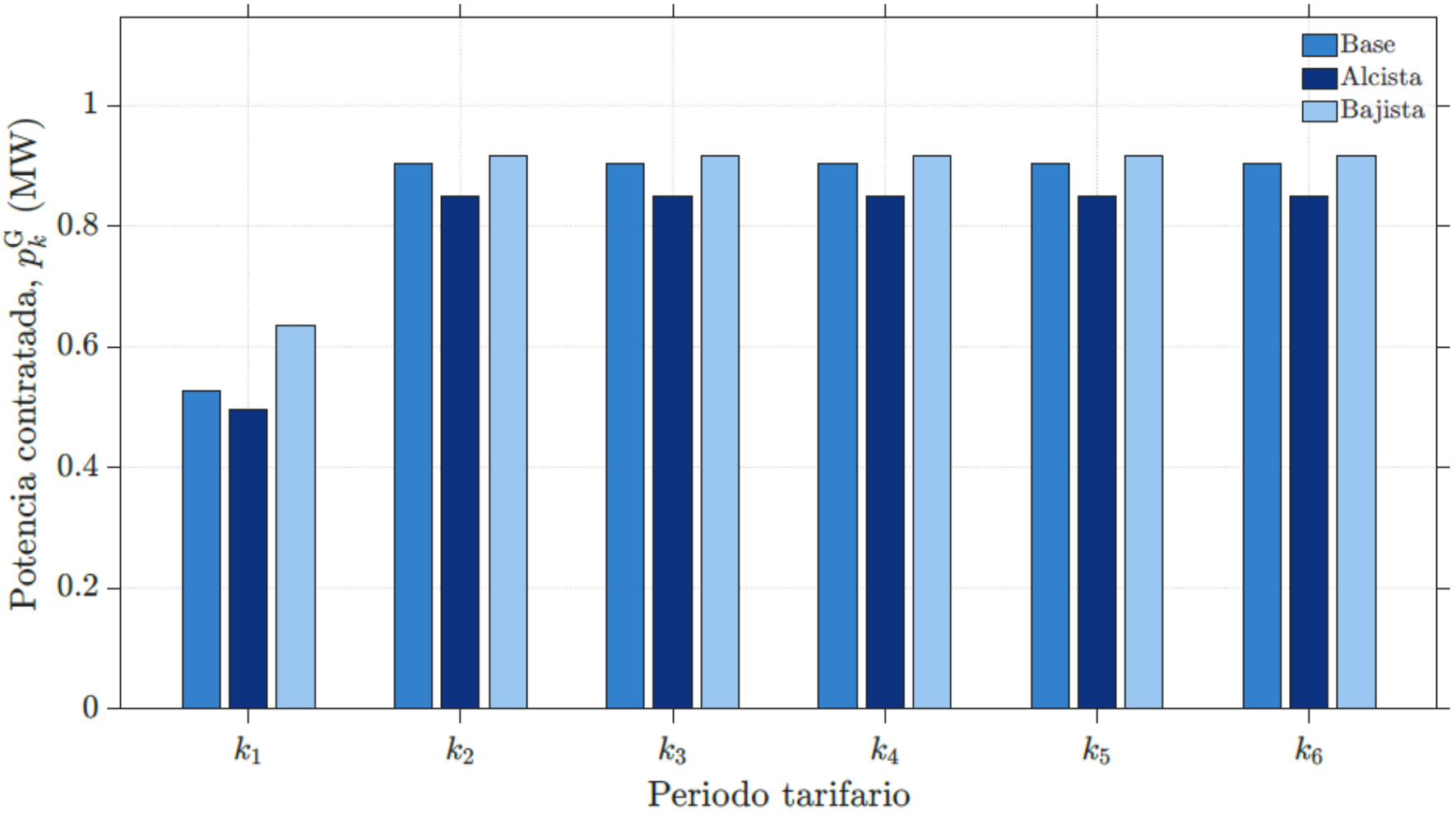
Este documento no es una certificación catastral, pero sus datos pueden ser verificados a través del "Acceso a datos catastrales no protegidos de la SEC"

Domingo , 1 de Febrero de 2026

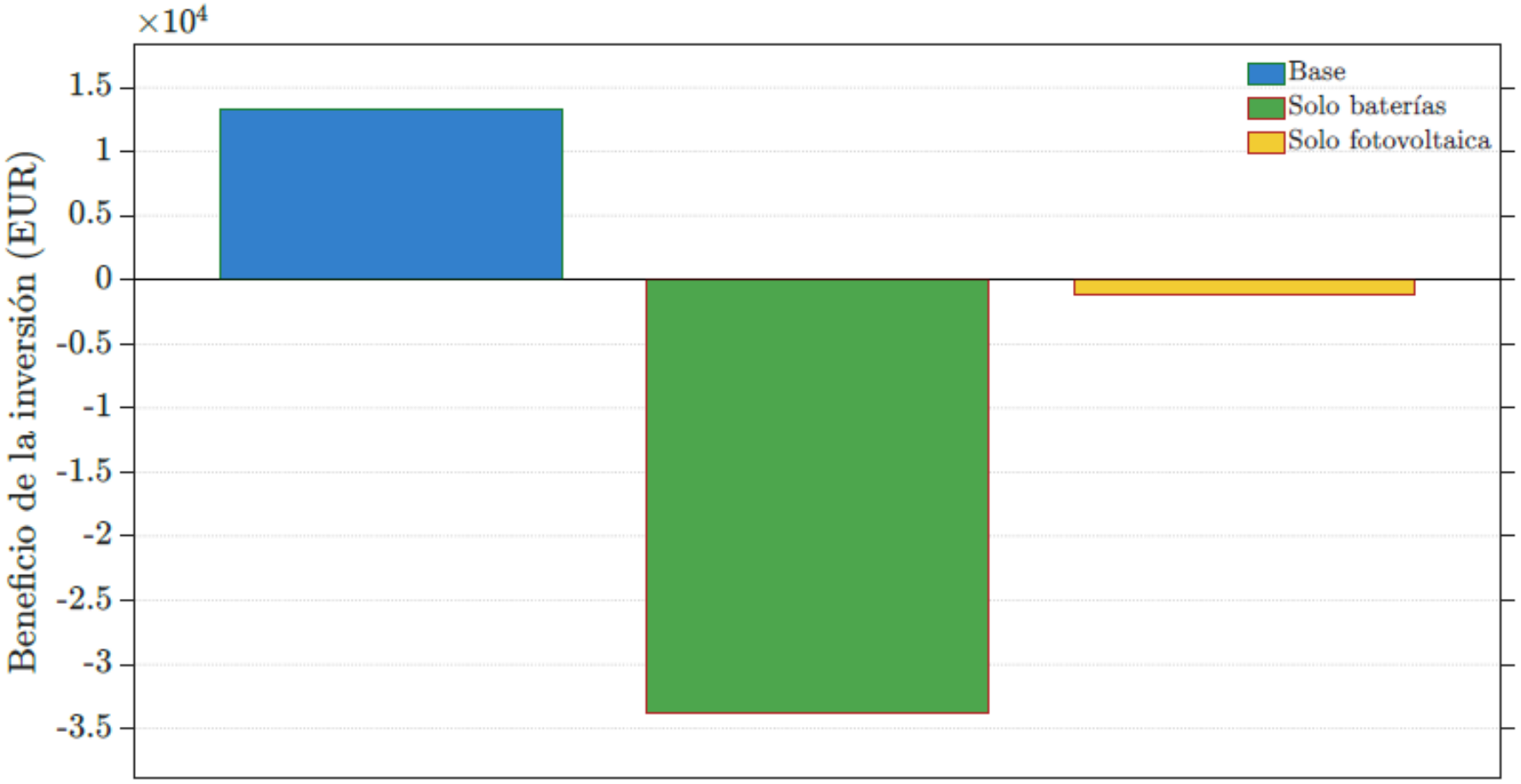
Students' Projects Supervised



Students' Projects Supervised



Students' Projects Supervised



Students' Projects Supervised

Author: José Miguel Fornell Delgado.

Project's Title: Operations research applied to the fleet assignment problem.

Supervisors: Álvaro García-Cerezo and Jesús Rosado Linares.

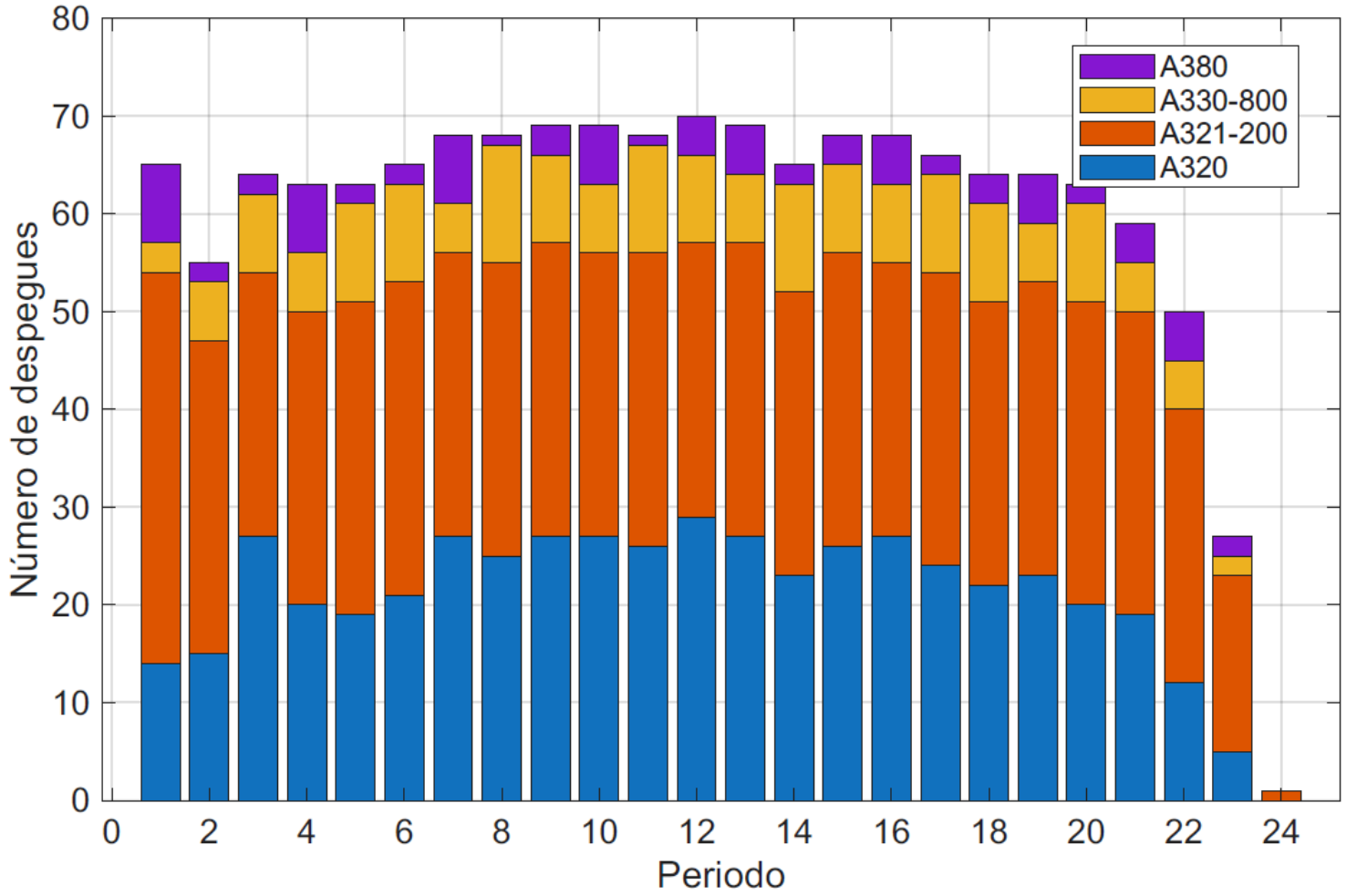
Bachelor's Final Project, Aerospace Engineering Degree, Universidad de Castilla-La Mancha.

Date of defense: July 2026.

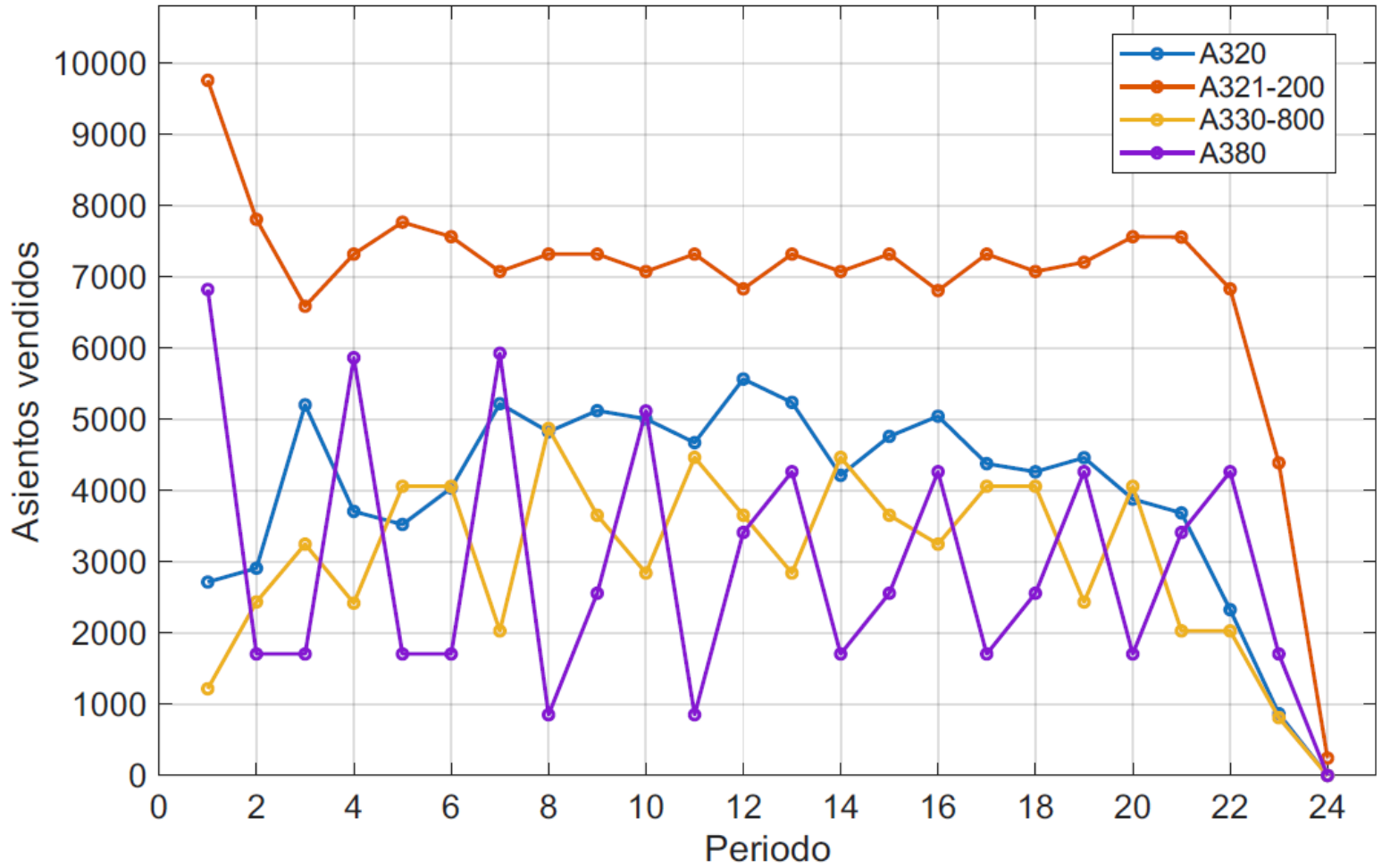
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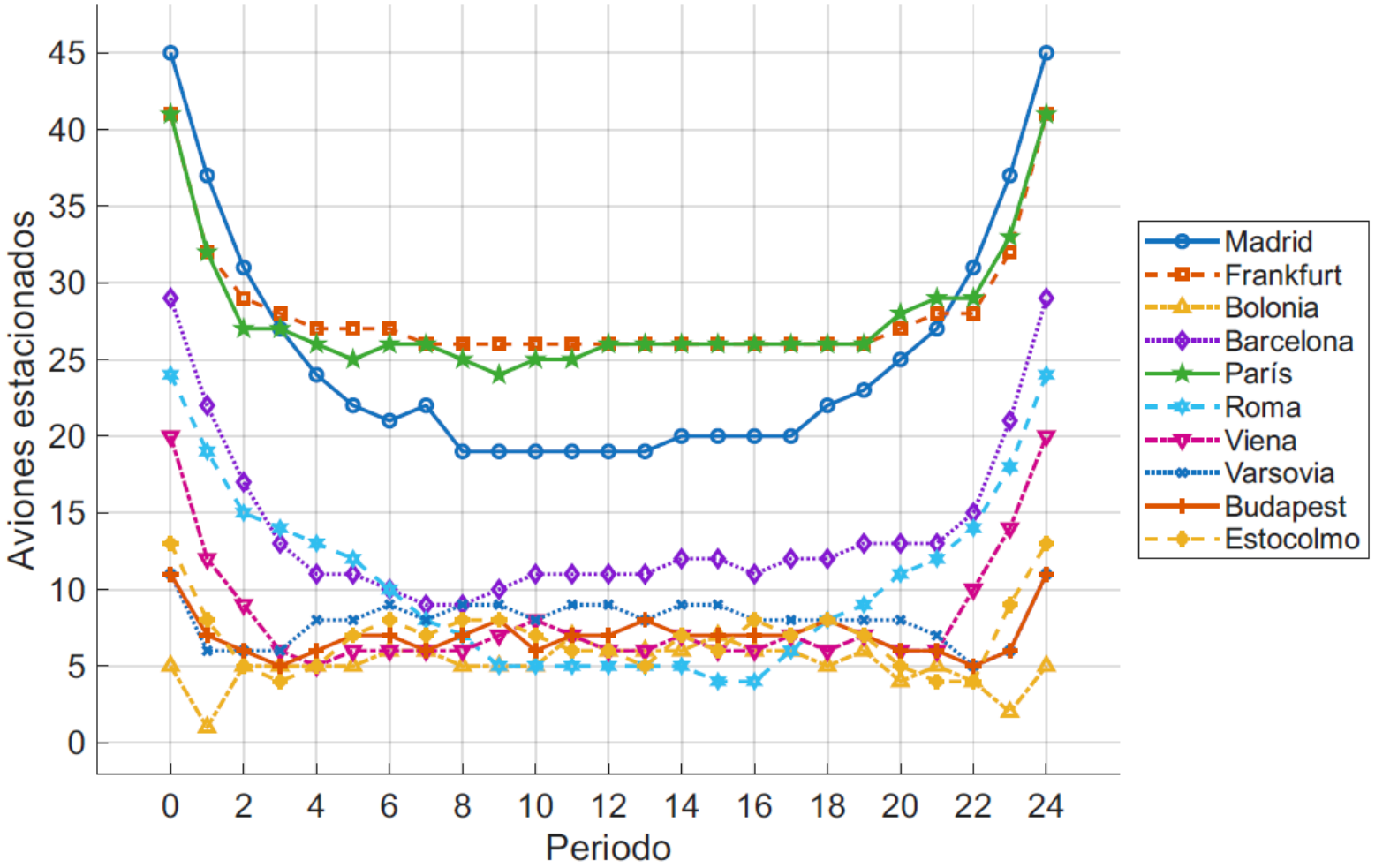
Students' Projects Supervised



Students' Projects Supervised



Students' Projects Supervised



4. Topics beyond this course

Topics beyond this course

Unit commitment with minimum up and down times:

M. Carrión and J. M. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem", *IEEE Transactions on Power Systems*, vol. 21, no. 3, pp. 1371–1378, Aug. 2006, DOI: 10.1109/TPWRS.2006.876672.

Topics beyond this course

Strategic generation investment:

S. J. Kazempour, A. J. Conejo, and C. Ruiz, "Strategic generation investment using a complementarity approach", *IEEE Transactions on Power Systems*, vol. 26, no. 2, pp. 940–948, May 2011, DOI: 10.1109/TPWRS.2010.2069573.

Topics beyond this course

Equilibrium models:

Y. Chen, A. S. Siddiqui, and M. Tanaka, *Analysis of Environmental Policy in the Power Sector: Equilibrium Methods and Bi-Level Modeling*. Springer, Cham, 2020. ISBN: 978-3-030-44866-0. DOI: 10.1007/978-3-030-44866-0.

Topics beyond this course

Coordinated transmission and generation expansion:

D. Pozo, E. Sauma, and J. Contreras, “A three-level static MILP model for generation and transmission expansion planning”, *IEEE Transactions on Power Systems*, vol. 28, no. 1, pp. 202–210, Feb. 2013, DOI: 10.1109/TPWRS.2012.2204073.

Topics beyond this course

Coordinated transmission and distribution generation expansion:

N.-U.-R. Chowdhury, Gregorio Muñoz-Delgado, N. Alguacil, and J. Contreras, “An ADMM-based optimization approach for coordinated transmission and distribution generation expansion planning”, *Sustainable Energy, Grids and Networks*, vol. 43, art. no. 101805, Sep. 2025, DOI: 10.1016/j.segan.2025.101805.

Topics beyond this course

Participation in reserve markets:

S. Maiz, L. Baringo, and R. García-Bertrand, “Expansion planning of a price-maker virtual power plant in energy and reserve markets”, *Sustainable Energy, Grids and Networks*, vol. 32, art. no. 100832, Dec. 2022, DOI: 10.1016/j.segan.2022.100832.

Topics beyond this course

Power-purchase agreements:

J. Arellano and M. Carrión, “Electricity procurement of large consumers considering power-purchase agreements”, *Energy Reports*, vol. 9, pp. 5384–5396, Dec. 2023, DOI: 10.1016/j.egyr.2023.04.371.

Topics beyond this course

Wind farm repowering:

A. Fernández-Guillamón, I. C. Gil-García, R. Zárate-Miñano, M Cañas-Carretón, and M. Carrión, "Optimal wind farm repowering under uncertainty with hydrogen hybridization", *IEEE Access*, vol. 14, pp. 89120–89140, Jun. 2026, DOI: 10.1109/ACCESS.2026.3702373.

Topics beyond this course

Coalition of distributed energy resources:

S. Maiz, L. Baringo, and R. García-Bertrand, “Dynamic expansion planning of a commercial virtual power plant through coalition with distributed energy resources considering rival competitors”, *Applied Energy*, vol. 377, part D, art. no. 124665, Jan. 2025, DOI: 10.1016/j.apenergy.2024.124665.

Topics beyond this course

Reliability and distribution networks:

A. Tabares, G. Muñoz-Delgado, J. F. Franco, J. M. Arroyo, and J. Contreras, "Multistage reliability-based expansion planning of ac distribution networks using a mixed-integer linear programming model", *International Journal of Electrical Power & Energy Systems*, vol. 138, art. no. 107916, Jun. 2022, DOI: 10.1016/j.segan.2022.100832.

Topics beyond this course

Accurate charging modeling of batteries (research group led by Hrvoje Pandžić):

https://doi.org/10.1109/TPWRS.2018.2876466

https://doi.org/10.1109/TPWRS.2024.3420215

https://doi.org/10.1109/EEM68581.2026.11589817

Topics beyond this course

Hydrogen in power systems:

J. Barba, M. Cañas-Carretón, M. Carrión, G. R. Hernández-Labrado, C. Merino, J. I. Muñoz, and R. Zárate-Miñano, "Integrating hydrogen into power systems: a comprehensive review", *Sustainability*, vol. 17, art. no. 6117, Jul. 2025, DOI: 10.3390/su17136117.

Topics beyond this course

Scenario generation:

A. J. Conejo, M. Carrión, and J. M. Morales, *Decision Making Under Uncertainty in Electricity Markets*. Springer New York, NY, 2010. ISBN: 978-1-4419-7421-1. DOI: 10.1007/978-1-4419-7421-1.

YouTube content that you should watch

Channel of Prof. Jalal Kazempour (DTU University):
https://www.youtube.com/@jalalkazempour1429

DTU: Lectures on Power & Energy Systems:
https://www.youtube.com/@dtulecturesonpowerenergysy8133

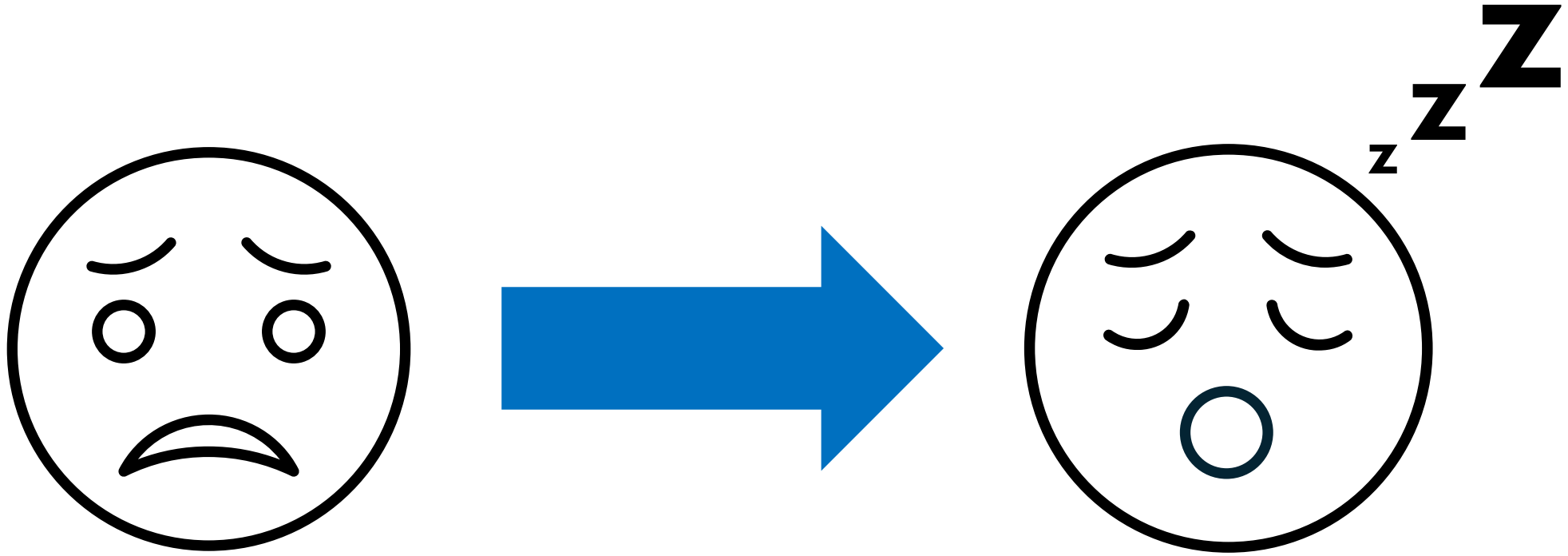
DTU PES Summer School 2026:

https://www.youtube.com/playlist?list=PLTe3_gzTpOcA

Now, the ball is in your court.



Start of
the project



Start of
the project

End of
the project

PEARL Power and Energy Analysis and Research Laboratory

Escuela de Ingeniería Industrial y Aeroespacial Toledo
Universidad de Castilla-La Mancha

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Thank you very much for your attention!

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